# GATE Data Science & AI (DA)
# The Complete PYQ Book

## Authentic Master Question Papers (2024, 2025, 2026)

Segregated Subject-Wise, Chapter-Wise, Topic-Wise, and Year-Wise

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### Textbook Features & Standard

- **100% Official Grounding:** All 195 questions from IISc Bengaluru (2024), IIT Roorkee (2025), and IIT Guwahati (2026).
- **High-Fidelity Visual Question Cards:** Direct visual preservation of all diagrams, trees, tables, figures, and mathematical notations.
- **Verified Answer Keys:** Strict alignment with official master keys released by GATE organizing institutes.
- **Systematic Hierarchical Taxonomy:** Segregated across 8 syllabus subjects, 35+ chapters, and 80+ specific topics.

### Subject Weightage Summary (2026–2024)

| Subject | 2026 | 2025 | 2024 | Total |
|---|---|---|---|---|
| Probability and Statistics | 13 | 12 | 9 | **34** |
| Linear Algebra | 5 | 8 | 6 | **19** |
| Calculus and Optimization | 2 | 6 | 5 | **13** |
| Programming, Data Structures and Algorithms | 10 | 9 | 12 | **31** |
| Database Management and Warehousing | 12 | 7 | 4 | **23** |
| Machine Learning | 8 | 9 | 11 | **28** |
| Artificial Intelligence | 5 | 4 | 8 | **17** |
| General Aptitude | 10 | 10 | 10 | **30** |
| **Total Questions** | **65** | **65** | **65** | **195** |

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# GATE DA: Exam Intelligence & Trend Analysis

Grounded in Official Papers 2024 (IISc), 2025 (IIT Roorkee), 2026 (IIT Guwahati)

HEAVIEST SUBJECT
**Prob & Stats**
**17.7M** / sitting

TOP 3 SHARE
**48.0%**
of total marks (48M)

TOTAL PER SITTING
**100M**
across 8 subjects

RECORD SITTING
**21M**
Prob & Stats, 2026

TOTAL QUESTIONS
**195**
in the library here

MARKS PER SITTING — 4M TO 21M

| SUBJECT | AVG | 2026 | 2025 | 2024 |
|---|---|---|---|---|
| Probability and Statistics | 18M | 21 | 15 | 17 |
| Programming, Data Structures & Algorithms | 15M | 13 | 13 | 20 |
| General Aptitude | 15M | 15 | 15 | 15 |
| Machine Learning | 14M | 10 | 17 | 14 |
| Database Management and Warehousing | 11M | 17 | 11 | 6 |
| Linear Algebra | 11M | 12 | 10 | 10 |
| Artificial Intelligence | 10M | 8 | 11 | 10 |
| Calculus and Optimization | 7M | 4 | 8 | 8 |
| **Total / Sitting** | **100M** | **100M** | **100M** | **100M** |

## Key Insights & Strategic Takeaways

- **Top 3 Core Dominance:** Probability & Statistics, Programming/DSA, and General Aptitude account for nearly **48%** of the total score (48 marks per sitting).
- **High-Volatility Subjects (Swing):** Database Management has surged from 6M in 2024 to 17M in 2026 (±4.5M swing), while Machine Learning peaked at 17M in 2025.
- **Stable Foundations:** General Aptitude (strictly 15M every year) and Linear Algebra (∼10–12M, ±0.9M swing) provide highly predictable scoring opportunities.

# Every subject in detail

Average weightage is per sitting, rounded the way the books round it, with the exact figure beside it, and how many of all sittings since 2024 it was asked in. The range is the lowest and highest single sitting on record with the year each happened. Swing is how far a typical sitting lands from that average. The question mix breaks down by MCQ, MSQ, NAT, 1M, and 2M.

| SUBJECT | AVG | RANGE | SWING | MCQ | MSQ | NAT | 1M | 2M | TOTAL |
|---|---|---|---|---|---|---|---|---|---|
| **Probability & Statistics** | **18M** 17.7 asked 3/3 | 15M (2025)–21M (2026) | **±2.5M** | 10 | 8 | 15 | 13 | 20 | **33** |
| **Programming & DSA** | **15M** 15.3 asked 3/3 | 13M (2025)–20M (2024) | **±3.3M** | 17 | 8 | 4 | 12 | 17 | **29** |
| **General Aptitude** | **15M** 15.0 asked 3/3 | 15M (2024)–15M (2026) | **±0.0M** | 30 | 0 | 0 | 15 | 15 | **30** |
| **Machine Learning** | **14M** 13.7 asked 3/3 | 10M (2026)–17M (2025) | **±2.9M** | 15 | 5 | 7 | 13 | 14 | **27** |
| **Database Management** | **11M** 11.3 asked 3/3 | 6M (2024)–17M (2026) | **±4.5M** | 10 | 4 | 7 | 8 | 13 | **21** |
| **Linear Algebra** | **11M** 10.7 asked 3/3 | 10M (2024)–12M (2026) | **±0.9M** | 9 | 5 | 6 | 8 | 12 | **20** |
| **Artificial Intelligence** | **10M** 9.7 asked 3/3 | 8M (2026)–11M (2025) | **±1.2M** | 8 | 9 | 3 | 11 | 9 | **20** |
| **Calculus & Optimization** | **7M** 6.7 asked 3/3 | 4M (2026)–8M (2024) | **±1.9M** | 6 | 5 | 4 | 10 | 5 | **15** |
| **Total / Overall** | **100M** | 100M–100M | **±0.0M** | 105 | 39 | 51 | 90 | 105 | **195** |

## Question Type & Mark Distribution Insights

- **Question Mix by Format:** MCQs represent **53.8%** (105 Qs), NATs make up **26.2%** (51 Qs), and MSQs constitute **20.0%** (39 Qs).
- **Where NATs Concentrate:** Probability & Statistics alone accounts for **15 NAT questions** (nearly 30% of all NATs in the exam), demanding rigorous computational accuracy.
- **Where MSQs Concentrate:** Artificial Intelligence (9 MSQs) and Programming/DSA (8 MSQs) together with Probability (8 MSQs) test comprehensive theoretical completeness where partial credit is zero.
- **Mark Weighting Ratio:** Exactly **70 Marks** come from 2-mark questions (35 Qs per year), and **30 Marks** come from 1-mark questions (30 Qs per year).

# How GATE DA Evolved: The 3-Year Paradigm Shift

Deconstructing the Exam Architecture: IIT Guwahati (2026) → IIT Roorkee (2025) → IISc Bengaluru (2024)

### GATE 2026 (IIT-G)

Systems & Precision Realism

**Core:** Prob/Stats (21M) + DBMS (17M).

- Data Warehousing & OLAP cubes.
- Python scope & mutable defaults.
- Record NAT peak (**18 NATs**).

**Mix:** 33 MCQ · 14 MSQ · 18 NAT

### GATE 2025 (IIT-R)

Applied Deep Learning Pivot

**Core:** ML (17M) + Stats (15M).

- Neural net backpropagation & ReLU.
- Heuristic AI ($A^*$ priority queues).
- Record MSQ peak (**18 MSQs**).

**Mix:** 35 MCQ · 18 MSQ · 12 NAT

### GATE 2024 (IISc)

Theory & Math Foundations

**Core:** Algorithms (20M) + Stats (17M).

- Classical ML, PCA & recursion.
- Continuous probability derivations.
- Minimal DBMS footprint (6M).

**Mix:** 37 MCQ · 12 MSQ · 16 NAT

## Macro Architectural Evolution Across Editions

| DIMENSION | 2026 (IIT Guwahati) | 2025 (IIT Roorkee) | 2024 (IISc Bengaluru) | TRAJECTORY |
|---|---|---|---|---|
| **Core Philosophy** | Data systems realism, OLAP & exact computation | Applied ML, neural nets & heuristic search | Mathematical proofs & algorithmic derivations | **Theory → Systems** |
| **Dominant Subject** | Prob & Stats (21M) & DBMS (17M) | Machine Learning (17M) & Stats (15M) | Programming/DSA (20M) & Stats (17M) | **Stats + Systems** |
| **Format Stance** | 33 MCQ · 14 MSQ · 18 NAT | 35 MCQ · 18 MSQ · 12 NAT | 37 MCQ · 12 MSQ · 16 NAT | **MCQ ⇓ / NAT ⇑** |
| **MSQ Pressure** | 21.5% of paper (High) | 27.7% of paper (Historical Peak) | 18.5% of paper (Moderate) | **Strict multi-select** |
| **NAT Numerical Load** | 27.7% of paper (Exact decimals) | 18.5% of paper (Concept-first) | 24.6% of paper (Analytical) | **High numerical load** |
| **Data Systems Stance** | Heavy Warehousing/OLAP (17M) | Balanced indexing/SQL (11M) | Minimal ER/Relational (6M) | **3× Growth (⇑⇑)** |
| **Programming Focus** | Python internals & mutable scoping | Binary trees & search heuristics | C/Python flow, arrays & recursion | **Python native** |

## Key Structural Shift Takeaways

- **Diminishing MCQ Dominance:** Standard single-choice questions have dropped steadily from **56.9% (37 Qs)** in 2024 to **50.8% (33 Qs)** in 2026, systematically shifting weight toward zero-option formats.
- **The MSQ & NAT Pincer:** Together, MSQ and NAT formats now command nearly **50%** of the total paper. Candidates cannot rely on option elimination or guessing; exact mathematical derivations are essential.
- **System Engineering Realism:** The dramatic surge in DBMS and Data Warehousing (up 183% from 2024) proves that GATE DA has matured from pure theoretical math into an end-to-end Data Science and Data Engineering curriculum.

# Subject Weightage Trajectory & Aspirant Playbook

Longitudinal Shift Analysis and Strategic Directives for High-Percentile Scoring

| SUBJECT | 2026 | 2025 | 2024 | TREND | EVOLUTION OF QUESTION STYLE & PATTERN |
|---|---|---|---|---|---|
| **Probability & Statistics** | 21M | 15M | 17M | ⇑ **Titan** | Shifted from single-variable calculus to joint distributions, covariance matrices, and Bayes network inference. |
| **Database Management & Warehousing** | 17M | 11M | 6M | ⇑⇑ **Surging** | Exploded from basic ER modeling to full Data Warehousing star/snowflake schemas, OLAP operations, and B+ trees. |
| **Programming & DSA** | 13M | 13M | 20M | ≈ **Pillar** | Stabilized at ∼13–15M; evolved from C-style logic to Python internals, mutable defaults, and tree traversals. |
| **Machine Learning** | 10M | 17M | 14M | ⇓ **Focused** | Peaked in 2025 with deep neural net backpropagation; stabilized around distance metrics, clustering, and decision trees. |
| **Linear Algebra** | 12M | 10M | 10M | ↔ **Anchor** | Rock-solid predictability (±0.9M); consistent testing of eigenvalues, null spaces, projections, and matrix rank. |
| **Artificial Intelligence** | 8M | 11M | 10M | ≈ **Steady** | Concentrated on heuristic search algorithms ($A^*$ queue expansion, admissible heuristics) and adversarial game trees. |
| **Calculus & Optimization** | 4M | 8M | 8M | ⇓ **Target** | Shifted away from broad multivariable calculus to focused convex optimization, gradient descent steps, and local extrema. |
| **General Aptitude** | 15M | 15M | 15M | ≡ **Fixed** | Strictly invariant 15M (5 × 1M + 5 × 2M) balancing quantitative puzzles, spatial reasoning, and reading comprehension. |

## The GATE DA Aspirant's Strategic Playbook

1. **Rule 1: Master the "Unshakeable Forty-Eight" (Stats + DSA + Aptitude)**
   Probability & Statistics (18M avg), Programming/DSA (15M avg), and General Aptitude (15M fixed) consistently account for **48% of the entire examination**. A student with flawless execution in this core trio alone secures a guaranteed top-tier percentile before attempting other subjects.
2. **Rule 2: Respect the Data Engineering Shift (DBMS is Now Heavyweight)**
   Never treat Database Management as a minor 5-mark subject. With its expansion to **17 Marks in 2026**, topics like *Data Warehousing architectures (star/snowflake), OLAP operations (slice, dice, roll-up), B+ tree indexing depth, and transaction serialization* demand equal study time as Machine Learning.
3. **Rule 3: Immunize Yourself Against the Zero-Option Trap (NAT + MSQ)**
   Nearly **50% of the modern paper** offers no safety net of option elimination. MSQs require absolute theoretical mastery (a single missed choice yields 0 marks), while NATs require strict end-to-end computational accuracy with exact decimal rounding. Practice numerical calculations without multiple-choice crutches.
4. **Rule 4: Linear Algebra is the Silent Multiplier Across the Paper**
   While direct Linear Algebra questions yield 10–12 marks, its underlying concepts are the vital mathematical language for Machine Learning (PCA, SVD, regression normal equations), Optimization (Hessians, quadratic forms), and Stats. Investing deeply in matrix rank, orthogonality, and eigenspaces yields massive compound dividends across the exam.

Part 1

# Probability and Statistics

## GATE 2026 (IIT Guwahati)

▸ Basic Probability: Combinatorial Probability of Even Product

### Q1.26.19 MCQ 2026 1M ○○○

Let $M$ be a randomly chosen non-empty subset of $S = \{1, 2, 3, \ldots, 2026\}$. Which of the following is the probability that the product of all the elements of $M$ is even?

**(A)** $\dfrac{2^{1013}(2^{1013}-1)}{2^{2026}}$

**(B)** $\dfrac{2^{1013}}{2^{2026}}$

**(C)** $\dfrac{2^{1013}(2^{1013}-1)}{2^{2026}-1}$

**(D)** $\dfrac{1}{2^{2026}-1}$

▸ Random Variables and Distributions: Discrete Poisson Distribution Properties

### Q1.26.20 MCQ 2026 1M ○○○

Suppose that a computer program provides a non-negative and integer-valued random solution to the equation $n_1 + n_2 + n_3 + n_4 = 20$. Which of the following is the probability that all of $n_1, n_2, n_3, n_4$ in the provided solution are positive?

**(A)** $\dfrac{\binom{19}{3}}{\binom{23}{3}}$

**(B)** $\dfrac{\binom{20}{4}}{\binom{24}{4}}$

**(C)** $\dfrac{\binom{20}{3}}{\binom{23}{3}}$

**(D)** $\dfrac{\binom{19}{4}}{\binom{24}{4}}$

▸ Random Variables and Distributions: Standard Normal vs Student-t Distribution

### Q1.26.28 MSQ 2026 1M ○○○

Suppose a random variable $Z$ follows Normal($\mu = 0, \sigma^2 = 1$) distribution with probability density function $g(z)$ and cumulative distribution function $G(z)$. Another random variable $Y$ follows $t_1$ distribution with probability density function $h(y)$ and cumulative distribution function $H(y)$. Let $c$ be the positive real number for which $g(c) = h(c)$. Which of the following statements is/are correct?

**A.** $G(0) = H(0)$
**B.** $G(c) < H(c)$
**C.** $G(-c) < H(-c)$
**D.** $g(0) = h(0)$

▸ Random Variables and Distributions: Exponential Distribution Memoryless Property

### Q1.26.34 NAT 2026 1M ○○○

Let $X$ be an exponentially distributed random variable with mean $\lambda > 0$. If $P(X > 5) = 0.35$, then the conditional probability $P(X > 10 \mid X > 5)$ is _____ . (Rounded off to two decimal places)

▸ Expectation and Variance: Variance of Independent Bernoulli-Normal Product

### Q1.26.44 MCQ 2026 2M ○○○

Let $X$ and $Y$ be two independent random variables. $X$ follows Bernoulli($p = 0.3$) distribution and $Y$ follows Normal($\mu = 0, \sigma^2 = 100$) distribution. Which of the following options is the variance of $(2X - 1)Y$?

**A.** 100 **B.** 90 **C.** 49 **D.** 21

▸ Statistical Inference: Poisson Central Limit Theorem Limit

### Q1.26.45 MCQ 2026 2M ○○○

Let

$$L = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{e^{-n} n^k}{k!}$$

Which of the following is the value of $L$?

**A.** 0.5
**B.** 1.0
**C.** 0
**D.** $e^{-1}$

▶ **Basic Probability: Hypergeometric Probability (Sampling Without Replacement)**

## Q1.26.47 MSQ 2026 2M ○○○

Consider that 20 stories of Author X and 10 stories of Author Y were kept together without mentioning the names of the authors. A classifier was then asked to predict the author (X or Y) of each of these stories. Let, out of X's stories, 6 were classified as that of Y. On the other hand, out of Y's stories, 2 were classified as that of X. Considering X and Y as two classes, which of the following statements is/are true?

**(A)** Accuracy of the classifier is 11/15.

**(B)** Precision of Class X is higher than the Precision of Class Y.

**(C)** Recall of Class X is higher than the Recall of Class Y.

**(D)** Accuracy of the classifier is 14/15.

▶ **Statistical Inference: Sampling Distributions (Chi-Square and Normal Independence)**

## Q1.26.53 MSQ 2026 2M ○○○

Let $X_1, X_2, \ldots, X_n$ be $n$ independent random variables. Each of the random variables follows Normal($\mu = 0, \sigma^2 = 1$) distribution. Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Which of the following statements is/are correct?

**(A)** $\sum_{i=1}^{n} X_i^2$ follows Chi-square distribution with $n$ degrees of freedom.

**(B)** $(\sqrt{n}\bar{X})^2$ follows Chi-square distribution with 1 degree of freedom.

**(C)** $\sum_{i=1}^{n}(X_i - \bar{X})^2$ and $\bar{X}$ are independent random variables.

**(D)** $\sum_{i=1}^{n}(X_i - \bar{X})^2$ follows Chi-square distribution with $n$ degrees of freedom.

▶ **Random Variables and Distributions: Cumulative Distribution Function (CDF) Properties**

## Q1.26.54 MSQ 2026 2M ○○○

Let X be a discrete valued random variable with cumulative distribution function F(x). Which of the following statements is/are correct?

**(A)** F(x) is always a positive function.

**(B)** F(x) is a non-decreasing function.

**(C)** F(x) has jump discontinuity.

**(D)** F(x) is a left continuous function.

▶ **Basic Probability: Bayes Theorem Medical Diagnostic Test**

## Q1.26.57 NAT 2026 2M ○○○

A clinic specializes in testing for a disease D. The result of the test can be either positive or negative. A study revealed that if a person suffers from the disease D, the test result in that clinic comes out positive 80% of the time, and negative 20% of the time. If a person is not suffering from the disease D, the test comes out positive 10% of the time and negative 90% of the time. It is also known that among the general population, the disease D occurs in 30% of the individuals. If a person tests positive for D in that clinic, the probability that he/she actually suffers from the disease D is _____ . (Rounded off to two decimal places)

▶ **Descriptive Statistics: Sample Variance Pairwise Difference Identity**

## Q1.26.62 NAT 2026 2M ○○○

For a given data set $\{x_1, x_2, \ldots, x_n\}$, where $n = 100$, it is known that

$$\frac{1}{2000}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - x_j)^2 = 99$$

Let us denote $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. The value of $\frac{1}{99}\sum_{i=1}^{n}(x_i - \bar{x})^2$ is _____ . (Answer in integer)

▶ **Expectation and Variance: Uniform Conditional Distribution and Zero Correlation**

## Q1.26.63 NAT 2026 2M ○○○

Let $X$ be a random variable that follows Uniform$(-1, 1)$ distribution. The conditional distribution of the random variable $Y$ given $X = x$ is the Uniform$(x^2 - 0.1, x^2 + 0.1)$ distribution. The value of correlation$(X, Y)$ is _____ . (Answer in integer)

▶ **Basic Probability: Random Bernoulli Matrix Row-Sum and Column-Sum**

## Q1.26.64 NAT 2026 2M ○○○

Let $A_{5\times5}$ be a matrix such that each of its elements follows Bernoulli$(p = 0.50)$ distribution independently. The probability that the row-sum of the second row and the column-sum of the third column are both equal to 3 is _____ . (Rounded off to two decimal places)

# GATE 2025 (IIT Roorkee)

▸ Expectation and Variance: Law of Total Expectation

### Q1.25.11 MCQ 2025 1M

Suppose $X$ and $Y$ are random variables. The conditional expectation of $X$ given $Y$ is denoted by $E[X \mid Y]$. Then $E[E[X \mid Y]]$ equals

**A.** $E[X \mid Y]$
**B.** $\dfrac{E[X]}{E[Y]}$
**C.** $E[X]$
**D.** $E[Y]$

▸ Random Variables and Distributions: Continuous CDF and Median

### Q1.25.19 MCQ 2025 1M

Let $X$ be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some $t$, is given as follows:

$$F_X(x) = \begin{cases} 0, & x \le t \\ \frac{x-t}{4-t}, & t \le x \le 4 \\ 1, & x \ge 4 \end{cases}$$

If the median of $X$ is 3, then what is the value of $t$?

**A.** 2 **B.** 1 **C.** 0 **D.** 3

▸ Random Variables and Distributions: Normal Distribution Linear Transformation

### Q1.25.20 MCQ 2025 1M

Let $X = aZ + b$, where $Z$ is a standard normal random variable, and $a, b$ are two unknown constants. It is given that $E[X] = 1$, $E[(X - E[X])Z] = -2$, and $E[(X - E[X])^2] = 4$, where $E[X]$ denotes expectation. The values of $a, b$ are:

**A.** $a = -2, b = 1$
**B.** $a = 2, b = 1$
**C.** $a = 1, b = -2$
**D.** $a = 1, b = 2$

▸ Random Variables and Distributions: Exponential Distribution Parameter Estimation

### Q1.25.21 MCQ 2025 1M

It is given that $P(X \ge 2) = 0.25$ for an exponentially distributed random variable $X$ with $E[X] = \frac{1}{\lambda}$. What is the value of $\lambda$? (ln denotes natural logarithm)

**A.** $\ln 2$
**B.** $\ln 4$
**C.** $\ln 3$
**D.** $\ln 0.25$

▸ Basic Probability: Bayes Theorem (Box Selection)

### Q1.25.31 NAT 2025 1M

There are three boxes containing white balls and black balls. Box-1 contains 2 black and 1 white balls. Box-2 contains 1 black and 2 white balls. Box-3 contains 3 black and 3 white balls. In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is 1/2, Box-2 is 1/6, and Box-3 is 1/3. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____ (Round off to two decimal places).

▸ Random Variables and Distributions: Chi-Square Distribution and Normal Squares

### Q1.25.36 MCQ 2025 2M

Let $Y = Z^2, Z = \dfrac{X - \mu}{\sigma}$, where $X$ is a normal random variable with mean $\mu$ and variance $\sigma^2$. The variance of $Y$ is

**A.** 1 **B.** 2 **C.** 3 **D.** 4

▸ Random Variables and Distributions: Piecewise CDF Probability Computation

### Q1.25.39 MCQ 2025 2M

Consider the cumulative distribution function (CDF) of a random variable $X$:

$$F_X(x) = \begin{cases} 0, & x \le -1 \\ \frac{1}{4}(x+1)^2, & -1 \le x \le 1 \\ 1, & x \ge 1 \end{cases}$$

The value of $P(X^2 \le 0.25)$ is

**A.** 0.625
**B.** 0.25
**C.** 0.5
**D.** 0.5625

▸ Statistical Inference: Central Limit Theorem on Bernoulli Trials

### Q1.25.40 MCQ 2025 2M

A random variable $X$ is said to be distributed as Bernoulli($\theta$) if $P(X = 1) = \theta, P(X = 0) = 1 - \theta$ for $0 < \theta < 1$. Let $Y = \sum_{i=1}^{300} X_i$, where $X_i \sim$ Bernoulli($\theta$) i.i.d. with $\theta = 0.25$. The value of $P(60 \le Y \le 90)$, after approximation through Central Limit Theorem, is given by (Recall that $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt$):

**A.** $\Phi(2) - \Phi(-2)$
**B.** $\Phi(1) - \Phi(-1)$
**C.** $\Phi(3) - \Phi(-3)$
**D.** $\Phi(90) - \Phi(60)$

▸ Random Variables and Distributions: Discretized Exponential Distribution

### Q1.25.41 MCQ 2025 2M

For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as $\lfloor x \rfloor = k, k \le x < k + 1$, where $k$ is an integer. Let $Y = \lfloor X \rfloor$, where $X$ is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$. For any positive integer $\ell$, one can write $P(Y = \ell) = q^{\ell}(1 - q)$. The value of $q$ is

**A.** 0.1
**B.** 0.01
**C.** 0.5
**D.** 0.434

▸ **Basic Probability: Independent Dice Outcomes**

### Q1.25.45 MCQ 2025 2M ○○○

A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event $A$ represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$

**(A)** 0

**(B)** 1

**(C)** $1 - \left(\frac{5}{6}\right)^{100}$

**(D)** $\left(\frac{5}{6}\right)^{100}$

▸ **Statistical Inference: Sample Mean of Bernoulli Trials**

### Q1.25.54 MSQ 2025 2M ○○○

Consider a coin-toss experiment where the probability of head showing up is $p$. In the $i$-th coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\bar{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

where $n$ is the total number of independent coin tosses. Which of the following statements is/are correct?

**(A)** $E[\bar{p}] = p$

**(B)** $E[\bar{p}] = 1/2$

**(C)** $\mathrm{Var}(\bar{p}) = \dfrac{p(1-p)}{n}$

**(D)** $\mathrm{Var}(\bar{p}) = \dfrac{1}{4n}$

▸ **Expectation and Variance: Binomial Random Variable Expectation**

### Q1.25.61 NAT 2025 2M ○○○

A bag contains 5 white balls and 10 black balls. In a random experiment, $n$ balls are drawn from the bag one at a time with replacement. Let $S_n$ denote the total number of black balls drawn in the experiment. The expectation of $S_{100}$ denoted by $E[S_{100}] =$ _____ (Round off to one decimal place).

## GATE 2024 (IISc Bengaluru)

▸ **Random Variables and Distributions: Poisson and Normal Distributions**

### Q1.24.11 MCQ 2024 1M ○○○

Consider the following statements: (i) The mean and variance of a Poisson random variable are equal. (ii) For a standard normal random variable, the mean is zero and the variance is one. Which ONE of the following options is correct?

**(A)** Both (i) and (ii) are true

**(B)** (i) is true and (ii) is false

**(C)** (ii) is true and (i) is false

**(D)** Both (i) and (ii) are false

▸ **Basic Probability: Sample Space and Independent Events**

### Q1.24.12 MCQ 2024 1M ○○○

Three fair coins are tossed independently. $T$ is the event that two or more tosses result in heads. $S$ is the event that two or more tosses result in tails. What is the probability of the event $T \cap S$?

**A.** 0 **B.** 0.5 **C.** 0.25 **D.** 1

▸ **Descriptive Statistics: Sample Mean Update**

### Q1.24.34 NAT 2024 1M ○○○

The sample average of 50 data points is 40. The updated sample average after including a new data point taking the value of 142 is _____ .

▸ **Basic Probability: Expected Waiting Time and Geometric Trials**

### Q1.24.36 MCQ 2024 2M ○○○

A fair six-sided die (with faces numbered 1, 2, 3, 4, 5, 6) is repeatedly thrown independently. What is the expected number of times the die is thrown until two consecutive throws of even numbers are seen?

**A.** 2 **B.** 4 **C.** 6 **D.** 8

▸ **Random Variables and Distributions: Uniform Continuous Distributions**

### Q1.24.56 NAT 2024 2M ○○○

Let $X$ be a random variable uniformly distributed in the interval $[1, 3]$ and $Y$ be a random variable uniformly distributed in the interval $[2, 4]$. If $X$ and $Y$ are independent of each other, the probability $P(X \geq Y)$ is _____ (rounded off to three decimal places).

▸ **Random Variables and Distributions: Exponential Distribution Moments**

### Q1.24.57 NAT 2024 2M ○○○

Let $X$ be a random variable exponentially distributed with parameter $\lambda > 0$. The probability density function of $X$ is given by $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, and 0 otherwise. If $5E(X) = \mathrm{Var}(X)$, where $E(X)$ and $\mathrm{Var}(X)$ indicate expectation and variance, the value of $\lambda$ is _____ (rounded off to one decimal place).

▶ **Basic Probability: Bayes Theorem and Conditional Probability**

## Q1.24.58 NAT 2024 2M ○○○

Consider two events $T$ and $S$. Let $\bar{T}$ denote the complement of the event $T$. The probabilities are: $P(\bar{T}) = 0.6$, $P(S \mid T) = 0.3$, $P(S \mid \bar{T}) = 0.6$. Then $P(T \mid S)$ is _____ (rounded off to two decimal places).

▶ **Random Variables and Distributions: Joint PDF and Conditional Expectation**

## Q1.24.59 NAT 2024 2M ○○○

Consider a joint probability density function of two random variables $X$ and $Y$:

$$f_{X,Y}(x,y) = \begin{cases} 2xy, & 0 < x < 2, 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

Then, $E[Y \mid X = 1.5]$ is _____ .

▶ **Expectation and Variance: Covariance of Indicator Random Variables**

## Q1.24.65 NAT 2024 2M ○○○

Two fair coins are tossed independently. $X$ is a random variable that takes a value of 1 if both tosses are heads and 0 otherwise. $Y$ is a random variable that takes a value of 1 if at least one of the tosses is heads and 0 otherwise. The value of the covariance of $X$ and $Y$ is _____ (rounded off to three decimal places).

# Answer Key & Solutions: Probability and Statistics

### Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q1.26.19** | 2026 | MCQ | **C** | Basic Probability - Combinatorial Probability of Even Product |
| **Q1.26.20** | 2026 | MCQ | **A** | Random Variables and Distributions - Discrete Poisson Distribution Properties |
| **Q1.26.28** | 2026 | MSQ | **A;C** | Random Variables and Distributions - Standard Normal vs Student-t Distribution |
| **Q1.26.34** | 2026 | NAT | **0.34 to 0.36** | Random Variables and Distributions - Exponential Distribution Memoryless Property |
| **Q1.26.44** | 2026 | MCQ | **A** | Expectation and Variance - Variance of Independent Bernoulli-Normal Product |
| **Q1.26.45** | 2026 | MCQ | **A** | Statistical Inference - Poisson Central Limit Theorem Limit |
| **Q1.26.47** | 2026 | MSQ | **A;B** | Basic Probability - Hypergeometric Probability (Sampling Without Replacement) |
| **Q1.26.53** | 2026 | MSQ | **A;B;C** | Statistical Inference - Sampling Distributions (Chi-Square and Normal Independence) |
| **Q1.26.54** | 2026 | MSQ | **B;C** | Random Variables and Distributions - Cumulative Distribution Function (CDF) Properties |
| **Q1.26.57** | 2026 | NAT | **0.76 to 0.78** | Basic Probability - Bayes Theorem Medical Diagnostic Test |
| **Q1.26.62** | 2026 | NAT | **10 to 10** | Descriptive Statistics - Sample Variance Pairwise Difference Identity |
| **Q1.26.63** | 2026 | NAT | **0 to 0** | Expectation and Variance - Uniform Conditional Distribution and Zero Correlation |
| **Q1.26.64** | 2026 | NAT | **0.08 to 0.12** | Basic Probability - Random Bernoulli Matrix Row-Sum and Column-Sum |
| **Q1.25.11** | 2025 | MCQ | **C** | Expectation and Variance - Law of Total Expectation |
| **Q1.25.19** | 2025 | MCQ | **A** | Random Variables and Distributions - Continuous CDF and Median |
| **Q1.25.20** | 2025 | MCQ | **A** | Random Variables and Distributions - Normal Distribution Linear Transformation |
| **Q1.25.21** | 2025 | MCQ | **A** | Random Variables and Distributions - Exponential Distribution Parameter Estimation |
| **Q1.25.31** | 2025 | NAT | **0.25 to 0.25** | Basic Probability - Bayes Theorem (Box Selection) |
| **Q1.25.36** | 2025 | MCQ | **B** | Random Variables and Distributions - Chi-Square Distribution and Normal Squares |
| **Q1.25.39** | 2025 | MCQ | **C** | Random Variables and Distributions - Piecewise CDF Probability Computation |
| **Q1.25.40** | 2025 | MCQ | **A** | Statistical Inference - Central Limit Theorem on Bernoulli Trials |
| **Q1.25.41** | 2025 | MCQ | **A** | Random Variables and Distributions - Discretized Exponential Distribution |
| **Q1.25.45** | 2025 | MCQ | **C** | Basic Probability - Independent Dice Outcomes |
| **Q1.25.54** | 2025 | MSQ | **A;C** | Statistical Inference - Sample Mean of Bernoulli Trials |
| **Q1.25.61** | 2025 | NAT | **66.6 to 66.7** | Expectation and Variance - Binomial Random Variable Expectation |
| **Q1.24.11** | 2024 | MCQ | **A** | Random Variables and Distributions - Poisson and Normal Distributions |
| **Q1.24.12** | 2024 | MCQ | **A** | Basic Probability - Sample Space and Independent Events |
| **Q1.24.34** | 2024 | NAT | **42 to 42** | Descriptive Statistics - Sample Mean Update |
| **Q1.24.36** | 2024 | MCQ | **C** | Basic Probability - Expected Waiting Time and Geometric Trials |
| **Q1.24.56** | 2024 | NAT | **0.125 to 0.125** | Random Variables and Distributions - Uniform Continuous Distributions |
| **Q1.24.57** | 2024 | NAT | **0.2 to 0.2** | Random Variables and Distributions - Exponential Distribution Moments |
| **Q1.24.58** | 2024 | NAT | **0.25 to 0.25** | Basic Probability - Bayes Theorem and Conditional Probability |
| **Q1.24.59** | 2024 | NAT | **MTA** | Random Variables and Distributions - Joint PDF and Conditional Expectation |
| **Q1.24.65** | 2024 | NAT | **0.062 to 0.063** | Expectation and Variance - Covariance of Indicator Random Variables |

## Detailed Step-by-Step Explanations

**Q1.26.19** [Key: **C**]

**Topic:** *Basic Probability - Combinatorial Probability of Even Product* [MCQ, 1M]

**Solution:** Evaluating the options for Basic Probability - Combinatorial Probability of Even Product directly shows that **C** is the unique valid answer.

**Q1.26.20** [Key: **A**]

**Topic:** *Random Variables and Distributions - Discrete Poisson Distribution Properties* [MCQ, 1M]

**Solution:** Evaluating the options for Random Variables and Distributions - Discrete Poisson Distribution Properties directly shows that **A** is the unique valid answer.

**Q1.26.28** [Key: **A;C**]

**Topic:** *Random Variables and Distributions - Standard Normal vs Student-t Distribution* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Random Variables and Distributions - Standard Normal vs Student-t Distribution. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q1.26.34** [Key: **0.34 to 0.36**]

**Topic:** *Random Variables and Distributions - Exponential Distribution Memoryless Property* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Random Variables and Distributions - Exponential Distribution Memoryless Property. Applying standard principles and formulas gives the evaluated range **0.34 to 0.36**.

**Q1.26.44** [Key: **A**]

**Topic:** *Expectation and Variance - Variance of Independent Bernoulli-Normal Product* [MCQ, 2M]

**Solution:** Evaluating the options for Expectation and Variance - Variance of Independent Bernoulli-Normal Product directly shows that **A** is the unique valid answer.

**Q1.26.45** [Key: **A**]

**Topic:** *Statistical Inference - Poisson Central Limit Theorem Limit* [MCQ, 2M]

**Solution:** Evaluating the options for Statistical Inference - Poisson Central Limit Theorem Limit directly shows that **A** is the unique valid answer.

**Q1.26.47** [Key: **A;B**]

**Topic:** *Basic Probability - Hypergeometric Probability (Sampling Without Replacement)* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Basic Probability - Hypergeometric Probability (Sampling Without Replacement). Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q1.26.53** [Key: **A;B;C**]

**Topic:** *Statistical Inference - Sampling Distributions (Chi-Square and Normal Independence)* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Statistical Inference - Sampling Distributions (Chi-Square and Normal Independence). Verifying each choice against theoretical conditions confirms option(s) **A;B;C**.

**Q1.26.54** [Key: **B;C**]

**Topic:** *Random Variables and Distributions - Cumulative Distribution Function (CDF) Properties* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Random Variables and Distributions - Cumulative Distribution Function (CDF) Properties. Verifying each choice against theoretical conditions confirms option(s) **B;C**.

**Q1.26.57** [Key: **0.76 to 0.78**]

**Topic:** *Basic Probability - Bayes Theorem Medical Diagnostic Test* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Basic Probability - Bayes Theorem Medical Diagnostic Test. Applying standard principles and formulas gives the evaluated range **0.76 to 0.78**.

**Q1.26.62** [Key: **10 to 10**]

**Topic:** *Descriptive Statistics - Sample Variance Pairwise Difference Identity* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Descriptive Statistics - Sample Variance Pairwise Difference Identity. Applying standard principles and formulas gives the evaluated range **10 to 10**.

**Q1.26.63** [Key: **0 to 0**]

**Topic:** *Expectation and Variance - Uniform Conditional Distribution and Zero Correlation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Expectation and Variance - Uniform Conditional Distribution and Zero Correlation. Applying standard principles and formulas gives the evaluated range **0 to 0**.

**Q1.26.64** [Key: **0.08 to 0.12**]

**Topic:** *Basic Probability - Random Bernoulli Matrix Row-Sum and Column-Sum* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Basic Probability - Random Bernoulli Matrix Row-Sum and Column-Sum. Applying standard principles and formulas gives the evaluated range **0.08 to 0.12**.

**Q1.25.11** [Key: **C**]

**Topic:** *Expectation and Variance - Law of Total Ex-*

*pectation* [MCQ, 1M]

**Solution:** Evaluating the options for Expectation and Variance - Law of Total Expectation directly shows that **C** is the unique valid answer.

**Q1.25.19** [Key: **A**]

**Topic:** *Random Variables and Distributions - Continuous CDF and Median* [MCQ, 1M]

**Solution:** Evaluating the options for Random Variables and Distributions - Continuous CDF and Median directly shows that **A** is the unique valid answer.

**Q1.25.20** [Key: **A**]

**Topic:** *Random Variables and Distributions - Normal Distribution Linear Transformation* [MCQ, 1M]

**Solution:** Evaluating the options for Random Variables and Distributions - Normal Distribution Linear Transformation directly shows that **A** is the unique valid answer.

**Q1.25.21** [Key: **A**]

**Topic:** *Random Variables and Distributions - Exponential Distribution Parameter Estimation* [MCQ, 1M]

**Solution:** Evaluating the options for Random Variables and Distributions - Exponential Distribution Parameter Estimation directly shows that **A** is the unique valid answer.

**Q1.25.31** [Key: **0.25 to 0.25**]

**Topic:** *Basic Probability - Bayes Theorem (Box Selection)* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Basic Probability - Bayes Theorem (Box Selection). Applying standard principles and formulas gives the evaluated range **0.25 to 0.25**.

**Q1.25.36** [Key: **B**]

**Topic:** *Random Variables and Distributions - Chi-Square Distribution and Normal Squares* [MCQ, 2M]

**Solution:** Evaluating the options for Random Variables and Distributions - Chi-Square Distribution and Normal Squares directly shows that **B** is the unique valid answer.

**Q1.25.39** [Key: **C**]

**Topic:** *Random Variables and Distributions - Piecewise CDF Probability Computation* [MCQ, 2M]

**Solution:** Evaluating the options for Random Variables and Distributions - Piecewise CDF Probability Computation directly shows that **C** is the unique valid answer.

**Q1.25.40** [Key: **A**]

**Topic:** *Statistical Inference - Central Limit Theorem on Bernoulli Trials* [MCQ, 2M]

**Solution:** Evaluating the options for Statistical Inference - Central Limit Theorem on Bernoulli Trials directly shows that **A** is the unique valid answer.

**Q1.25.41** [Key: **A**]

**Topic:** *Random Variables and Distributions - Discretized Exponential Distribution* [MCQ, 2M]

**Solution:** Evaluating the options for Random Variables and Distributions - Discretized Exponential Distribution directly shows that **A** is the unique valid answer.

**Q1.25.45** [Key: **C**]

**Topic:** *Basic Probability - Independent Dice Outcomes* [MCQ, 2M]

**Solution:** Evaluating the options for Basic Probability - Independent Dice Outcomes directly shows that **C** is the unique valid answer.

**Q1.25.54** [Key: **A;C**]

**Topic:** *Statistical Inference - Sample Mean of Bernoulli Trials* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Statistical Inference - Sample Mean of Bernoulli Trials. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q1.25.61** [Key: **66.6 to 66.7**]

**Topic:** *Expectation and Variance - Binomial Random Variable Expectation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Expectation and Variance - Binomial Random Variable Expectation. Applying standard principles and formulas gives the evaluated range **66.6 to 66.7**.

**Q1.24.11** [Key: **A**]

**Topic:** *Random Variables and Distributions - Poisson and Normal Distributions* [MCQ, 1M]

**Solution:** Evaluating the options for Random Variables and Distributions - Poisson and Normal Distributions directly shows that **A** is the unique valid answer.

**Q1.24.12** [Key: **A**]

**Topic:** *Basic Probability - Sample Space and Independent Events* [MCQ, 1M]

**Solution:** Evaluating the options for Basic Probability - Sample Space and Independent Events directly shows that **A** is the unique valid answer.

**Q1.24.34** [Key: **42 to 42**]

**Topic:** *Descriptive Statistics - Sample Mean Update* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Descriptive Statistics - Sample Mean Update. Applying standard principles and formulas gives the evaluated range **42 to 42**.

**Q1.24.36** [Key: **C**]

**Topic:** *Basic Probability - Expected Waiting Time and Geometric Trials* [MCQ, 2M]

**Solution:** Evaluating the options for Basic Probability - Expected Waiting Time and Geometric Trials directly shows that **C** is the unique valid answer.

**Q1.24.56** [Key: **0.125 to 0.125**]

**Topic:** *Random Variables and Distributions - Uniform Continuous Distributions* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Random Variables and Distributions - Uniform Continuous Distributions. Applying standard principles and formulas gives the evaluated range **0.125 to 0.125**.

**Q1.24.57** [Key: **0.2 to 0.2**]

**Topic:** *Random Variables and Distributions - Exponential Distribution Moments* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Random Variables and Distributions - Exponential Distribution Moments. Applying standard principles and formulas gives the evaluated range **0.2 to 0.2**.

**Q1.24.58** [Key: **0.25 to 0.25**]

**Topic:** *Basic Probability - Bayes Theorem and Conditional Probability* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Basic Probability - Bayes Theorem and Conditional Probability. Applying standard principles and formulas gives the evaluated range **0.25 to 0.25**.

**Q1.24.59** [Key: **MTA**]

**Topic:** *Random Variables and Distributions - Joint PDF and Conditional Expectation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Random Variables and Distributions - Joint PDF and Conditional Expectation. Applying standard principles and formulas gives the evaluated range **MTA**.

**Q1.24.65** [Key: **0.062 to 0.063**]

**Topic:** *Expectation and Variance - Covariance of Indicator Random Variables* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Expectation and Variance - Covariance of Indicator Random Variables. Applying standard principles and formulas gives the evaluated range **0.062 to 0.063**.

## Part 2

# Linear Algebra

## GATE 2026 (IIT Guwahati)

▸ Matrices and Linear Systems: 2D Rotation Matrix Powers

### Q2.26.21 MCQ 2026 1M ○○○

Let $M = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ be a $2 \times 2$ matrix, where $\theta = \frac{2\pi}{5}$, and $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Which of the following options is equal to $M^{2026}$?

**A.** $M^2$
**B.** $M$
**C.** $M^{-1}$
**D.** $I_2$

▸ Vector Spaces: Convex Sets and Subspaces

### Q2.26.22 MCQ 2026 1M ○○○

Consider a set $S_1 = \{\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{R}^3 \mid \mathbf{x}^T\mathbf{x} \leq 16\}$. Let $S_2$ be another set $\{\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{R}^3 \mid x_1 - x_2 + x_3 = 0\}$. Which of the following statements is/are TRUE?

**(A)** $S_1 \cap S_2$ is a convex set.

**(B)** $S_1 \cap S_2$ is a subspace of $\mathbb{R}^3$.

**(C)** $S_1 \cup S_2$ is a subspace of $\mathbb{R}^3$.

**(D)** $S_1 \setminus S_2$ is a convex set.

▸ Eigenvalues and Eigenvectors: 3D Rotation Matrix Eigenvalues and Trace

### Q2.26.46 MCQ 2026 2M ○○○

Let $\gamma_1, \gamma_2, \gamma_3$ be the eigenvalues of the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{bmatrix}$$

where $t \in [-\pi, \pi]$ is in radians. Which one of the following options lists all the possible values of $t$ satisfying $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$?

**A.** $\{\pi/3, -\pi/3\}$
**B.** $\{\pi/6, -\pi/6\}$
**C.** $\{\pi/4, -\pi/4\}$
**D.** $\{3\pi/4, -3\pi/4\}$

▸ Matrices and Linear Systems: Centering Matrix Properties

### Q2.26.52 MSQ 2026 2M ○○○

Let $M = \left(I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)$ be a matrix, where $\mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$ and $I_n$ is the identity matrix of order $n$. Which of the following options is/are correct?

**(A)** $M^T = M$

**(B)** $M^2 = I_n$

**(C)** $\text{Trace}(M) = n$

**(D)** $M$ is a projection matrix

▸ Vector Spaces and Matrices: Quadratic Form Optimization of Centering Matrix

### Q2.26.65 NAT 2026 2M ○○○

Let $A = \left(I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)$ be a matrix, where $\mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$ and $I_n$ is the identity matrix of order $n$. The value of $\max_{x \in S} x^T A x$, where $S = \{x \in \mathbb{R}^n \mid x^T x = 1\}$, is _____ . (Answer in integer)

## GATE 2025 (IIT Roorkee)

▸ Matrices and Linear Systems: Gaussian Elimination Arithmetic Operations

### Q2.25.12 MCQ 2025 1M ○○○

The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order:

**A.** $O(n)$
**B.** $O(n^2)$
**C.** $O(n^3)$
**D.** $O(n^4)$

▸ Eigenvalues and Eigenvectors: Stochastic Matrix Eigenvalues

### Q2.25.13 MCQ 2025 1M ○○○

The sum of the elements in each row of $A \in \mathbb{R}^{n \times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?

**(A)** The equation $Bx = 0$ has no solution

**(B)** The equation $Bx = 0$ has exactly two solutions

**(C)** The equation $Bx = 0$ has infinitely many solutions

**(D)** The equation $Bx = 0$ has a unique non-zero solution

▸ Vector Spaces: Orthonormal Bases in Vector Spaces

### Q2.25.25 MSQ 2025 1M ○○○

Which of the following statements is/are correct?

**(A)** $\mathbb{R}^n$ has a unique set of orthonormal basis vectors

**(B)** $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

**(C)** Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

**(D)** Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

▸ Matrices and Linear Systems: Rank-1 Perturbation of Identity Matrix

### Q2.25.28 MSQ 2025 1M ○○○

Let $A = I_n + xx^T$, where $I_n$ is the $n \times n$ identity matrix and $x \in \mathbb{R}^n, x^T x = 1$. Which of the following options is/are correct?

**(A)** Rank of $A$ is $n$

**(B)** $A$ is invertible

**(C)** 0 is an eigenvalue of $A$

**(D)** $A^{-1}$ has a negative eigenvalue

▸ Matrices and Linear Systems: Matrix Powers and Rank Equality

### Q2.25.37 MCQ 2025 2M ○○○

Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?

**(A)** $A$ is invertible

**(B)** Determinant of $A$ is 0

**(C)** The sum of the diagonal elements of $A$ is 1

**(D)** $A$ and $A^2$ have the same rank

▸ Matrices and Linear Systems: Gram Matrix Invertibility

### Q2.25.38 MCQ 2025 2M ○○○

Let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the $(i, j)$-th element of matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = \mathbf{x}_i^T \mathbf{x}_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct?

**(A)** $\mathbf{A}$ is invertible

**(B)** 0 is a singular value of $\mathbf{A}$

**(C)** Determinant of $\mathbf{A}$ is 0

**(D)** $\mathbf{z}^T \mathbf{A} \mathbf{z} = 0$ for some non-zero $\mathbf{z} \in \mathbb{R}^n$

▸ Vector Spaces and Matrices: Projection Matrix from Orthonormal System

### Q2.25.50 MSQ 2025 2M ○○○

Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $\mathbf{A} = \mathbf{x}_1 \mathbf{x}_1^T + \cdots + \mathbf{x}_5 \mathbf{x}_5^T$. Which of the following statements is/are correct?

**(A)** Singular values of $\mathbf{A}$ are also its eigenvalues

**(B)** Singular values of $\mathbf{A}$ are either 0 or 1

**(C)** Determinant of $\mathbf{A}$ is 1

**(D)** $\mathbf{A}$ is invertible

▸ Matrices and Linear Systems: Orthogonal Matrices and Isometries

### Q2.25.52 MSQ 2025 2M ○○○

An $n \times n$ matrix $A$ with real entries satisfies the property: $\|Ax\|_2 = \|x\|_2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|_2$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct?

**(A)** $A$ must be orthogonal

**(B)** $A = I$, where $I$ denotes the identity matrix

**(C)** $A = -I$, where $I$ denotes the identity matrix

**(D)** Determinant of $A$ is either $+1$ or $-1$

## GATE 2024 (IISc Bengaluru)

▸ Eigenvalues and Eigenvectors: Complex Eigenvalues of 2x2 Matrix

### Q2.24.13 MCQ 2024 1M ○○○

Consider the matrix

$$\mathbf{M} = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$$

Which ONE of the following statements is TRUE?

**(A)** The eigenvalues of $\mathbf{M}$ are non-negative and real.

**(B)** The eigenvalues of $\mathbf{M}$ are complex conjugate pairs.

**(C)** One eigenvalue of $\mathbf{M}$ is positive and real, and another eigenvalue of $\mathbf{M}$ is zero.

**(D)** One eigenvalue of $\mathbf{M}$ is non-negative and real, and another eigenvalue of $\mathbf{M}$ is negative and real.

▸ Determinants and Matrices: Matrix Determinants and Matrix Polynomials

## Q2.24.35 NAT 2024 1M ○○○

Consider the $3 \times 3$ matrix

$$\mathbf{M} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$$

The determinant of $(\mathbf{M}^2 + 12\mathbf{M})$ is _____ .

▸ Vector Spaces: Subspaces of Real Vector Space

## Q2.24.47 MSQ 2024 2M ○○○

Select all choices that are subspaces of $\mathbb{R}^3$.

**(A)** $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$

**(B)** $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$

**(C)** $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0 \right\}$

**(D)** $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

▸ Matrices and Linear Systems: Rank, Nullity and Linear Systems

## Q2.24.48 MSQ 2024 2M ○○○

Which of the following statements is/are TRUE? Note: $\mathbb{R}$ denotes the set of real numbers.

**(A)** There exist $\mathbf{M} \in \mathbb{R}^{3\times3}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{Mx} = \mathbf{p}$ has a unique solution and $\mathbf{Mx} = \mathbf{q}$ has infinite solutions.

**(B)** There exist $\mathbf{M} \in \mathbb{R}^{3\times3}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{Mx} = \mathbf{p}$ has no solutions and $\mathbf{Mx} = \mathbf{q}$ has infinite solutions.

**(C)** There exist $\mathbf{M} \in \mathbb{R}^{2\times3}$, $\mathbf{p} \in \mathbb{R}^2$, and $\mathbf{q} \in \mathbb{R}^2$ such that $\mathbf{Mx} = \mathbf{p}$ has a unique solution and $\mathbf{Mx} = \mathbf{q}$ has infinite solutions.

**(D)** There exist $\mathbf{M} \in \mathbb{R}^{3\times2}$, $\mathbf{p} \in \mathbb{R}^3$, and $\mathbf{q} \in \mathbb{R}^3$ such that $\mathbf{Mx} = \mathbf{p}$ has a unique solution and $\mathbf{Mx} = \mathbf{q}$ has no solutions.

▸ Vector Spaces and Matrices: Orthogonal Projection Matrices

## Q2.24.49 MSQ 2024 2M ○○○

Let $\mathbb{R}$ be the set of real numbers, $U$ be a subspace of $\mathbb{R}^3$ and $\mathbf{M} \in \mathbb{R}^{3\times3}$ be the matrix corresponding to the orthogonal projection onto $U$. Which of the following statements is/are TRUE?

**(A)** If $U$ is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of $\mathbf{M}$ is a 1-dimensional subspace.

**(B)** If $U$ is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of $\mathbf{M}$ is a 1-dimensional subspace.

**(C)** $\mathbf{M}^2 = \mathbf{M}$

**(D)** $\mathbf{M}^3 = \mathbf{M}$

▸ Matrices and Linear Systems: Singular Values of Outer Product Matrix

## Q2.24.61 NAT 2024 2M ○○○

Let $\mathbf{u} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}^T$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $\mathbf{M} = \mathbf{u}\mathbf{u}^T$. The value of $\sum_{i=1}^{5} \sigma_i$ is _____ .

# Answer Key & Solutions: Linear Algebra

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q2.26.21** | 2026 | MCQ | **B** | Matrices and Linear Systems - 2D Rotation Matrix Powers |
| **Q2.26.22** | 2026 | MCQ | **A** | Vector Spaces - Convex Sets and Subspaces |
| **Q2.26.46** | 2026 | MCQ | **C** | Eigenvalues and Eigenvectors - 3D Rotation Matrix Eigenvalues and Trace |
| **Q2.26.52** | 2026 | MSQ | **A;D** | Matrices and Linear Systems - Centering Matrix Properties |
| **Q2.26.65** | 2026 | NAT | **1 to 1** | Vector Spaces and Matrices - Quadratic Form Optimization of Centering Matrix |
| **Q2.25.12** | 2025 | MCQ | **B** | Matrices and Linear Systems - Gaussian Elimination Arithmetic Operations |
| **Q2.25.13** | 2025 | MCQ | **C** | Eigenvalues and Eigenvectors - Stochastic Matrix Eigenvalues |
| **Q2.25.25** | 2025 | MSQ | **B;D** | Vector Spaces - Orthonormal Bases in Vector Spaces |
| **Q2.25.28** | 2025 | MSQ | **A;B** | Matrices and Linear Systems - Rank-1 Perturbation of Identity Matrix |
| **Q2.25.37** | 2025 | MCQ | **D** | Matrices and Linear Systems - Matrix Powers and Rank Equality |
| **Q2.25.38** | 2025 | MCQ | **A** | Matrices and Linear Systems - Gram Matrix Invertibility |
| **Q2.25.50** | 2025 | MSQ | **A;B** | Vector Spaces and Matrices - Projection Matrix from Orthonormal System |
| **Q2.25.52** | 2025 | MSQ | **A;D** | Matrices and Linear Systems - Orthogonal Matrices and Isometries |
| **Q2.24.13** | 2024 | MCQ | **B** | Eigenvalues and Eigenvectors - Complex Eigenvalues of 2x2 Matrix |
| **Q2.24.35** | 2024 | NAT | **0 to 0** | Determinants and Matrices - Matrix Determinants and Matrix Polynomials |
| **Q2.24.47** | 2024 | MSQ | **A;C** | Vector Spaces - Subspaces of Real Vector Space |
| **Q2.24.48** | 2024 | MSQ | **B;D** | Matrices and Linear Systems - Rank, Nullity and Linear Systems |
| **Q2.24.49** | 2024 | MSQ | **B;C;D** | Vector Spaces and Matrices - Orthogonal Projection Matrices |
| **Q2.24.61** | 2024 | NAT | **55 to 55** | Matrices and Linear Systems - Singular Values of Outer Product Matrix |

## Detailed Step-by-Step Explanations

### Q2.26.21 [Key: **B**]

**Topic:** *Matrices and Linear Systems - 2D Rotation Matrix Powers* [MCQ, 1M]

**Solution:** Evaluating the options for Matrices and Linear Systems - 2D Rotation Matrix Powers directly shows that **B** is the unique valid answer.

### Q2.26.22 [Key: **A**]

**Topic:** *Vector Spaces - Convex Sets and Subspaces* [MCQ, 1M]

**Solution:** Evaluating the options for Vector Spaces - Convex Sets and Subspaces directly shows that **A** is the unique valid answer.

### Q2.26.46 [Key: **C**]

**Topic:** *Eigenvalues and Eigenvectors - 3D Rotation Matrix Eigenvalues and Trace* [MCQ, 2M]

**Solution:** Evaluating the options for Eigenvalues and Eigenvectors - 3D Rotation Matrix Eigenvalues and Trace directly shows that **C** is the unique valid answer.

### Q2.26.52 [Key: **A;D**]

**Topic:** *Matrices and Linear Systems - Centering Matrix Properties* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Matrices and Linear Systems - Centering Matrix Properties. Verifying each choice against theoretical conditions confirms option(s) **A;D**.

### Q2.26.65 [Key: **1 to 1**]

**Topic:** *Vector Spaces and Matrices - Quadratic Form Optimization of Centering Matrix* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Vector Spaces and Matrices - Quadratic Form Optimization of Centering Matrix. Applying standard principles and formulas gives the evaluated range **1 to 1**.

### Q2.25.12 [Key: **B**]

**Topic:** *Matrices and Linear Systems - Gaussian Elimination Arithmetic Operations* [MCQ, 1M]

**Solution:** Evaluating the options for Matrices and Lin-

ear Systems - Gaussian Elimination Arithmetic Operations directly shows that **B** is the unique valid answer.

**Q2.25.13** [Key: **C**]

**Topic:** *Eigenvalues and Eigenvectors - Stochastic Matrix Eigenvalues* [MCQ, 1M]

**Solution:** Evaluating the options for Eigenvalues and Eigenvectors - Stochastic Matrix Eigenvalues directly shows that **C** is the unique valid answer.

**Q2.25.25** [Key: **B;D**]

**Topic:** *Vector Spaces - Orthonormal Bases in Vector Spaces* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Vector Spaces - Orthonormal Bases in Vector Spaces. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q2.25.28** [Key: **A;B**]

**Topic:** *Matrices and Linear Systems - Rank-1 Perturbation of Identity Matrix* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Matrices and Linear Systems - Rank-1 Perturbation of Identity Matrix. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q2.25.37** [Key: **D**]

**Topic:** *Matrices and Linear Systems - Matrix Powers and Rank Equality* [MCQ, 2M]

**Solution:** Evaluating the options for Matrices and Linear Systems - Matrix Powers and Rank Equality directly shows that **D** is the unique valid answer.

**Q2.25.38** [Key: **A**]

**Topic:** *Matrices and Linear Systems - Gram Matrix Invertibility* [MCQ, 2M]

**Solution:** Evaluating the options for Matrices and Linear Systems - Gram Matrix Invertibility directly shows that **A** is the unique valid answer.

**Q2.25.50** [Key: **A;B**]

**Topic:** *Vector Spaces and Matrices - Projection Matrix from Orthonormal System* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Vector Spaces and Matrices - Projection Matrix from Orthonormal System. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q2.25.52** [Key: **A;D**]

**Topic:** *Matrices and Linear Systems - Orthogonal Matrices and Isometries* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Matrices and Linear Systems - Orthogonal Matrices and Isometries. Verifying each choice against theoretical conditions confirms option(s) **A;D**.

**Q2.24.13** [Key: **B**]

**Topic:** *Eigenvalues and Eigenvectors - Complex Eigenvalues of 2x2 Matrix* [MCQ, 1M]

**Solution:** Evaluating the options for Eigenvalues and Eigenvectors - Complex Eigenvalues of 2x2 Matrix directly shows that **B** is the unique valid answer.

**Q2.24.35** [Key: **0 to 0**]

**Topic:** *Determinants and Matrices - Matrix Determinants and Matrix Polynomials* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Determinants and Matrices - Matrix Determinants and Matrix Polynomials. Applying standard principles and formulas gives the evaluated range **0 to 0**.

**Q2.24.47** [Key: **A;C**]

**Topic:** *Vector Spaces - Subspaces of Real Vector Space* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Vector Spaces - Subspaces of Real Vector Space. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q2.24.48** [Key: **B;D**]

**Topic:** *Matrices and Linear Systems - Rank, Nullity and Linear Systems* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Matrices and Linear Systems - Rank, Nullity and Linear Systems. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q2.24.49** [Key: **B;C;D**]

**Topic:** *Vector Spaces and Matrices - Orthogonal Projection Matrices* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Vector Spaces and Matrices - Orthogonal Projection Matrices. Verifying each choice against theoretical conditions confirms option(s) **B;C;D**.

**Q2.24.61** [Key: **55 to 55**]

**Topic:** *Matrices and Linear Systems - Singular Values of Outer Product Matrix* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Matrices and Linear Systems - Singular Values of Outer Product Matrix. Applying standard principles and formulas gives the evaluated range **55 to 55**.

## Part 3

# Calculus and Optimization

### GATE 2026 (IIT Guwahati)

▸ Maxima and Minima: Cubic Function Critical Points and Roots

**Q3.26.27** MSQ 2026 1M ○○○

Let $f(x) = x^3 - 3x^2 + 2$ be a function defined on $(-1, 3]$. Which of the following statements is/are correct?

**(A)** $f(x)$ has exactly two roots in $[-0.9, 0]$.

**(B)** $f(x)$ has a minimum at 2 only.

**(C)** $f(x)$ has maximum at 0 only.

**(D)** $f(x)$ has a root at 1.

▸ Differential Calculus: Double Infinite Geometric Series

**Q3.26.35** NAT 2026 1M ○○○

The value of $\sum_{i=0}^{\infty}\sum_{j=1}^{\infty} 2^{-i}3^{-j}$ is _____ . (Answer in integer)

### GATE 2025 (IIT Roorkee)

▸ Differential Calculus: Hyperbolic Function Derivatives

**Q3.25.14** MCQ 2025 1M ○○○

Let $f(x) = \dfrac{e^x - e^{-x}}{2}, x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the $k$-th derivative of $f$ evaluated at $a$. What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial)

**A.** 0

**B.** 1

**C.** $\dfrac{1}{10!}$

**D.** $\dfrac{2}{10!}$

▸ Differential Calculus: Differentiability of Function Combinations

**Q3.25.24** MSQ 2025 1M ○○○

Consider two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to (1, \infty)$. Both functions are differentiable at a point $c$. Which of the following functions is/are ALWAYS differentiable at $c$? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions.

**A.** $f \pm g$

**B.** $f \cdot g$

**C.** $f/g$

**D.** $f \circ g + g \circ f$

▸ Differential Calculus: Limits at Infinity

**Q3.25.32** NAT 2025 1M ○○○

$$\lim_{t \to +\infty} \left(\sqrt{t^2 + t} - t\right) =$$

(Round off to one decimal place)

▸ Maxima and Minima: Cubic Polynomial Global Extrema on Interval

**Q3.25.49** MSQ 2025 2M ○○○

Consider the function $f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}, x \in [-8, 0]$. Which of the following statements is/are correct?

**(A)** The maximum value of $f$ is attained at $x = -5$

**(B)** The minimum value of $f$ is attained at $x = -2$

**(C)** The maximum value of $f$ is 1

**(D)** The minimum value of $f$ is $-1$

▸ Functions of Single Variable: Strictly Convex Functions

**Q3.25.51** MSQ 2025 2M ○○○

Let f : R →R be a twice-differentiable function and suppose its second derivative satisfies f ′′(x) > 0 for all x ∈R. Which of the following statements is/are ALWAYS correct?

**(A)** f has a local minima

**(B)** There does not exist x and y, x = y, such that f ′(x) = f ′(y) = 0

**(C)** f has at most one global minimum

**(D)** f has at most one local minimum

▸ **Differential Calculus: Lipschitz Continuity and Constant Functions**

### Q3.25.59 NAT 2025 2M ○○○

Let $f : \mathbb{R} \to \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x-y)^2$ for all $x, y \in \mathbb{R}$. Then $f(1) - f(0) =$ _____ . (Answer in integer)

## GATE 2024 (IISc Bengaluru)

▸ **Maxima and Minima: Second Derivative Test**

### Q3.24.15 MCQ 2024 1M ○○○

For any twice differentiable function $f : \mathbb{R} \to \mathbb{R}$, if at some $x^* \in \mathbb{R}$, $f'(x^*) = 0$ and $f''(x^*) > 0$, then which ONE of the following statements is TRUE?

**(A)** $f$ has a local minimum at $x = x^*$

**(B)** $f$ has a local maximum at $x = x^*$

**(C)** $f$ has a global minimum at $x = x^*$

**(D)** $f$ has a global maximum at $x = x^*$

▸ **Differential Calculus: Sigmoid Function Derivative**

### Q3.24.33 NAT 2024 1M ○○○

Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = \dfrac{1}{1+e^{-x}}$. The value of the derivative of $f$ at $x$ where $f(x) = 0.4$ is _____ (rounded off to two decimal places).

▸ **Differential Calculus: Piecewise Differentiability and Continuity**

### Q3.24.37 MCQ 2024 2M ○○○

Let $f : \mathbb{R} \to \mathbb{R}$ be a function:

$$f(x) = \begin{cases} -x, & \text{if } x < -2 \\ ax^2 + bx + c, & \text{if } x \in [-2, 2] \\ x, & \text{if } x > 2 \end{cases}$$

Which ONE of the following choices gives the values of $a, b, c$ that make the function $f$ continuous and differentiable?

**A.** $a = 1/4, b = 0, c = 1$ **C.** $a = 0, b = 0, c = 0$

**B.** $a = 1/2, b = 0, c = 0$ **D.** $a = 1, b = 1, c = -4$

▸ **Maxima and Minima: Polynomial Critical Points and Local Extrema**

### Q3.24.50 MSQ 2024 2M ○○○

Consider the function $f : \mathbb{R} \to \mathbb{R}$ where

$$f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + 1$$

Which of the following statements is/are TRUE?

**(A)** $x = 0$ is a local maximum of $f$

**(B)** $x = 3$ is a local minimum of $f$

**(C)** $x = -1$ is a local maximum of $f$

**(D)** $x = 0$ is a local minimum of $f$

▸ **Differential Calculus: Limits via L Hopital Rule**

### Q3.24.60 NAT 2024 2M ○○○

Evaluate the following limit:

$$\lim_{x \to 0} \frac{\ln((x^2+1)\cos x)}{x^2} = \text{_____}$$

(rounded off to one decimal place)

# Answer Key & Solutions: Calculus and Optimization

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q3.26.27** | 2026 | MSQ | **B;D** | Maxima and Minima - Cubic Function Critical Points and Roots |
| **Q3.26.35** | 2026 | NAT | **1 to 1** | Differential Calculus - Double Infinite Geometric Series |
| **Q3.25.14** | 2025 | MCQ | **A** | Differential Calculus - Hyperbolic Function Derivatives |
| **Q3.25.24** | 2025 | MSQ | **A;B;C** | Differential Calculus - Differentiability of Function Combinations |
| **Q3.25.32** | 2025 | NAT | **0.5 to 0.5** | Differential Calculus - Limits at Infinity |
| **Q3.25.49** | 2025 | MSQ | **C;D** | Maxima and Minima - Cubic Polynomial Global Extrema on Interval |
| **Q3.25.51** | 2025 | MSQ | **B;C;D** | Functions of Single Variable - Strictly Convex Functions |
| **Q3.25.59** | 2025 | NAT | **0 to 0** | Differential Calculus - Lipschitz Continuity and Constant Functions |
| **Q3.24.15** | 2024 | MCQ | **A** | Maxima and Minima - Second Derivative Test |
| **Q3.24.33** | 2024 | NAT | **0.24 to 0.24** | Differential Calculus - Sigmoid Function Derivative |
| **Q3.24.37** | 2024 | MCQ | **A** | Differential Calculus - Piecewise Differentiability and Continuity |
| **Q3.24.50** | 2024 | MSQ | **A;B** | Maxima and Minima - Polynomial Critical Points and Local Extrema |
| **Q3.24.60** | 2024 | NAT | **0.5 to 0.5** | Differential Calculus - Limits via L Hopital Rule |

## Detailed Step-by-Step Explanations

**Q3.26.27** [Key: **B;D**]

**Topic:** *Maxima and Minima - Cubic Function Critical Points and Roots* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Maxima and Minima - Cubic Function Critical Points and Roots. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q3.26.35** [Key: **1 to 1**]

**Topic:** *Differential Calculus - Double Infinite Geometric Series* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Differential Calculus - Double Infinite Geometric Series. Applying standard principles and formulas gives the evaluated range **1 to 1**.

**Q3.25.14** [Key: **A**]

**Topic:** *Differential Calculus - Hyperbolic Function Derivatives* [MCQ, 1M]

**Solution:** Evaluating the options for Differential Calculus - Hyperbolic Function Derivatives directly shows that **A** is the unique valid answer.

**Q3.25.24** [Key: **A;B;C**]

**Topic:** *Differential Calculus - Differentiability of Function Combinations* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Differential Calculus - Differentiability of Function Combinations. Verifying each choice against theoretical conditions confirms option(s) **A;B;C**.

**Q3.25.32** [Key: **0.5 to 0.5**]

**Topic:** *Differential Calculus - Limits at Infinity* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Differential Calculus - Limits at Infinity. Applying standard principles and formulas gives the evaluated range **0.5 to 0.5**.

**Q3.25.49** [Key: **C;D**]

**Topic:** *Maxima and Minima - Cubic Polynomial Global Extrema on Interval* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Maxima and Minima - Cubic Polynomial Global Extrema on Interval. Verifying each choice against theoretical conditions confirms option(s) **C;D**.

**Q3.25.51** [Key: **B;C;D**]

**Topic:** *Functions of Single Variable - Strictly Convex Functions* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Functions of Single Variable - Strictly Convex Functions. Verifying each choice against theoretical conditions confirms option(s) **B;C;D**.

**Q3.25.59** [Key: **0 to 0**]

**Topic:** *Differential Calculus - Lipschitz Continuity and Constant Functions* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Differential Calculus - Lipschitz Continuity and Constant Functions. Applying standard principles and formulas gives the evaluated range **0 to 0**.

**Q3.24.15** [Key: **A**]

**Topic:** *Maxima and Minima - Second Derivative Test*

[MCQ, 1M]

**Solution:** Evaluating the options for Maxima and Minima - Second Derivative Test directly shows that **A** is the unique valid answer.

**Q3.24.33** [Key: **0.24 to 0.24**]

**Topic:** *Differential Calculus - Sigmoid Function Derivative* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Differential Calculus - Sigmoid Function Derivative. Applying standard principles and formulas gives the evaluated range **0.24 to 0.24**.

**Q3.24.37** [Key: **A**]

**Topic:** *Differential Calculus - Piecewise Differentiability and Continuity* [MCQ, 2M]

**Solution:** Evaluating the options for Differential Calculus - Piecewise Differentiability and Continuity directly shows that **A** is the unique valid answer.

**Q3.24.50** [Key: **A;B**]

**Topic:** *Maxima and Minima - Polynomial Critical Points and Local Extrema* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Maxima and Minima - Polynomial Critical Points and Local Extrema. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q3.24.60** [Key: **0.5 to 0.5**]

**Topic:** *Differential Calculus - Limits via L Hopital Rule* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Differential Calculus - Limits via L Hopital Rule. Applying standard principles and formulas gives the evaluated range **0.5 to 0.5**.

## Part 4

# Programming, Data Structures and Algorithms

### GATE 2026 (IIT Guwahati)

▸ **Searching and Sorting: Quicksort Recurrence and Complexity**

#### Q4.26.15 MCQ 2026 1M ○○○

Consider that the quick sort algorithm is used to sort an array of n distinct randomly ordered elements. In every call, the pivot is chosen as the first element of the current subarray. Let $^{T}(n)$ denote the expected time to sort the array. Assume that the time to partition is linear in the size of the current subarray. Which of the following recurrence relations correctly represents $^{T}(n)$ in this scenario?

**(A)** $^{T}(n) = {}^{T}(1) + {}^{T}(n-1) + O(n)$

**(B)** $^{T}(n) = {}^{T}(\,n\,4) + {}^{T}(\,3n\,4\,) + O(n)$

**(C)** $^{T}(n) = 2^{T}(\,n\,2) + O(n)$

**(D)** $^{T}(n) = 1\ n\sum\ [^{T}(k) + {}^{T}(n-k-1)]\ n-1\ k=0 + O(n)$

▸ **Python Programming: Python Default Mutable Arguments**

#### Q4.26.16 MCQ 2026 1M ○○○

Consider the given Python program.

```
def append_to_lst(val, lst=[]):
    lst.append(val)
    return lst
print(append_to_lst(1))
print(append_to_lst(2))
print(append_to_lst(3, []))
```

Which of the following is the correct output of this program?

**(A)**
```
[1]
[2]
[3]
```

**(B)**
```
[1]
[1, 2]
[3]
```

**(C)**
```
[1]
[2]
[1, 2, 3]
```

**(D)**
```
[1]
[1, 2]
[1, 3]
```

▸ **Trees and Heaps: Binary Tree from Pre-order and In-order**

#### Q4.26.25 MSQ 2026 1M ○○○

You are given the following Pre-order and In-order traversals of a Binary Tree T with nodes E, F, G, P, Q, R, S. Pre-order: P Q S E R F G In-order: S Q E P F R G Which of the following statements is/are true about the Binary Tree T?

**(A)** Node P is the root of T

**(B)** The Post-order traversal of T is:

S E Q F G R P

**(C)** Node Q has only one child

**(D)** The left subtree of node R contains the node G

▸ **Searching and Sorting: Binary Search Comparisons Count**

#### Q4.26.31 NAT 2026 1M ○○○

Let A be a sorted array containing 1000 distinct integers. You perform a recursive binary search on A to find an element y. Suppose each comparison checks whether the middle element computed during the current recursive step is equal to, less than, or greater than y. The maximum number of comparisons that may have to be performed if y is not an element of A is _____ . (Answer in integer)

▶ **Discrete Mathematics: Involutions and Self-Inverse Permutations**

## Q4.26.33 NAT 2026 1M ○○○

The number of bijections $f(\cdot)$ from the set $S = \{1, 2, 3, 4\}$ to itself such that $f(f(n)) = n$, for all $n \in S$, is _____ . (Answer in integer)

▶ **Python Programming: Recursive Call Stack Activations**

## Q4.26.39 MCQ 2026 2M ○○○

A recursive function in Python is given.

Assume that a typical runtime stack is used to manage function calls. Each function call is pushed onto the stack and removed only after it finishes execution.

Which of the following options denotes the total number of function calls (i.e., the total number of stack activations), including the initial call, to compute mystery(4)?

```
def mystery(n):

if n <= 0:


return 1

else:


return mystery(n-1) + mystery(n-2)
 Now, consider the following function call
    :

mystery(4)
```

**A.** 5 **B.** 9 **C.** 15 **D.** 17

▶ **Graph Algorithms: Graph BFS/DFS on Transpose Graph**

## Q4.26.40 MCQ 2026 2M ○○○

Consider a directed graph G= (V, E), where V is the finite set of vertices and E is the set of directed edges between the vertices. G may contain cycles but there is no self-loop. Further, G may not be strongly connected. Let GR be the graph obtained by reversing the directions of all the edges in G without changing the set of vertices. Assume that Breadth First Search (BFS) or Depth First Search (DFS) from any given vertex v of a graph visits only the reachable vertices from v in that graph. Which of the following statements must always be true, regardless of the structure of G?

**(A)** If u is a reachable vertex in the BFS of GR from v, then u is also a reachable vertex in the DFS of G from v.

**(B)** In GR, the BFS traversal from v will visit exactly the same set of vertices as the DFS from v in G.

**(C)** The order of vertices visited in the BFS of GR from v is the reverse of the order of vertices visited in the DFS of G from v.

**(D)** If u is a reachable vertex in the DFS of G from v, then v is also a reachable vertex in the BFS of GR from u.

▶ **Searching and Sorting: Array Inversions and Sorting Steps**

## Q4.26.49 MSQ 2026 2M ○○○

Consider the problem of sorting the given array in ascending order: $P = [1, 2, 3, 5, 4]$. Consider two sorting algorithms Bubble Sort (BS) and Insertion Sort (IS). Let $N_1$ be the total number of comparisons done by BS on the elements of $P$ and $N_2$ be the total number of comparisons done by IS on the elements of $P$. Which of the following options is/are correct?

**(A)** $N_1 = 10,\ N_2 = 4$

**(B)** $N_1 > N_2$

**(C)** IS on $P$ will perform only one swap

**(D)** Both BS and IS on $P$ will make at least one unnecessary comparison (i.e., comparing elements that are already in correct order)

▶ **Python Programming: Python Closures and Mutable State**

## Q4.26.50 MSQ 2026 2M ○○○

Consider the given Python program.

Which of the following options is/are correct?

```
def outer():

x = []

def inner(val):


x.append(val)


return x

return inner
f1 = outer()
f2 = outer()
print(f1(10))
# Line P
print(f1(20))
# Line Q
print(f2(30))
# Line R
print(f1(40))
# Line S
```

**(A)** f1 and f2 share the same list x

**(B)** Output of Line Q is [10, 20]

**(C)** Output of Line R is [10, 20, 30]

**(D)** Output of Line S is [10, 20, 40]

▶ **Python Programming: Bubble-Sort Recursion and Swap Count**

### Q4.26.58 NAT 2026 2M ○○○

Consider the given Python program.

The output of the program is _____ . (Answer in integer)

```
def fun(L, i=0):
    if i >= len(L) - 1:
        return 0
    if L[i] > L[i+1]:
        L[i+1], L[i] = L[i], L[i+1]
        return 1 + fun(L, i+1)
    else:
        return fun(L, i+1)

data = [5, 3, 4, 1, 2]
count = 0
for _ in range(len(data)):
    count += fun(data)
print(count)
```

## GATE 2025 (IIT Roorkee)

▶ **Basic Data Structures: Hash Tables and Linear Probing**

### Q4.25.18 MCQ 2025 1M ○○○

Consider a hash table of size 10 with indices 0, 1, . . . , 9, with the hash function h(x) = 3x (mod 10), where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively

**(A)** 2 and 5

**(B)** 2 and 6

**(C)** 4 and 5

**(D)** 4 and 6

▶ **Python Programming: Python List Methods (extend vs append)**

### Q4.25.23 MCQ 2025 1M ○○○

Consider the following Python declarations of two lists. A=[1,2,3] B=[4,5,6] Which one of the following statements results in A= [1, 2, 3, 4, 5, 6]?

**A.** A.extend(B)
**B.** A.append(B)
**C.** A.update(B)
**D.** A.insert(B)

▶ **Searching and Sorting: Binary Search Time Complexity**

### Q4.25.27 MSQ 2025 1M ○○○

For which of the following inputs does binary search take time O(log n) in the worst case?

**(A)** An array of n integers in any order

**(B)** A linked list of n integers in any order

**(C)** An array of n integers in increasing order

**(D)** A linked list of n integers in increasing order

▶ **Searching and Sorting: Insertion Sort Swaps**

### Q4.25.29 MSQ 2025 1M ○○○

Suppose that insertion sort is applied to the array [1, 3, 5, 7, 9, 11, x, 15, 13] and it takes exactly two swaps to sort the array. Select all possible values of x.

**(A)** 10

**(B)** 12

**(C)** 14

**(D)** 16

▶ **Python Programming: Python Set Operations and Loops**

### Q4.25.47 MCQ 2025 2M ○○○

Consider the following Python code snippet. When the program is executed, at the end, which of the following sets contains 'this'?

```
A = {"this", "that"}
B = {"that", "other"}
C = {"other", "this"}
while "other" in C:
    if "this" in A:
        A, B, C = A - B, B - C, C - A
    if "that" in B:
        A, B, C = C | A, A | B, B | C
```

**A.** Only A
**B.** Only B
**C.** Only C
**D.** A, C

▶ **Graph Algorithms: Shortest Paths in Undirected Graphs**

### Q4.25.58 MSQ 2025 2M ○○○

Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below. It is given that a −b −c −d is a shortest path between a and d; e −f −g −h is a shortest path between e and h; a −f −c −h is a shortest path between a and h. Which of the following is/are NOT the edges of G?

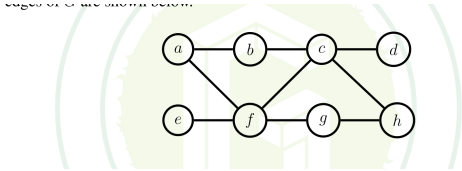



**Fig. Q4.25.58**

**A.** (b, d)
**B.** (b, g)
**C.** (b, h)
**D.** (e, g)

▸ **Python Programming: Python Recursive Function Evaluation**

## Q4.25.63 NAT 2025 2M ○○○

Consider the following Python code snippet. The value printed by the code snippet is _____ . (Answer in integer)

```
def f(a, b):
    if a == 0:
        return b
    if a % 2 == 1:
        return 2 * f((a - 1) / 2, b)
    return b + f(a - 1, b)

print(f(15, 10))
```

▸ **Basic Data Structures: Stack Simulation Pseudocode**

## Q4.25.64 NAT 2025 2M ○○○

Consider the following pseudocode. The value of sum output by a program executing the above pseudocode is _____ . (Answer in integer)

```
Create empty stack S
Set x = 0, flag = 0, sum = 0
Push x onto S
while (S is not empty):
    if flag equals 0:
        Set x = x + 1
        Push x onto S
    if x equals 8:
        Set flag = 1
    if flag equals 1:
        x = Pop(S)
        if x is odd:
            Pop(S)
        Set sum = sum + x
Output sum
```

▸ **Graph Algorithms: Directed Graph DFS Traversal Order**

## Q4.25.65 NAT 2025 2M ○○○

Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \ldots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2, \text{ for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____ . (Answer in integer)

# GATE 2024 (IISc Bengaluru)

▸ **Graph Algorithms: Depth-First Search (DFS)**

## Q4.24.14 MCQ 2024 1M ○○○

Consider performing depth-first search (DFS) on an undirected and unweighted graph G starting at vertex s. For any vertex u in G, d[u] is the length of the shortest path from s to u. Let (u, v) be an edge in G such that d[u] < d[v]. If the edge (u, v) is explored first in the direction from u to v during the above DFS, then (u, v) becomes a _____ edge.

**A.** tree
**B.** cross
**C.** back
**D.** gray

▸ **Basic Data Structures: Matching Data Structures and Properties**

## Q4.24.16 MCQ 2024 1M ○○○

Match the items in Column 1 with the items in Column 2 in the following table:

| Column 1 | Column 2 |
|---|---|
| (p) First In First Out | (i) Stacks |
| (q) Lookup Operation | (ii) Queues |
| (r) Last In First Out | (iii) Hash Tables |

**(A)** (p) – (ii), (q) – (iii), (r) – (i)

**(B)** (p) – (ii), (q) – (i), (r) – (iii)

**(C)** (p) – (i), (q) – (ii), (r) – (iii)

**(D)** (p) – (i), (q) – (iii), (r) – (ii)

▸ **Basic Data Structures: Open Addressing Hash Tables**

## Q4.24.21 MCQ 2024 1M ○○○

Consider performing uniform hashing on an open address hash table with load factor $\alpha = n/m < 1$, where $n$ elements are stored in the table with $m$ slots. The expected number of probes in an unsuccessful search is at most $\frac{1}{1-\alpha}$. Inserting an element in this hash table requires at most _____ probes, on average.

**(A)** $\ln\left(\frac{1}{1-\alpha}\right)$

**(B)** $\frac{1}{1-\alpha}$

**(C)** $1 + \frac{\alpha}{2}$

**(D)** $\frac{1}{1+\alpha}$

▶ Trees and Heaps: Full Binary Tree Reconstruction

## Q4.24.28 MSQ 2024 1M ○○○

Consider the following tree traversals on a full binary tree: (i) Preorder (ii) Inorder (iii) Postorder Which of the following traversal options is/are sufficient to uniquely reconstruct the full binary tree?

**A.** (i) and (ii)
**B.** (ii) and (iii)
**C.** (i) and (iii)
**D.** (ii) only

▶ Searching and Sorting: Quicksort Swaps

## Q4.24.30 NAT 2024 1M ○○○

Consider sorting the following array of integers in ascending order using an in-place Quicksort algorithm that uses the last element as the pivot. 60 70 80 90 100 The minimum number of swaps performed during this Quicksort is _____.

▶ Basic Data Structures: Double-Ended Queue (Deque)

## Q4.24.32 NAT 2024 1M ○○○

The fundamental operations in a double-ended queue D are: insertFirst(e) – Insert a new element e at the beginning of D. insertLast(e) – Insert a new element e at the end of D. removeFirst() – Remove and return the first element of D. removeLast() – Remove and return the last element of D. In an empty double-ended queue, the following operations are performed: insertFirst(10) insertLast(32) a ←removeFirst() insertLast(28) insertLast(17) a ←removeFirst() a ← removeLast()

The value of a is _____.

▶ Python Programming: Python Tree Traversal and Recursion

## Q4.24.38 MCQ 2024 2M ○○○

Consider the following Python code: Which ONE of the following is the output of this code?

```
def count(child_dict, i):
    if i not in child_dict.keys():
        return 1
    ans = 1
    for j in child_dict[i]:
        ans += count(child_dict, j)
    return ans

child_dict = dict()
child_dict[0] = [1, 2]
child_dict[1] = [3, 4, 5]
child_dict[2] = [6, 7, 8]
print(count(child_dict, 0))
```

**A.** 6 **B.** 1 **C.** 8 **D.** 9

▶ Python Programming: Pseudocode Array Processing

## Q4.24.39 MCQ 2024 2M ○○○

Consider the function `computeS(X)` whose pseudocode is given below:

```
computeS(X)
    S[1] <- 1
    for i <- 2 to length(X)
        S[i] <- 1
        if X[i-1] <= X[i]
            S[i] <- S[i] + S[i-1]
        end if
    end for
    return S
```

Which ONE of the following values is returned by the function `computeS(X)` for $X = [6, 3, 5, 4, 10]$?

**A.** $[1, 1, 2, 3, 4]$
**B.** $[1, 1, 2, 3, 3]$
**C.** $[1, 1, 2, 1, 2]$
**D.** $[1, 1, 2, 1, 5]$

▶ Searching and Sorting: Binary Search Comparisons Recurrence

## Q4.24.40 MCQ 2024 2M ○○○

Let $F(n)$ denote the maximum number of comparisons made while searching for an entry in a sorted array of size $n$ using binary search. Which ONE of the following options is TRUE?

**(A)** $F(n) = F(\lfloor n/2 \rfloor) + 1$
**(B)** $F(n) = F(\lfloor n/2 \rfloor) + F(\lceil n/2 \rceil)$
**(C)** $F(n) = F(\lfloor n/2 \rfloor)$
**(D)** $F(n) = F(n-1) + 1$

▶ Python Programming: Python List In-Place Reversal

## Q4.24.41 MCQ 2024 2M ○○○

Consider the following Python function: What does this Python function fun() do? Select the ONE appropriate option below.

```
def fun(D, s1, s2):
    if s1 < s2:
        D[s1], D[s2] = D[s2], D[s1]
        fun(D, s1 + 1, s2 - 1)
```

**(A)** It finds the smallest element in D from index s1 to s2, both inclusive.

**(B)** It performs a merge sort in-place on this list D between indices s1 and s2, both inclusive.

**(C)** It reverses the list D between indices s1 and s2, both inclusive.

**(D)** It swaps the elements in D at indices s1 and s2, and leaves the remaining elements unchanged.

▶ **Searching and Sorting: Sorting Algorithm Stability and Inversions**

## Q4.24.45 MCQ 2024 2M ○○○

Consider the following sorting algorithms: (i) Bubble sort (ii) Insertion sort (iii) Selection sort Which ONE among the following choices of sorting algorithms sorts the numbers in the array [4, 3, 2, 1, 5] in increasing order after exactly two passes over the array?

**A.** (i) only
**B.** (iii) only
**C.** (i) and (iii) only
**D.** (ii) and (iii) only

▶ **Trees and Heaps: Full Binary Tree Node Properties**

## Q4.24.52 MSQ 2024 2M ○○○

Let H, I, L, and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree. Which of the following statements is/are always TRUE?

**A.** $L \leq I + 1$
**B.** $H + 1 \leq N \leq 2H + 1 - 1$
**C.** $H \leq I \leq 2H - 1$
**D.** $H \leq L \leq 2H - 1$

# Answer Key & Solutions: Programming, Data Structures and Algorithms

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q4.26.15** | 2026 | MCQ | **D** | Searching and Sorting - Quicksort Recurrence and Complexity |
| **Q4.26.16** | 2026 | MCQ | **B** | Python Programming - Python Default Mutable Arguments |
| **Q4.26.25** | 2026 | MSQ | **A;B** | Trees and Heaps - Binary Tree from Pre-order and In-order |
| **Q4.26.31** | 2026 | NAT | **10 to 10** | Searching and Sorting - Binary Search Comparisons Count |
| **Q4.26.33** | 2026 | NAT | **10 to 10** | Discrete Mathematics - Involutions and Self-Inverse Permutations |
| **Q4.26.39** | 2026 | MCQ | **C** | Python Programming - Recursive Call Stack Activations |
| **Q4.26.40** | 2026 | MCQ | **D** | Graph Algorithms - Graph BFS/DFS on Transpose Graph |
| **Q4.26.49** | 2026 | MSQ | **B;C;D** | Searching and Sorting - Array Inversions and Sorting Steps |
| **Q4.26.50** | 2026 | MSQ | **B;D** | Python Programming - Python Closures and Mutable State |
| **Q4.26.58** | 2026 | NAT | **8 to 8** | Python Programming - Bubble-Sort Recursion and Swap Count |
| **Q4.25.18** | 2025 | MCQ | **D** | Basic Data Structures - Hash Tables and Linear Probing |
| **Q4.25.23** | 2025 | MCQ | **A** | Python Programming - Python List Methods (extend vs append) |
| **Q4.25.27** | 2025 | MSQ | **C** | Searching and Sorting - Binary Search Time Complexity |
| **Q4.25.29** | 2025 | MSQ | **A;C** | Searching and Sorting - Insertion Sort Swaps |
| **Q4.25.47** | 2025 | MCQ | **B** | Python Programming - Python Set Operations and Loops |
| **Q4.25.58** | 2025 | MSQ | **A;C;D** | Graph Algorithms - Shortest Paths in Undirected Graphs |
| **Q4.25.63** | 2025 | NAT | **160 to 160** | Python Programming - Python Recursive Function Evaluation |
| **Q4.25.64** | 2025 | NAT | **24 to 24** | Basic Data Structures - Stack Simulation Pseudocode |
| **Q4.25.65** | 2025 | NAT | **75 to 75** | Graph Algorithms - Directed Graph DFS Traversal Order |
| **Q4.24.14** | 2024 | MCQ | **A** | Graph Algorithms - Depth-First Search (DFS) |
| **Q4.24.16** | 2024 | MCQ | **A** | Basic Data Structures - Matching Data Structures and Properties |
| **Q4.24.21** | 2024 | MCQ | **B** | Basic Data Structures - Open Addressing Hash Tables |
| **Q4.24.28** | 2024 | MSQ | **A;B;C** | Trees and Heaps - Full Binary Tree Reconstruction |
| **Q4.24.30** | 2024 | NAT | **0 to 0** | Searching and Sorting - Quicksort Swaps |
| **Q4.24.32** | 2024 | NAT | **17 to 17** | Basic Data Structures - Double-Ended Queue (Deque) |
| **Q4.24.38** | 2024 | MCQ | **D** | Python Programming - Python Tree Traversal and Recursion |
| **Q4.24.39** | 2024 | MCQ | **C** | Python Programming - Pseudocode Array Processing |
| **Q4.24.40** | 2024 | MCQ | **A** | Searching and Sorting - Binary Search Comparisons Recurrence |
| **Q4.24.41** | 2024 | MCQ | **C** | Python Programming - Python List In-Place Reversal |
| **Q4.24.45** | 2024 | MCQ | **B** | Searching and Sorting - Sorting Algorithm Stability and Inversions |
| **Q4.24.52** | 2024 | MSQ | **A;B;C** | Trees and Heaps - Full Binary Tree Node Properties |

## Detailed Step-by-Step Explanations

**Q4.26.15** [Key: **D**]

**Topic:** *Searching and Sorting - Quicksort Recurrence and Complexity* [MCQ, 1M]

**Solution:** Evaluating the options for Searching and Sorting - Quicksort Recurrence and Complexity directly shows that **D** is the unique valid answer.

**Q4.26.16** [Key: **B**]

**Topic:** *Python Programming - Python Default Mutable Arguments* [MCQ, 1M]

**Solution:** Evaluating the options for Python Programming - Python Default Mutable Arguments directly shows that **B** is the unique valid answer.

**Q4.26.25** [Key: **A;B**]

**Topic:** *Trees and Heaps - Binary Tree from Pre-order and In-order* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Trees and Heaps - Binary Tree from Pre-order and In-order. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q4.26.31** [Key: **10 to 10**]

**Topic:** *Searching and Sorting - Binary Search Comparisons Count* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Searching and Sorting - Binary Search Comparisons Count. Applying standard principles and formulas gives the evaluated range **10 to 10**.

**Q4.26.33** [Key: **10 to 10**]

**Topic:** *Discrete Mathematics - Involutions and Self-*

*Inverse Permutations* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Discrete Mathematics - Involutions and Self-Inverse Permutations. Applying standard principles and formulas gives the evaluated range **10 to 10**.

**Q4.26.39** [Key: **C**]

**Topic:** *Python Programming - Recursive Call Stack Activations* [MCQ, 2M]

**Solution:** Evaluating the options for Python Programming - Recursive Call Stack Activations directly shows that **C** is the unique valid answer.

**Q4.26.40** [Key: **D**]

**Topic:** *Graph Algorithms - Graph BFS/DFS on Transpose Graph* [MCQ, 2M]

**Solution:** Evaluating the options for Graph Algorithms - Graph BFS/DFS on Transpose Graph directly shows that **D** is the unique valid answer.

**Q4.26.49** [Key: **B;C;D**]

**Topic:** *Searching and Sorting - Array Inversions and Sorting Steps* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Searching and Sorting - Array Inversions and Sorting Steps. Verifying each choice against theoretical conditions confirms option(s) **B;C;D**.

**Q4.26.50** [Key: **B;D**]

**Topic:** *Python Programming - Python Closures and Mutable State* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Python Programming - Python Closures and Mutable State. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q4.26.58** [Key: **8 to 8**]

**Topic:** *Python Programming - Bubble-Sort Recursion and Swap Count* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Python Programming - Bubble-Sort Recursion and Swap Count. Applying standard principles and formulas gives the evaluated range **8 to 8**.

**Q4.25.18** [Key: **D**]

**Topic:** *Basic Data Structures - Hash Tables and Linear Probing* [MCQ, 1M]

**Solution:** Evaluating the options for Basic Data Structures - Hash Tables and Linear Probing directly shows that **D** is the unique valid answer.

**Q4.25.23** [Key: **A**]

**Topic:** *Python Programming - Python List Methods (extend vs append)* [MCQ, 1M]

**Solution:** Evaluating the options for Python Programming - Python List Methods (extend vs append) directly shows that **A** is the unique valid answer.

**Q4.25.27** [Key: **C**]

**Topic:** *Searching and Sorting - Binary Search Time Complexity* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Searching and Sorting - Binary Search Time Complexity. Verifying each choice against theoretical conditions confirms option(s) **C**.

**Q4.25.29** [Key: **A;C**]

**Topic:** *Searching and Sorting - Insertion Sort Swaps* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Searching and Sorting - Insertion Sort Swaps. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q4.25.47** [Key: **B**]

**Topic:** *Python Programming - Python Set Operations and Loops* [MCQ, 2M]

**Solution:** Evaluating the options for Python Programming - Python Set Operations and Loops directly shows that **B** is the unique valid answer.

**Q4.25.58** [Key: **A;C;D**]

**Topic:** *Graph Algorithms - Shortest Paths in Undirected Graphs* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Graph Algorithms - Shortest Paths in Undirected Graphs. Verifying each choice against theoretical conditions confirms option(s) **A;C;D**.

**Q4.25.63** [Key: **160 to 160**]

**Topic:** *Python Programming - Python Recursive Function Evaluation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Python Programming - Python Recursive Function Evaluation. Applying standard principles and formulas gives the evaluated range **160 to 160**.

**Q4.25.64** [Key: **24 to 24**]

**Topic:** *Basic Data Structures - Stack Simulation Pseudocode* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Basic Data Structures - Stack Simulation Pseudocode. Applying standard principles and formulas gives the evaluated range **24 to 24**.

**Q4.25.65** [Key: **75 to 75**]

**Topic:** *Graph Algorithms - Directed Graph DFS Traversal Order* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Graph Algorithms - Directed Graph DFS Traversal Order. Applying standard principles and formulas gives the evaluated range **75 to 75**.

**Q4.24.14** [Key: **A**]

**Topic:** *Graph Algorithms - Depth-First Search (DFS)*

[MCQ, 1M]

**Solution:** Evaluating the options for Graph Algorithms - Depth-First Search (DFS) directly shows that **A** is the unique valid answer.

**Q4.24.16** [Key: **A**]

**Topic:** *Basic Data Structures - Matching Data Structures and Properties* [MCQ, 1M]

**Solution:** Evaluating the options for Basic Data Structures - Matching Data Structures and Properties directly shows that **A** is the unique valid answer.

**Q4.24.21** [Key: **B**]

**Topic:** *Basic Data Structures - Open Addressing Hash Tables* [MCQ, 1M]

**Solution:** Evaluating the options for Basic Data Structures - Open Addressing Hash Tables directly shows that **B** is the unique valid answer.

**Q4.24.28** [Key: **A;B;C**]

**Topic:** *Trees and Heaps - Full Binary Tree Reconstruction* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Trees and Heaps - Full Binary Tree Reconstruction. Verifying each choice against theoretical conditions confirms option(s) **A;B;C**.

**Q4.24.30** [Key: **0 to 0**]

**Topic:** *Searching and Sorting - Quicksort Swaps* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Searching and Sorting - Quicksort Swaps. Applying standard principles and formulas gives the evaluated range **0 to 0**.

**Q4.24.32** [Key: **17 to 17**]

**Topic:** *Basic Data Structures - Double-Ended Queue (Deque)* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Basic Data Structures - Double-Ended Queue (Deque). Applying standard principles and formulas gives the evaluated range **17 to 17**.

**Q4.24.38** [Key: **D**]

**Topic:** *Python Programming - Python Tree Traversal and Recursion* [MCQ, 2M]

**Solution:** Evaluating the options for Python Programming - Python Tree Traversal and Recursion directly shows that **D** is the unique valid answer.

**Q4.24.39** [Key: **C**]

**Topic:** *Python Programming - Pseudocode Array Processing* [MCQ, 2M]

**Solution:** Evaluating the options for Python Programming - Pseudocode Array Processing directly shows that **C** is the unique valid answer.

**Q4.24.40** [Key: **A**]

**Topic:** *Searching and Sorting - Binary Search Comparisons Recurrence* [MCQ, 2M]

**Solution:** Evaluating the options for Searching and Sorting - Binary Search Comparisons Recurrence directly shows that **A** is the unique valid answer.

**Q4.24.41** [Key: **C**]

**Topic:** *Python Programming - Python List In-Place Reversal* [MCQ, 2M]

**Solution:** Evaluating the options for Python Programming - Python List In-Place Reversal directly shows that **C** is the unique valid answer.

**Q4.24.45** [Key: **B**]

**Topic:** *Searching and Sorting - Sorting Algorithm Stability and Inversions* [MCQ, 2M]

**Solution:** Evaluating the options for Searching and Sorting - Sorting Algorithm Stability and Inversions directly shows that **B** is the unique valid answer.

**Q4.24.52** [Key: **A;B;C**]

**Topic:** *Trees and Heaps - Full Binary Tree Node Properties* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Trees and Heaps - Full Binary Tree Node Properties. Verifying each choice against theoretical conditions confirms option(s) **A;B;C**.

# Part 5

# Database Management and Warehousing

## GATE 2026 (IIT Guwahati)

▸ **Transactions and Concurrency: ACID Properties and Serializability**

### Q5.26.14 MCQ 2026 1M ○○○

Which of the following statements is NOT true? (The names of the predicates are intuitive.)

**(A)** ∀x ∀y Classmate(x, y) ⇒Classmate(y, x)

**(B)** ∀x Likes(x, Icecream) ⇒ ¬∃x ¬Likes(x, Icecream)

**(C)** "Each king is a person" is equivalent to ∀x IsKing(x) ∧ IsPerson(x)

**(D)** "All humans are mortal" is equivalent to ∀x IsHuman(x) ⇒IsMortal(x)

▸ **Relational Model and Normalization: Functional Dependencies and Normal Forms**

### Q5.26.17 MCQ 2026 1M ○○○

Let R(A,B,C,D,E) be a relational schema with functional dependency set F=A→BC, CD→E, E→A. Which of the following statements is correct?

**(A)** AD, ED and CD are the only candidate keys of R.

**(B)** AD and ED are the only candidate keys of R.

**(C)** A, E and CD are the only candidate keys of R.

**(D)** A and CD are the only candidate keys of R.

▸ **Data Warehousing: OLAP Cube Operations (Roll-up, Drill-down, Slice, Dice)**

### Q5.26.18 MCQ 2026 1M ○○○

Consider that the visualization of a 3-dimensional data cube is showing Sales Quantity for each combination of the attributes Product Type, Month and Country. From this, if we want to further visualize the Sales Quantity for each combination of Product Type, Month and State, which of the following OLAP operations should be performed?

**A.** Slicing
**B.** Dicing
**C.** Roll-up
**D.** Drill-down

▸ **Relational Model and Normalization: Foreign Key Referential Integrity Constraints**

### Q5.26.26 MSQ 2026 1M ○○○

Consider two relations r and s defined on the relational schemas R(A,B) and S(E,C), respectively. A is the primary key of R and E is a foreign key of S referencing A in R. Which of the following operations will NEVER violate the foreign key constraint?

**(A)** Inserting records into relation r

**(B)** Deleting records from relation s

**(C)** Deleting records from relation r

**(D)** Inserting records into relation s

▸ **Indexing and Storage: B+ Tree Fan-out and Node Capacity**

### Q5.26.32 NAT 2026 1M ○○○

In a relational database, a B+ Tree Index is to be constructed for a relation on a key field. In a B+ Tree, a Node Pointer points to a sub-tree and a Data Record Pointer points to a block of database records. Let, Node size = 4096 bytes, Node Pointer size = 10 bytes, Search Key Field size = 11 bytes and Data Record Pointer size = 12 bytes. The maximum number of Node Pointers that can be present in a non-leaf node of the B+ Tree is _____ . (Answer in integer)

▸ **Indexing and Storage: B+ Tree Node Split and Key Promotion**

### Q5.26.41 MCQ 2026 2M ○○○

Consider a B+ Tree where the maximum number of key values in each leaf node is 2 and the maximum number of pointers in each non-leaf node is 3. Let the content of the B+ Tree be as shown in the figure. Which of the following options denotes the key value(s) stored in the root node after inserting a key value 3 in the given B+ Tree?

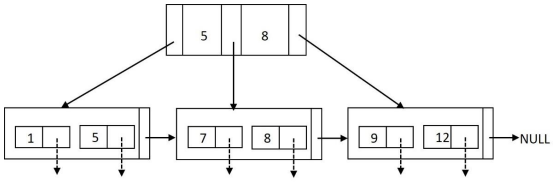



**Fig. Q5.26.41**

**A.** 5
**B.** 8
**C.** 3 and 5
**D.** 3, 5 and 8

▸ **Relational Algebra: Relational Algebra Natural Join and Selection**

## Q5.26.42 MCQ 2026 2M ○○○

Consider the given relations $X, Y$ and $Z$:

| $X$ | | |
|---|---|---|
| **P** | **Q** | **R** |
| P1 | Q1 | R1 |
| P2 | Q2 | R2 |
| P3 | Q3 | R2 |

| $Y$ | | |
|---|---|---|
| **P** | **Q** | **S** |
| P1 | Q1 | 2 |
| P1 | Q2 | 5 |
| P2 | Q1 | 6 |
| P3 | Q3 | 1 |

| $Z$ | |
|---|---|
| **P** | **T** |
| P1 | T1 |
| P3 | T2 |
| P4 | T3 |
| P4 | NULL |

Consider the relational algebra expression:

$$\Pi_{PRS}\left[(\sigma_{Q=Q_3 \vee R=R_2}(X \bowtie Y)) \bowtie (\sigma_{S>1}(Y \bowtie Z))\right]$$

where $\bowtie$ denotes natural join. Which of the following is the correct number of tuples returned?

**A.** 0 **B.** 1 **C.** 2 **D.** 3

▸ **Data Warehousing: Concept Hierarchies and Roll-Up Dimensions**

## Q5.26.43 MCQ 2026 2M ○○○

Consider the concept hierarchies as shown in the figure. Which of the following options denotes the total number of possible data cuboids from these concept hierarchies?

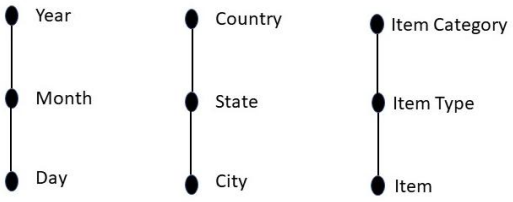



**Fig. Q5.26.43**

**A.** $4^3$
**B.** $2^3$
**C.** 2!
**D.** 4!

▸ **SQL: SQL Self-Join and Aggregation**

## Q5.26.51 MSQ 2026 2M ○○○

Consider a table `Employee(EmpID, TeamID)`, where the column `EmpID` (ID of an employee) is the primary key. The column `TeamID` denotes the team ID of the team of which the employee is a member. `TeamID` is a `NOT NULL` column. We want to display the size of the team (denoted as `TeamSize`) in which each employee is a member by using SQL. As an example, the desired output for the given `Employee` table is also shown in tabular form.

| Employee | |
|---|---|
| **EmpID** | **TeamID** |
| 1 | 8 |
| 2 | 8 |
| 3 | 8 |
| 4 | 7 |
| 5 | 7 |
| 6 | 9 |

| Output | |
|---|---|
| **EmpID** | **TeamSize** |
| 1 | 3 |
| 2 | 3 |
| 3 | 3 |
| 4 | 2 |
| 5 | 2 |
| 6 | 1 |

Which of the following is/are correct?

**(A)**
```
SELECT E.EmpID, B.TeamSize FROM Employee
AS E, (SELECT TeamID, COUNT(TeamID) AS
TeamSize FROM Employee GROUP BY TeamID)
AS B WHERE E.TeamID = B.TeamID
```

**(B)**
```
SELECT A.EmpID, COUNT(B.TeamID) AS TeamSize
FROM Employee AS A, Employee AS B WHERE
A.TeamID = B.TeamID AND A.EmpID = B.EmpID
GROUP BY A.EmpID
```

**(C)**
```
SELECT B.EmpID, B.TeamSize FROM (SELECT
EmpID, COUNT(TeamID) AS TeamSize FROM
Employee GROUP BY EmpID) AS B
```

**(D)**
```
SELECT A.EmpID, B.TeamSize FROM Employee
AS A, (SELECT COUNT(TeamID) AS TeamSize
FROM Employee GROUP BY TeamID) AS B
WHERE A.TeamID = B.TeamID
```

▸ **Relational Algebra: Tuple Relational Calculus (TRC) Evaluation**

## Q5.26.59 NAT 2026 2M ○○○

Let there be two relations $X$ and $Y$ as shown:

| Relation $X$ | | |
|---|---|---|
| **P** | **Q** | **R** |
| P1 | Q1 | R1 |
| P2 | Q2 | R2 |
| P3 | Q3 | R2 |

| Relation $Y$ | |
|---|---|
| **P** | **S** |
| P1 | 10 |
| P1 | 15 |
| P2 | 20 |
| P3 | 1 |

Consider that the following tuple relational calculus expression is evaluated:

$$\{t \mid t \in X \wedge \exists z \in X(t[P] = z[P]) \wedge \exists m \in Y(m[P] = t[P] \wedge m[S] > 1$$

The number of tuples that will be returned is _____ . (Answer in integer)

▸ **SQL: SQL Correlated Subquery with Comparison Counts**

## Q5.26.60 NAT 2026 2M ○○○

Let Account be a relation as shown:

| AccNo | Balance |
|---|---|
| A1 | 5000 |
| A2 | 5000 |
| A3 | 10000 |
| A4 | 15000 |
| A5 | 18000 |

Consider the given SQL query:

```
SELECT AccNo FROM Account AS A
WHERE (SELECT COUNT(*) FROM Account AS B
       WHERE A.Balance < B.Balance) >=
      (SELECT COUNT(*) FROM Account AS C
       WHERE A.Balance > C.Balance);
```

The number of rows returned by the SQL query is _____ . (Answer in integer)

▶ Relational Model and Normalization: ER Diagram to 3NF Relational Mapping

### Q5.26.61 NAT 2026 2M ○○○

Consider an ER model with the entities E1(A11,A12,A13) and E2(A21,A22,A23), where, A11, A12, A13 are the attributes of E1, and A21, A22, A23 are the attributes of E2. Let A22 be a multi-valued attribute. A11 and A21 are the primary keys of E1 and E2, respectively. Let R12 be a many-to-many relationship between E1 and E2. Participation of both E1 and E2 in R12 is total. The minimum number of relations required to convert the ER model into relational model (assuming there is no other functional dependency) where each relation is in third normal form (3NF) is _____ . (Answer in integer)

## GATE 2025 (IIT Roorkee)

▶ Relational Model and Normalization: Dependency-Preserving Decompositions

### Q5.25.16 MCQ 2025 1M ○○○

If a relational decomposition is not dependency-preserving, which one of the following relational operators will be executed more frequently in order to maintain the depen- dencies?

**(A)** Selection

**(B)** Projection

**(C)** Join

**(D)** Set union

▶ Relational Algebra: Relational Algebra Expression Evaluation

### Q5.25.17 MCQ 2025 1M ○○○

Consider the following three relations: Car (model, year, serial, color) Make (maker, model) Own (owner, serial) A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ($\bowtie$denotes natural join) $\pi$owner(Own $\bowtie$($\sigma$color="red"(Car $\bowtie$($\sigma$maker="ABC"Make)))) Which one of the following options describes what the above expression computes?

**(A)** All owners of a red car, a car made by ABC, or a red car made by ABC

**(B)** All owners of more than one car, where at least one car is red and made by ABC

**(C)** All owners of a red car made by ABC

**(D)** All red cars made by ABC

▶ SQL: SQL Correlated Subquery on Bank Loan

### Q5.25.33 NAT 2025 1M ○○○

On a relation named Loan of a bank:

| loan_number | branch_name | amount |
|---|---|---|
| L11 | Banjara Hills | 90000 |
| L14 | Kondapur | 50000 |
| L15 | SR Nagar | 40000 |
| L22 | SR Nagar | 25000 |
| L23 | Balanagar | 80000 |
| L25 | Kondapur | 70000 |
| L19 | SR Nagar | 65000 |

the following SQL query is executed:

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
                   FROM Loan L2
                   WHERE L2.branch_name =
                       'SR Nagar');
```

The number of rows returned by the query is _____ . (Answer in integer)

▶ Data Warehousing: OLAP CUBE Aggregate Tuples

### Q5.25.46 MCQ 2025 2M ○○○

Consider a fact table in an OLAP application: Facts(D1, D2, val), where D1 and D2 are its dimension attributes and val is a dependent attribute. Suppose attribute D1 takes 3 values and D2 takes 2 values, and all combinations of these values are present in the table Facts. How many tuples are there in the result of the following query? SELECT D1, D2, sum(val) FROM Facts GROUP BY CUBE (D1, D2);

**(A)** 1

**(B)** 6

**(C)** 9

**(D)** 12

▶ **Relational Model and Normalization: Primary Keys, Foreign Keys and Nullability**

## Q5.25.56 MSQ 2025 2M ○○○

Consider the following two relations, named `Customer` and `Person`, in a database: `Person(aadhaar CHAR(12) PRIMARY KEY, name VARCHAR(32));` `Customer(name VARCHAR(32), email VARCHAR(32) PRIMARY KEY, phone CHAR(10), aadhaar CHAR(12), FOREIGN KEY (aadhaar) REFERENCES Person(aadhaar))`; Which of the following statements is/are correct?

**(A)** `aadhaar` is a candidate key in the `Customer` relation

**(B)** `phone` can be `NULL` in the `Customer` relation

**(C)** `aadhaar` is a candidate key in the `Person` relation

**(D)** `aadhaar` can be `NULL` in the `Person` relation

▶ **Relational Model and Normalization: Functional Dependencies and BCNF**

## Q5.25.57 MSQ 2025 2M ○○○

Consider a database relation R with attributes ABCDEFG, and having the following functional dependencies: A →BCEF E →DG BC →A Which of the following statements is/are correct?

**(A)** A is the only candidate key of R

**(B)** A, BC are the candidate keys of R

**(C)** A, BC, E are the candidate keys of R

**(D)** Relation R is not in Boyce-Codd Normal Form (BCNF)

▶ **Relational Algebra: Relational Division Operator**

## Q5.25.62 NAT 2025 2M ○○○

Consider the following tables, Loan and Borrower, of a bank:

| Loan | | |
|---|---|---|
| loan_num | branch_name | amount |
| L11 | Banjara Hills | 90000 |
| L14 | Kondapur | 50000 |
| L15 | SR Nagar | 40000 |
| L22 | SR Nagar | 25000 |
| L23 | Balanagar | 80000 |
| L25 | Kondapur | 70000 |
| L19 | SR Nagar | 65000 |

| Borrower | |
|---|---|
| customer_name | loan_num |
| Anand | L11 |
| Karteek | L11 |
| Karteek | L14 |
| Ankita | L15 |
| Gopal | L19 |
| Karteek | L22 |
| Karteek | L23 |
| Sunil | L23 |
| Sunil | L25 |

Query: $\pi_{\text{branch_name,customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan})$ where $\bowtie$ denotes natural join. The number of tuples returned by the query is _____ . (Answer in integer)

# GATE 2024 (IISc Bengaluru)

▶ **Relational Algebra: Relational Set Operations and Joins**

## Q5.24.26 MCQ 2024 1M ○○○

Consider a database that includes the following relations: Defender(name, rating, side, goals) Forward(name, rating, assists, goals) Team(name, club, price) Which ONE of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where $\phi$ denotes the empty set?

**(A)** $\Pi$name(Team) \ ($\Pi$name(Defender)$\cap\Pi$name(Forward)) $= \phi$

**(B)** ($\Pi$name(Defender)$\cap\Pi$name(Forward))\$\Pi$name(Team) $= \phi$

**(C)** $\Pi$name(Team) \ ($\Pi$name(Defender)$\cup\Pi$name(Forward)) $= \phi$

**(D)** ($\Pi$name(Defender)$\cup\Pi$name(Forward)) \$\Pi$name(Team) $= \phi$

▶ **SQL: SQL Join and Filter Query**

## Q5.24.31 NAT 2024 1M ○○○

Consider the following two tables named Raider and Team in a relational database maintained by a Kabaddi league. The attribute ID in table Team references the primary key of the Raider table, ID.

| Raider | | | |
|---|---|---|---|
| ID | Name | Raids | RaidPoints |
| 1 | Arjun | 200 | 250 |
| 2 | Ankush | 190 | 219 |
| 3 | Sunil | 150 | 200 |
| 4 | Reza | 150 | 190 |
| 5 | Pratham | 175 | 220 |
| 6 | Gopal | 193 | 215 |

| Team | | |
|---|---|---|
| City | ID | BidPoints |
| Jaipur | 2 | 200 |
| Patna | 3 | 195 |
| Hyderabad | 5 | 175 |
| Jaipur | 1 | 250 |
| Patna | 4 | 200 |
| Jaipur | 6 | 200 |

The SQL query described below is executed on this database:

```
SELECT *
FROM Raider, Team
WHERE Raider.ID=Team.ID AND City='Jaipur'
  AND RaidPoints > 200;
```

The number of rows returned by this query is _____ .

▶ **Relational Model and Normalization: Candidate Keys and Normal Forms**

## Q5.24.46 MSQ 2024 2M ○○○

Given the relational schema R= (U,V, W, X, Y, Z) and the set of functional dependencies: U→V, U→W, WX→Y, WX→Z, V→X Which of the following functional dependencies can be derived from the above set?

**A.** VW→YZ
**B.** WX→YZ
**C.** VW→U
**D.** VW→Y

▶ **Indexing and Storage: B+ Tree vs Hash Index Optimization**

## Q5.24.55 MSQ 2024 2M ○○○

An OTT company is maintaining a large disk-based relational database of different movies with the following schema: Movie(ID, CustomerRating) Genre(ID, Name) Movie_Genre(MovieID, GenreID) Consider the following SQL query on the relation database above: SELECT * FROM Movie, Genre, Movie_Genre WHERE

Movie.CustomerRating > 3.4 AND

Genre.Name = "Comedy" AND

Movie_Genre.MovieID = Movie.ID AND

Movie_Genre.GenreID = Genre.ID; This SQL query can be sped up using which of the following indexing options?

**(A)** B+ tree on all the attributes.

**(B)** Hash index on Genre.Name and B+ tree on the remaining attributes.

**(C)** Hash index on Movie.CustomerRating and B+ tree on the remaining attributes.

**(D)** Hash index on all the attributes.

# Answer Key & Solutions: Database Management and Warehousing

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q5.26.14** | 2026 | MCQ | **C** | Transactions and Concurrency - ACID Properties and Serializability |
| **Q5.26.17** | 2026 | MCQ | **A** | Relational Model and Normalization - Functional Dependencies and Normal Forms |
| **Q5.26.18** | 2026 | MCQ | **D** | Data Warehousing - OLAP Cube Operations (Roll-up, Drill-down, Slice, Dice) |
| **Q5.26.26** | 2026 | MSQ | **A;B** | Relational Model and Normalization - Foreign Key Referential Integrity Constraints |
| **Q5.26.32** | 2026 | NAT | **195 to 195** | Indexing and Storage - B+ Tree Fan-out and Node Capacity |
| **Q5.26.41** | 2026 | MCQ | **A** | Indexing and Storage - B+ Tree Node Split and Key Promotion |
| **Q5.26.42** | 2026 | MCQ | **D** | Relational Algebra - Relational Algebra Natural Join and Selection |
| **Q5.26.43** | 2026 | MCQ | **A** | Data Warehousing - Concept Hierarchies and Roll-Up Dimensions |
| **Q5.26.51** | 2026 | MSQ | **A** | SQL - SQL Self-Join and Aggregation |
| **Q5.26.59** | 2026 | NAT | **2 to 3** | Relational Algebra - Tuple Relational Calculus (TRC) Evaluation |
| **Q5.26.60** | 2026 | NAT | **3 to 3** | SQL - SQL Correlated Subquery with Comparison Counts |
| **Q5.26.61** | 2026 | NAT | **4 to 4** | Relational Model and Normalization - ER Diagram to 3NF Relational Mapping |
| **Q5.25.16** | 2025 | MCQ | **C** | Relational Model and Normalization - Dependency-Preserving Decompositions |
| **Q5.25.17** | 2025 | MCQ | **C** | Relational Algebra - Relational Algebra Expression Evaluation |
| **Q5.25.33** | 2025 | NAT | **3 to 3** | SQL - SQL Correlated Subquery on Bank Loan |
| **Q5.25.46** | 2025 | MCQ | **D** | Data Warehousing - OLAP CUBE Aggregate Tuples |
| **Q5.25.56** | 2025 | MSQ | **B;C** | Relational Model and Normalization - Primary Keys, Foreign Keys and Nullability |
| **Q5.25.57** | 2025 | MSQ | **B;D** | Relational Model and Normalization - Functional Dependencies and BCNF |
| **Q5.25.62** | 2025 | NAT | **1 to 1** | Relational Algebra - Relational Division Operator |
| **Q5.24.26** | 2024 | MCQ | **C** | Relational Algebra - Relational Set Operations and Joins |
| **Q5.24.31** | 2024 | NAT | **3 to 3** | SQL - SQL Join and Filter Query |
| **Q5.24.46** | 2024 | MSQ | **A;B;D** | Relational Model and Normalization - Candidate Keys and Normal Forms |
| **Q5.24.55** | 2024 | MSQ | **A;B** | Indexing and Storage - B+ Tree vs Hash Index Optimization |

## Detailed Step-by-Step Explanations

### Q5.26.14 [Key: **C**]

**Topic:** *Transactions and Concurrency - ACID Properties and Serializability* [MCQ, 1M]

**Solution:** Evaluating the options for Transactions and Concurrency - ACID Properties and Serializability directly shows that **C** is the unique valid answer.

### Q5.26.17 [Key: **A**]

**Topic:** *Relational Model and Normalization - Functional Dependencies and Normal Forms* [MCQ, 1M]

**Solution:** Evaluating the options for Relational Model and Normalization - Functional Dependencies and Normal Forms directly shows that **A** is the unique valid answer.

### Q5.26.18 [Key: **D**]

**Topic:** *Data Warehousing - OLAP Cube Operations (Roll-up, Drill-down, Slice, Dice)* [MCQ, 1M]

**Solution:** Evaluating the options for Data Warehousing - OLAP Cube Operations (Roll-up, Drill-down, Slice, Dice) directly shows that **D** is the unique valid answer.

### Q5.26.26 [Key: **A;B**]

**Topic:** *Relational Model and Normalization - Foreign Key Referential Integrity Constraints* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Relational Model and Normalization - Foreign Key Referential Integrity Constraints. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

**Q5.26.32** [Key: **195 to 195**]

**Topic:** *Indexing and Storage - B+ Tree Fan-out and Node Capacity* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Indexing and Storage - B+ Tree Fan-out and Node Capacity. Applying standard principles and formulas gives the evaluated range **195 to 195**.

**Q5.26.41** [Key: **A**]

**Topic:** *Indexing and Storage - B+ Tree Node Split and Key Promotion* [MCQ, 2M]

**Solution:** Evaluating the options for Indexing and Storage - B+ Tree Node Split and Key Promotion directly shows that **A** is the unique valid answer.

**Q5.26.42** [Key: **D**]

**Topic:** *Relational Algebra - Relational Algebra Natural Join and Selection* [MCQ, 2M]

**Solution:** Evaluating the options for Relational Algebra - Relational Algebra Natural Join and Selection directly shows that **D** is the unique valid answer.

**Q5.26.43** [Key: **A**]

**Topic:** *Data Warehousing - Concept Hierarchies and Roll-Up Dimensions* [MCQ, 2M]

**Solution:** Evaluating the options for Data Warehousing - Concept Hierarchies and Roll-Up Dimensions directly shows that **A** is the unique valid answer.

**Q5.26.51** [Key: **A**]

**Topic:** *SQL - SQL Self-Join and Aggregation* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing SQL - SQL Self-Join and Aggregation. Verifying each choice against theoretical conditions confirms option(s) **A**.

**Q5.26.59** [Key: **2 to 3**]

**Topic:** *Relational Algebra - Tuple Relational Calculus (TRC) Evaluation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Relational Algebra - Tuple Relational Calculus (TRC) Evaluation. Applying standard principles and formulas gives the evaluated range **2 to 3**.

**Q5.26.60** [Key: **3 to 3**]

**Topic:** *SQL - SQL Correlated Subquery with Comparison Counts* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing SQL - SQL Correlated Subquery with Comparison Counts. Applying standard principles and formulas gives the evaluated range **3 to 3**.

**Q5.26.61** [Key: **4 to 4**]

**Topic:** *Relational Model and Normalization - ER Diagram to 3NF Relational Mapping* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Relational Model and Normalization - ER Diagram to 3NF Relational Mapping. Applying standard principles and formulas gives the evaluated range **4 to 4**.

**Q5.25.16** [Key: **C**]

**Topic:** *Relational Model and Normalization - Dependency-Preserving Decompositions* [MCQ, 1M]

**Solution:** Evaluating the options for Relational Model and Normalization - Dependency-Preserving Decompositions directly shows that **C** is the unique valid answer.

**Q5.25.17** [Key: **C**]

**Topic:** *Relational Algebra - Relational Algebra Expression Evaluation* [MCQ, 1M]

**Solution:** Evaluating the options for Relational Algebra - Relational Algebra Expression Evaluation directly shows that **C** is the unique valid answer.

**Q5.25.33** [Key: **3 to 3**]

**Topic:** *SQL - SQL Correlated Subquery on Bank Loan* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing SQL - SQL Correlated Subquery on Bank Loan. Applying standard principles and formulas gives the evaluated range **3 to 3**.

**Q5.25.46** [Key: **D**]

**Topic:** *Data Warehousing - OLAP CUBE Aggregate Tuples* [MCQ, 2M]

**Solution:** Evaluating the options for Data Warehousing - OLAP CUBE Aggregate Tuples directly shows that **D** is the unique valid answer.

**Q5.25.56** [Key: **B;C**]

**Topic:** *Relational Model and Normalization - Primary Keys, Foreign Keys and Nullability* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Relational Model and Normalization - Primary Keys, Foreign Keys and Nullability. Verifying each choice against theoretical conditions confirms option(s) **B;C**.

**Q5.25.57** [Key: **B;D**]

**Topic:** *Relational Model and Normalization - Functional Dependencies and BCNF* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Relational Model and Normalization - Functional Dependencies and BCNF. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q5.25.62** [Key: **1 to 1**]

**Topic:** *Relational Algebra - Relational Division Operator* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Relational Algebra - Relational Division Operator. Applying standard principles and formulas gives the evaluated range **1 to 1**.

**Q5.24.26** [Key: **C**]

**Topic:** *Relational Algebra - Relational Set Operations and Joins* [MCQ, 1M]

**Solution:** Evaluating the options for Relational Algebra - Relational Set Operations and Joins directly shows that **C** is the unique valid answer.

**Q5.24.31** [Key: **3 to 3**]

**Topic:** *SQL - SQL Join and Filter Query* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing SQL - SQL Join and Filter Query. Applying standard principles and formulas gives the evaluated range **3 to 3**.

**Q5.24.46** [Key: **A;B;D**]

**Topic:** *Relational Model and Normalization - Candidate Keys and Normal Forms* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Relational Model and Normalization - Candidate Keys and Normal Forms. Verifying each choice against theoretical conditions confirms option(s) **A;B;D**.

**Q5.24.55** [Key: **A;B**]

**Topic:** *Indexing and Storage - B+ Tree vs Hash Index Optimization* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Indexing and Storage - B+ Tree vs Hash Index Optimization. Verifying each choice against theoretical conditions confirms option(s) **A;B**.

# Part 6

# Machine Learning

## GATE 2026 (IIT Guwahati)

▸ **Linear Dimensionality Reduction: PCA Orthogonal Eigenvectors**

### Q6.26.11 MCQ 2026 1M ○○○

For a classification problem, Principal Component Analysis (PCA) has been used to reduce the dimensionality of a feature space from 100 to 10. Which of the following options is true about the angle $\theta$ between the first and the tenth principal components?

**(A)** $\theta = 0°$

**(B)** $\theta = 90°$

**(C)** $90° < \theta \leq 180°$

**(D)** $0° < \theta < 90°$

▸ **Model Evaluation and Selection: Leave-One-Out Cross-Validation (LOOCV)**

### Q6.26.12 MCQ 2026 1M ○○○

Consider that you are training a classifier for a 10-class classification problem. Each input is represented as a 512-dimensional vector. There are 1000 samples, out of which first 100 will be used for testing. Let Leave-One-Out-Cross- Validation (LOOCV) be used for selection of the classifier model before testing. Which of the following options is the correct number of validation splits that will be generated?

**A.** 10 **B.** 512 **C.** 900 **D.** 1000

▸ **Overview of Machine Learning: Matching ML Paradigms (Regression, Clustering, Classification)**

### Q6.26.23 MSQ 2026 1M ○○○

In the following table, the Task column lists a few tasks related to machine learning. The Algorithm column lists a few algorithms.

Each entry $t$ from the Task column is to be matched with an appropriate entry $a$ from the Algorithm column such that the task $t$ can be solved using the algorithm $a$. Denote such a match as $t : a$.

| Task | Algorithm |
|---|---|
| T1 – Clustering | A1 – Markov Chain Monte Carlo |
| T2 – Classification | A2 – K-Medoid |
| T3 – Sampling | A3 – Linear Discriminant Analysis |
| T4 – Feature Extraction | A4 – Naive Bayes |

Which of the following options is/are the correct matching(s)?

**(A)** T1:A4, T2:A3, T3:A1, T4:A2

**(B)** T1:A2, T2:A4, T3:A1, T4:A3

**(C)** T1:A3, T2:A4, T3:A1, T4:A2

**(D)** T1:A4, T2:A2, T3:A1, T4:A3

▸ **Optimization: Stochastic Gradient Descent (SGD) Parameter Update**

### Q6.26.29 NAT 2026 1M ○○○

Consider that for a supervised learning task, the objective function being minimized is fw(x) = wx, where x∈ℝ is the input and w∈ℝ is the parameter. Stochastic Gradient Descent with learning rate of 0.10 is used for parameter updates. Suppose that at the end of iteration i, the value of w becomes 10.00. Let x= 10.00 be the input for iteration (i+ 1). The value of w at the end of iteration (i+ 1) is _____ . (Rounded off to two decimal places)

▸ **Unsupervised Learning: Hierarchical Agglomerative Clustering (Manhattan Distance)**

### Q6.26.36 MCQ 2026 2M ○○○

Let four points in three-dimensional space be: P1: [2, 3, −1], P2: [3, 1, 1], P3: [5, −2, 3] and P4: [3, 3, 3]. Hierarchical Agglomerative Clustering is used to cluster the above points. If Manhattan Distance is used as the distance metric during clustering, which of the following options indicates the two points that will be merged first?

**A.** P1, P2 **C.** P3, P4
**B.** P2, P3 **D.** P2, P4

▶ Supervised Learning: Ridge Regression Regularization

### Q6.26.37 MCQ 2026 2M ○○○

Which of the following statements is true for Ridge Regression?

**(A)** The regularizer in the objective function of Ridge Regression is used to guard against scenarios where the model works well for the test data, but poorly for the training data.

**(B)** The regularizer of Ridge Regression uses L1 norm.

**(C)** Ridge Regression aims to reduce the number of parameters that have negative values.

**(D)** The regularizer of Ridge Regression may increase the bias of the model, but it helps in reducing the variance in predictions.

▶ Supervised Learning: Ridge Regression Regularized Loss Computation

### Q6.26.55 NAT 2026 2M ○○○

Consider that Linear Ridge Regression is being used to learn a prediction function ypred= $w^T x$, where w, x∈ℝ2 and Mean Absolute Error (MAE) is used to measure the prediction error. A weight of 0.20 is associated with the regularizer. At an intermediate step of the training process, assume that the parameter w= $[-3.00, 4.00]^T$. In the next step, for the input x= $[1.00, 2.00]^T$, the predicted value of y is noted. Let the relation between x= $[x1, x2]^T$ and the true value of y be ytrue= x1 + x2.

The value of the overall regularized loss function for this instance is _____ . (Rounded off to two decimal places)

▶ Neural Networks: Multilayer Perceptron Parameter Count

### Q6.26.56 NAT 2026 2M ○○○

Consider a fully-connected feed-forward multi-layer perceptron. It has 30 neurons in the input layer, followed by two hidden layers and an output layer. The first hidden layer has 4 neurons and the second hidden layer has 3 neurons. The output layer has only one neuron. Assume that no bias parameters are used. The number of learnable parameters in the multi-layer perceptron is _____ . (Answer in integer)

## GATE 2025 (IIT Roorkee)

▶ Supervised Learning: Perceptron Update Rule

### Q6.25.22 MCQ 2025 1M ○○○

Consider designing a linear classifier y = sign(f(x; w, b)), f(x; w, b) = $w^\top x$ + b on a dataset D = (x1, y1), (x2, y2), . . . , (xN, yN), xi ∈Rd, yi ∈+1, −1, i = 1, 2, . . . , N. Recall that the sign function outputs +1 if the argument is positive, and −1 if the argument is non-positive. The parameters w and b are updated as per the following training algorithm: wnew = wold + ynxn, bnew = bold + yn whenever sign(f(xn; wold, bold)) = yn. In other words, whenever the classifier wrongly predicts a sample (xn, yn) from the dataset, wold gets updated to wnew, and likewise bold gets updated to bnew. Consider the case (xn, +1), f(xn; wold, bold) < 0. Then

**(A)** f(xn; wnew, bnew) > f(xn; wold, bold)

**(B)** f(xn; wnew, bnew) < f(xn; wold, bold)

**(C)** f(xn; wnew, bnew) = f(xn; wold, bold)

**(D)** ynf(xn; wold, bold) > 1

▶ Unsupervised Learning: Hierarchical Clustering Linkage Metrics

### Q6.25.30 MSQ 2025 1M ○○○

Let C1 and C2 be two sets of objects. Let D(x, y) be a measure of dissimilarity between two objects x and y. Consider the following definitions of dissimilarity between C1 and C2. DIS-1(C1, C2) = max x∈C1, y∈C2D(x, y) DIS-2(C1, C2) = min x∈C1, y∈C2D(x, y) Which of the following statements is/are correct?

**(A)** Single Linkage Clustering uses DIS-1

**(B)** Single Linkage Clustering uses DIS-2

**(C)** Complete Linkage Clustering uses DIS-2

**(D)** Complete Linkage Clustering uses DIS-1

▶ Supervised Learning: Linear Least Squares Regression

### Q6.25.34 NAT 2025 1M ○○○

Given data $\{(-1,1),(2,-5),(3,5)\}$ of the form $(x, y)$, we fit a model $y = wx$ using linear least-squares regression. The optimal value of $w$ is _____ . (Round off to three decimal places)

▶ Supervised Learning: Naive Bayes Classification Error

### Q6.25.35 NAT 2025 1M ○○○

The naive Bayes classifier is used to solve a two-class classification problem with class-labels $y_1, y_2$. Suppose the prior probabilities are $P(y_1) = \frac{1}{3}$ and $P(y_2) = \frac{2}{3}$. Assuming a discrete feature space with $P(x \mid y_1) = \frac{3}{4}$ and $P(x \mid y_2) = \frac{1}{4}$ for a specific feature vector $x$. The probability of misclassifying $x$ is _____ . (Round off to two decimal places)

▶ Neural Networks: Feed-Forward Neural Network Backpropagation

### Q6.25.42 MCQ 2025 2M ○○○

Consider the neural network shown in the figure with: inputs: $u, v$; weights: $a, b, c, d, e, f$; and output: $y$.

$R$ denotes the ReLU function, $R(x) = \max(0, x)$.

Given $u = 2, v = 3$, $a = 1, b = 1, c = 1, d = -1, e = 4, f = -1$, which one of the following is correct?

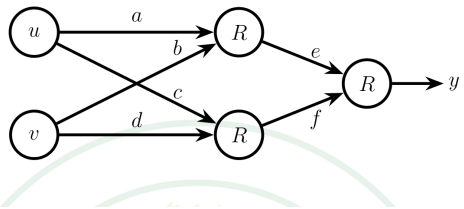



**Fig. Q6.25.42**

**(A)** $\frac{\partial y}{\partial a} = 8, \quad \frac{\partial y}{\partial f} = 0$

**(B)** $\frac{\partial y}{\partial a} = 1, \quad \frac{\partial y}{\partial f} = 0$

**(C)** $\frac{\partial y}{\partial a} = 1, \quad \frac{\partial y}{\partial f} = -1$

**(D)** $\frac{\partial y}{\partial a} = 2, \quad \frac{\partial y}{\partial f} = -1$

▶ **Neural Networks: ReLU Activation Continuity and Differentiability**

## Q6.25.48 MSQ 2025 2M ○○○

Which of the following statements is/are correct about the rectified linear unit (ReLU) activation function defined as ReLU(x) = max(x, 0), where x ∈R?

**(A)** ReLU is continuous everywhere

**(B)** ReLU is differentiable everywhere

**(C)** ReLU is not differentiable at x = 0

**(D)** ReLU(x) = ReLU(ax), for all a ∈R

▶ **Supervised Learning: Hard-Margin Support Vector Machine (SVM)**

## Q6.25.53 MSQ 2025 2M ○○○

Consider designing a linear binary classifier $f(\mathbf{x}) = \text{sign}(\mathbf{w}^T\mathbf{x} + b), \mathbf{x} \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1: } \left\{\begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}0\\2\end{bmatrix}, \begin{bmatrix}2\\2\end{bmatrix}\right\}, \quad \text{Class-2: } \left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}$$

Hard-margin support vector machine (SVM) formulation is used. Which of the following options is/are correct?

**(A)** $\mathbf{w} = \begin{bmatrix}1\\1\end{bmatrix}, b = -1$

**(B)** $\mathbf{w} = \begin{bmatrix}1\\1\end{bmatrix}, b = -2$

**(C)** The margin of separation is $\sqrt{2}$

**(D)** The margin of separation is $2\sqrt{2}$

▶ **Supervised Learning: Nearest Centroid Decision Boundary**

## Q6.25.55 MSQ 2025 2M ○○○

Consider a two-class problem in Rd with class labels red and green. Let $\mu$red and $\mu$green be the means of the two classes. Given test sample x ∈Rd, a classifier calculates the squared Euclidean distance (denoted by $\|\cdot\|2$) between x and the means of the two classes and assigns the class label that the sample x is closest to. That is, the classifier computes f(x) = $\|\mu$red −x$\|2$ −$\|\mu$green −x$\|2$ and assigns the label red to x if f(x) < 0, and green otherwise. Which of the following statements is/are correct?

**(A)** The sample x = 0 is assigned the label green if $\|\mu\text{red}\| < \|\mu\text{green}\|$

**(B)** f is a linear function of x

**(C)** f(x) = $\text{w}^\top$x + b, where w and b are functions of $\mu$red and $\mu$green

**(D)** f is a quadratic polynomial in x

▶ **Linear Dimensionality Reduction: Principal Component Analysis (Covariance Eigenvalues)**

## Q6.25.60 NAT 2025 2M ○○○

Let $D = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\}$ be a dataset of $n$ observations where each $\mathbf{x}^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^{n} \mathbf{x}^{(i)} = 0$. The covariance matrix computed from $D$ has eigenvalues $\lambda_i = 100^{2-i}, 1 \le i \le 100$. Let $\mathbf{u} \in \mathbb{R}^{100}$ be the direction of maximum variance with $\mathbf{u}^T\mathbf{u} = 1$. The value of $\frac{1}{n}\sum_{i=1}^{n}(\mathbf{u}^T\mathbf{x}^{(i)})^2 =$ _____ . (Answer in integer)

# GATE 2024 (IISc Bengaluru)

▶ **Supervised Learning: Support Vector Machines (SVM)**

## Q6.24.17 MCQ 2024 1M ○○○

Consider the dataset with six datapoints: $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_6, y_6)\}$, where

$$\mathbf{x}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix}0\\1\end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix}0\\-1\end{bmatrix}, \mathbf{x}_4 = \begin{bmatrix}-1\\0\end{bmatrix}, \mathbf{x}_5 = \begin{bmatrix}2\\2\end{bmatrix}, \mathbf{x}_6 =$$

and the labels are given by $y_1 = y_2 = y_5 = 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear support vector machine is trained on the above dataset. Which of the following is/are the support vector(s)?

**(A)** $\mathbf{x}_5$ and $\mathbf{x}_6$ only

**(B)** $\mathbf{x}_1$ and $\mathbf{x}_4$ only

**(C)** $\mathbf{x}_2$ and $\mathbf{x}_3$ only

**(D)** $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, and $\mathbf{x}_4$

▶ Overview of Machine Learning: Matching ML Paradigms (PCA, k-Means, Naive Bayes)

## Q6.24.18 MCQ 2024 1M ○○○

Match the items in Column 1 with the items in Column 2 in the following table:

| Column 1 | Column 2 |
|---|---|
| (p) Principal Component Analysis | (i) Discriminative Model |
| (q) Naive Bayes Classification | (ii) Dimensionality Reduction |
| (r) Logistic Regression | (iii) Generative Model |

**(A)** (p) – (iii), (q) – (i), (r) – (ii)

**(B)** (p) – (ii), (q) – (i), (r) – (iii)

**(C)** (p) – (ii), (q) – (iii), (r) – (i)

**(D)** (p) – (iii), (q) – (ii), (r) – (i)

▶ Unsupervised Learning: k-Means Clustering Convergence

## Q6.24.19 MCQ 2024 1M ○○○

Euclidean distance based k-means clustering algorithm was run on a dataset of 100 points with k= 3. If the points [1 1] and [−1 1 ] are both part of cluster 3, then which ONE of the following points is necessarily also part of cluster 3?

0 0

0 2

2 0

0 1

▶ Supervised Learning: Naive Bayes Binary Attributes

## Q6.24.20 MCQ 2024 1M ○○○

Given a dataset with K binary-valued attributes (where K> 2) for a two-class classification task, the number of parameters to be estimated for learning a naive Bayes classifier is

**A.** 2K+ 1 **C.** 2K+1 + 1

**B.** 2K+ 1 **D.** K2 + 1

▶ Linear Dimensionality Reduction: Fisher Linear Discriminant

## Q6.24.22 MCQ 2024 1M ○○○

For any binary classification dataset, let $S_B \in \mathbb{R}^{d\times d}$ and $S_W \in \mathbb{R}^{d\times d}$ be the between-class and within-class scatter matrices, respectively. If $S_W$ is nonsingular, Fisher's linear discriminant maximizes which ONE of the following objective functions?

**(A)** $\dfrac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$

**(B)** $\dfrac{\mathbf{w}^T S_W \mathbf{w}}{\mathbf{w}^T S_B \mathbf{w}}$

**(C)** $\dfrac{\mathbf{w}^T S_B S_W \mathbf{w}}{\mathbf{w}^T \mathbf{w}}$

**(D)** $\dfrac{\mathbf{w}^T (S_B + S_W) \mathbf{w}}{\mathbf{w}^T \mathbf{w}}$

▶ Data Preprocessing: z-Score Normalization

## Q6.24.27 MCQ 2024 1M ○○○

Let the minimum, maximum, mean and standard deviation values for the attribute income of data scientists be Rs. 46000, Rs. 170000, Rs. 96000, and Rs. 21000, respectively. The z-score normalized income value of Rs. 106000 is closest to which ONE of the following options?

**A.** 0.217 **C.** 0.623

**B.** 0.476 **D.** 2.304

▶ Unsupervised Learning: Single Linkage Hierarchical Clustering

## Q6.24.42 MCQ 2024 2M ○○○

Consider the table below, where the (i, j)th element of the table is the distance between points xi and xj. Single linkage clustering is performed on data points, x1,x2, x3,x4, x5.

Which ONE of the following is the correct representation of the clusters produced?

x1 x2 x3 x4 x5 x1 0 1 4 3 6 x2 1 0 3 5 3 x3 4 3 0 2 5 x4 3 5 2 0 1 x5 6 3 5 1 0 (A)

(B)

(C)

(D)

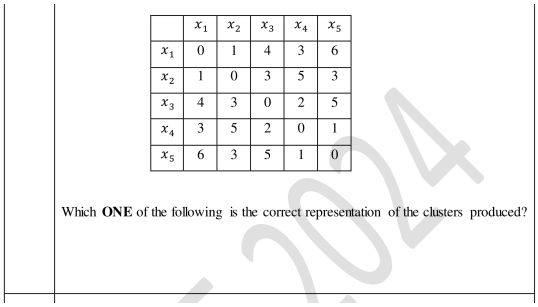



**Fig. Q6.24.42**

▶ Neural Networks: ReLU Multilayer Perceptron Equivalence

## Q6.24.43 MCQ 2024 2M ○○○

Consider the two neural networks (NNs) shown in Figures 1 and 2, with ReLU activation (ReLU(z) = max0, z , ∀z∈ℝ). ℝ denotes the set of real numbers. The connections and their corresponding weights are shown in the Figures. The biases at every neuron are set to 0. For what values of p, q, r in Figure 2 are the two NNs equivalent, when x1,x2, x3 are positive? Figure 1

Figure 2

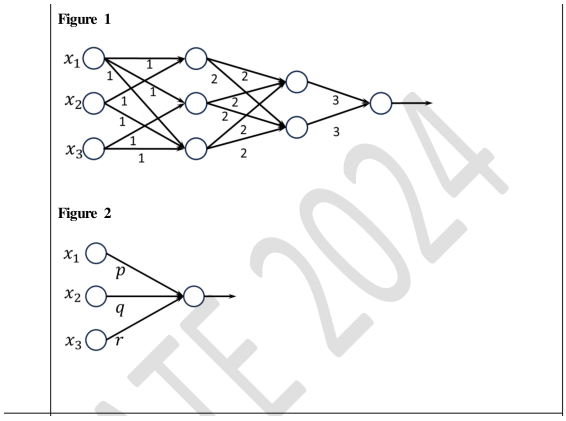



**Fig. Q6.24.43**

**A.** p= 36, q= 24,r= 24
**B.** p= 24, q= 24,r= 36
**C.** p= 18, q= 36,r= 24
**D.** p= 36, q= 36,r= 36

▶ **Supervised Learning: Linear Separability of 2D Datasets**

## Q6.24.53 MSQ 2024 2M ○○○

Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and squares.

(i)

(ii)

(iii)

(iv) Which of the following is/are TRUE?

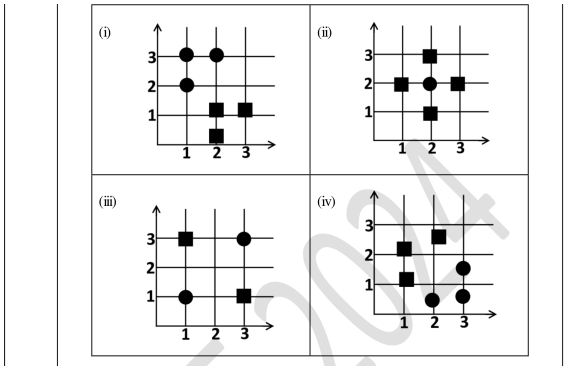



**Fig. Q6.24.53**

**(A)** (i) is linearly separable.

**(B)** (ii) is linearly separable.

**(C)** (iii) is linearly separable.

**(D)** (iv) is linearly separable.

▶ **Supervised Learning: Decision Tree Information Gain**

## Q6.24.62 NAT 2024 2M ○○○

Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute Pitch can take one of two values: spin-friendly (represented as $S$) or pace-friendly (represented as $F$). The attribute Format can take one of two values: one-day match (represented as $O$) or test match (represented as $T$).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

**Table C**

| Match | Pitch | Format | Winner (Target) |
|---|---|---|---|
| 1 | $S$ | $T$ | Green |
| 2 | $S$ | $T$ | Blue |
| 3 | $F$ | $O$ | Blue |
| 4 | $S$ | $O$ | Blue |
| 5 | $F$ | $T$ | Green |
| 6 | $F$ | $O$ | Blue |
| 7 | $S$ | $O$ | Green |
| 8 | $F$ | $T$ | Blue |
| 9 | $F$ | $O$ | Blue |
| 10 | $S$ | $O$ | Green |

To develop such a model, the computed InformationGain$(C, \text{Pitch})$ with respect to the Target is _____ (rounded off to two decimal places).

▶ **Supervised Learning: k-Nearest Neighbors Decision Boundary**

## Q6.24.63 NAT 2024 2M ○○○

Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k-nearest neighbor algorithm for which the diamond (◇) shaped data point is assigned the label square is _____.

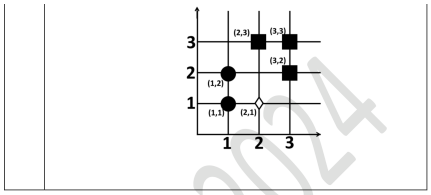



**Fig. Q6.24.63**

GATE 2024

# Answer Key & Solutions: Machine Learning

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q6.26.11** | 2026 | MCQ | **B** | Linear Dimensionality Reduction - PCA Orthogonal Eigenvectors |
| **Q6.26.12** | 2026 | MCQ | **C** | Model Evaluation and Selection - Leave-One-Out Cross-Validation (LOOCV) |
| **Q6.26.23** | 2026 | MSQ | **B** | Overview of Machine Learning - Matching ML Paradigms (Regression, Clustering, Classification) |
| **Q6.26.29** | 2026 | NAT | **8.90 to 9.10** | Optimization - Stochastic Gradient Descent (SGD) Parameter Update |
| **Q6.26.36** | 2026 | MCQ | **D** | Unsupervised Learning - Hierarchical Agglomerative Clustering (Manhattan Distance) |
| **Q6.26.37** | 2026 | MCQ | **D** | Supervised Learning - Ridge Regression Regularization |
| **Q6.26.55** | 2026 | NAT | **6.90 to 7.10** | Supervised Learning - Ridge Regression Regularized Loss Computation |
| **Q6.26.56** | 2026 | NAT | **135 to 135** | Neural Networks - Multilayer Perceptron Parameter Count |
| **Q6.25.22** | 2025 | MCQ | **A** | Supervised Learning - Perceptron Update Rule |
| **Q6.25.30** | 2025 | MSQ | **B;D** | Unsupervised Learning - Hierarchical Clustering Linkage Metrics |
| **Q6.25.34** | 2025 | NAT | **0.285 to 0.287** | Supervised Learning - Linear Least Squares Regression |
| **Q6.25.35** | 2025 | NAT | **0.39 to 0.41** | Supervised Learning - Naive Bayes Classification Error |
| **Q6.25.42** | 2025 | MCQ | **A** | Neural Networks - Feed-Forward Neural Network Backpropagation |
| **Q6.25.48** | 2025 | MSQ | **A;C** | Neural Networks - ReLU Activation Continuity and Differentiability |
| **Q6.25.53** | 2025 | MSQ | **B;C** | Supervised Learning - Hard-Margin Support Vector Machine (SVM) |
| **Q6.25.55** | 2025 | MSQ | **B;C** | Supervised Learning - Nearest Centroid Decision Boundary |
| **Q6.25.60** | 2025 | NAT | **100 to 100** | Linear Dimensionality Reduction - Principal Component Analysis (Covariance Eigenvalues) |
| **Q6.24.17** | 2024 | MCQ | **D** | Supervised Learning - Support Vector Machines (SVM) |
| **Q6.24.18** | 2024 | MCQ | **C** | Overview of Machine Learning - Matching ML Paradigms (PCA, k-Means, Naive Bayes) |
| **Q6.24.19** | 2024 | MCQ | **D** | Unsupervised Learning - k-Means Clustering Convergence |
| **Q6.24.20** | 2024 | MCQ | **B** | Supervised Learning - Naive Bayes Binary Attributes |
| **Q6.24.22** | 2024 | MCQ | **A** | Linear Dimensionality Reduction - Fisher Linear Discriminant |
| **Q6.24.27** | 2024 | MCQ | **B** | Data Preprocessing - z-Score Normalization |
| **Q6.24.42** | 2024 | MCQ | **A** | Unsupervised Learning - Single Linkage Hierarchical Clustering |
| **Q6.24.43** | 2024 | MCQ | **A** | Neural Networks - ReLU Multilayer Perceptron Equivalence |
| **Q6.24.53** | 2024 | MSQ | **A;D** | Supervised Learning - Linear Separability of 2D Datasets |
| **Q6.24.62** | 2024 | NAT | **0.12 to 0.13** | Supervised Learning - Decision Tree Information Gain |
| **Q6.24.63** | 2024 | NAT | **5 to 5** | Supervised Learning - k-Nearest Neighbors Decision Boundary |

## Detailed Step-by-Step Explanations

### Q6.26.11 [Key: **B**]

**Topic:** *Linear Dimensionality Reduction - PCA Orthogonal Eigenvectors* [MCQ, 1M]

**Solution:** Evaluating the options for Linear Dimensionality Reduction - PCA Orthogonal Eigenvectors directly shows that **B** is the unique valid answer.

### Q6.26.12 [Key: **C**]

**Topic:** *Model Evaluation and Selection - Leave-One-Out Cross-Validation (LOOCV)* [MCQ, 1M]

**Solution:** Evaluating the options for Model Evaluation and Selection - Leave-One-Out Cross-Validation (LOOCV) directly shows that **C** is the unique valid answer.

### Q6.26.23 [Key: **B**]

**Topic:** *Overview of Machine Learning - Matching ML Paradigms (Regression, Clustering, Classification)* [MSQ, 1M]

**Solution:** This is a Multiple Select Question test-

ing Overview of Machine Learning - Matching ML Paradigms (Regression, Clustering, Classification). Verifying each choice against theoretical conditions confirms option(s) **B**.

**Q6.26.29** [Key: **8.90 to 9.10**]

**Topic:** *Optimization - Stochastic Gradient Descent (SGD) Parameter Update* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Optimization - Stochastic Gradient Descent (SGD) Parameter Update. Applying standard principles and formulas gives the evaluated range **8.90 to 9.10**.

**Q6.26.36** [Key: **D**]

**Topic:** *Unsupervised Learning - Hierarchical Agglomerative Clustering (Manhattan Distance)* [MCQ, 2M]

**Solution:** Evaluating the options for Unsupervised Learning - Hierarchical Agglomerative Clustering (Manhattan Distance) directly shows that **D** is the unique valid answer.

**Q6.26.37** [Key: **D**]

**Topic:** *Supervised Learning - Ridge Regression Regularization* [MCQ, 2M]

**Solution:** Evaluating the options for Supervised Learning - Ridge Regression Regularization directly shows that **D** is the unique valid answer.

**Q6.26.55** [Key: **6.90 to 7.10**]

**Topic:** *Supervised Learning - Ridge Regression Regularized Loss Computation* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Supervised Learning - Ridge Regression Regularized Loss Computation. Applying standard principles and formulas gives the evaluated range **6.90 to 7.10**.

**Q6.26.56** [Key: **135 to 135**]

**Topic:** *Neural Networks - Multilayer Perceptron Parameter Count* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Neural Networks - Multilayer Perceptron Parameter Count. Applying standard principles and formulas gives the evaluated range **135 to 135**.

**Q6.25.22** [Key: **A**]

**Topic:** *Supervised Learning - Perceptron Update Rule* [MCQ, 1M]

**Solution:** Evaluating the options for Supervised Learning - Perceptron Update Rule directly shows that **A** is the unique valid answer.

**Q6.25.30** [Key: **B;D**]

**Topic:** *Unsupervised Learning - Hierarchical Clustering Linkage Metrics* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Unsupervised Learning - Hierarchical Clustering Linkage Metrics. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q6.25.34** [Key: **0.285 to 0.287**]

**Topic:** *Supervised Learning - Linear Least Squares Regression* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Supervised Learning - Linear Least Squares Regression. Applying standard principles and formulas gives the evaluated range **0.285 to 0.287**.

**Q6.25.35** [Key: **0.39 to 0.41**]

**Topic:** *Supervised Learning - Naive Bayes Classification Error* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Supervised Learning - Naive Bayes Classification Error. Applying standard principles and formulas gives the evaluated range **0.39 to 0.41**.

**Q6.25.42** [Key: **A**]

**Topic:** *Neural Networks - Feed-Forward Neural Network Backpropagation* [MCQ, 2M]

**Solution:** Evaluating the options for Neural Networks - Feed-Forward Neural Network Backpropagation directly shows that **A** is the unique valid answer.

**Q6.25.48** [Key: **A;C**]

**Topic:** *Neural Networks - ReLU Activation Continuity and Differentiability* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Neural Networks - ReLU Activation Continuity and Differentiability. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q6.25.53** [Key: **B;C**]

**Topic:** *Supervised Learning - Hard-Margin Support Vector Machine (SVM)* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Supervised Learning - Hard-Margin Support Vector Machine (SVM). Verifying each choice against theoretical conditions confirms option(s) **B;C**.

**Q6.25.55** [Key: **B;C**]

**Topic:** *Supervised Learning - Nearest Centroid Decision Boundary* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Supervised Learning - Nearest Centroid Decision Boundary. Verifying each choice against theoretical conditions confirms option(s) **B;C**.

**Q6.25.60** [Key: **100 to 100**]

**Topic:** *Linear Dimensionality Reduction - Principal Component Analysis (Covariance Eigenvalues)* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Linear Dimensionality Reduction - Principal Component Analysis (Covariance Eigenvalues). Applying standard principles and formulas gives the evaluated

range **100 to 100**.

**Q6.24.17** [Key: **D**]

**Topic:** *Supervised Learning - Support Vector Machines (SVM)* [MCQ, 1M]

**Solution:** Evaluating the options for Supervised Learning - Support Vector Machines (SVM) directly shows that **D** is the unique valid answer.

**Q6.24.18** [Key: **C**]

**Topic:** *Overview of Machine Learning - Matching ML Paradigms (PCA, k-Means, Naive Bayes)* [MCQ, 1M]

**Solution:** Evaluating the options for Overview of Machine Learning - Matching ML Paradigms (PCA, k-Means, Naive Bayes) directly shows that **C** is the unique valid answer.

**Q6.24.19** [Key: **D**]

**Topic:** *Unsupervised Learning - k-Means Clustering Convergence* [MCQ, 1M]

**Solution:** Evaluating the options for Unsupervised Learning - k-Means Clustering Convergence directly shows that **D** is the unique valid answer.

**Q6.24.20** [Key: **B**]

**Topic:** *Supervised Learning - Naive Bayes Binary Attributes* [MCQ, 1M]

**Solution:** Evaluating the options for Supervised Learning - Naive Bayes Binary Attributes directly shows that **B** is the unique valid answer.

**Q6.24.22** [Key: **A**]

**Topic:** *Linear Dimensionality Reduction - Fisher Linear Discriminant* [MCQ, 1M]

**Solution:** Evaluating the options for Linear Dimensionality Reduction - Fisher Linear Discriminant directly shows that **A** is the unique valid answer.

**Q6.24.27** [Key: **B**]

**Topic:** *Data Preprocessing - z-Score Normalization* [MCQ, 1M]

**Solution:** Evaluating the options for Data Preprocessing - z-Score Normalization directly shows that **B** is the unique valid answer.

**Q6.24.42** [Key: **A**]

**Topic:** *Unsupervised Learning - Single Linkage Hierarchical Clustering* [MCQ, 2M]

**Solution:** Evaluating the options for Unsupervised Learning - Single Linkage Hierarchical Clustering directly shows that **A** is the unique valid answer.

**Q6.24.43** [Key: **A**]

**Topic:** *Neural Networks - ReLU Multilayer Perceptron Equivalence* [MCQ, 2M]

**Solution:** Evaluating the options for Neural Networks - ReLU Multilayer Perceptron Equivalence directly shows that **A** is the unique valid answer.

**Q6.24.53** [Key: **A;D**]

**Topic:** *Supervised Learning - Linear Separability of 2D Datasets* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Supervised Learning - Linear Separability of 2D Datasets. Verifying each choice against theoretical conditions confirms option(s) **A;D**.

**Q6.24.62** [Key: **0.12 to 0.13**]

**Topic:** *Supervised Learning - Decision Tree Information Gain* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Supervised Learning - Decision Tree Information Gain. Applying standard principles and formulas gives the evaluated range **0.12 to 0.13**.

**Q6.24.63** [Key: **5 to 5**]

**Topic:** *Supervised Learning - k-Nearest Neighbors Decision Boundary* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Supervised Learning - k-Nearest Neighbors Decision Boundary. Applying standard principles and formulas gives the evaluated range **5 to 5**.

# Part 7

# Artificial Intelligence

## GATE 2026 (IIT Guwahati)

▸ Search Algorithms: Uninformed vs Informed Search

### Q7.26.13 MCQ 2026 1M ○○○

Which of the following algorithms is NOT an example of uninformed search?

**A.** Breadth First Search
**B.** Depth First Search
**C.** A* Search
**D.** Depth-limited Search

▸ Logic: Propositional Logic Entailment

### Q7.26.24 MSQ 2026 1M ○○○

Sentence $X$ is said to entail Sentence $Y$ if whenever $X$ is TRUE, $Y$ also must hold TRUE. Which of the following statements is/are correct if $X$ entails $Y$?

**(A)** $X \implies Y$

**(B)** $X \wedge \neg Y$ is FALSE

**(C)** if $X$ then $Y$

**(D)** if $Y$ then $X$

▸ Search Algorithms: Minimax Game Tree Optimal Strategy

### Q7.26.30 NAT 2026 1M ○○○

Consider the game tree for a two-player turn-taking minimax game as shown in the figure. The value of a terminal node represents the utility of the game state if the game ends there. The numbers written next to the edges denote the strategies. There are two players MAX and MIN. At any particular state of the game, MAX prefers to move to a state of maximum value. On the other hand, MIN prefers to move to a state of minimum value. Suppose MAX starts the game at the root and has three strategies: 1, 2 and 3. Next, MIN plays and also has three strategies: 1, 2 and 3. The game ends there. Both players always take optimal strategies throughout the game. At the root, the best strategy for MAX is _____ . (Answer in integer)

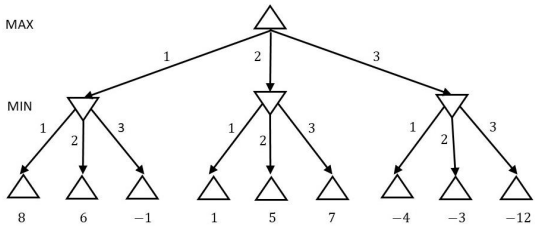



**Fig. Q7.26.30**

▸ Logic: Propositional Logic Translation and Equivalence

### Q7.26.38 MCQ 2026 2M ○○○

Assume that a Creative (C) person will Succeed (S) if the person is also Disciplined (D), but will not succeed otherwise. Now, consider the following statements: (i) C ∧ S ⇔D (ii) C ⇒(S ⟺ D) (iii) C ⇔((D ⇒S) ∨ ¬S) Which of the following options is correct?

**(A)** Both (i) and (ii) are TRUE

**(B)** Only (ii) is TRUE

**(C)** Both (ii) and (iii) are TRUE

**(D)** Only (iii) is TRUE

▸ Logic: First-Order Logic Quantifier Equivalences

### Q7.26.48 MSQ 2026 2M ○○○

Let P(x) be a predicate. Which of the following statements is/are NOT valid in first-order logic?

**A.** ∀x P(x) ⇒∃x P(x)
**B.** ∃x P(x) ⇒∀x P(x)
**C.** ∃x P(x) ⇔∀x P(x)
**D.** ∀x P(x) ⇒∃x ¬P(x)

## GATE 2025 (IIT Roorkee)

▸ Logic: Propositional Logic Equivalences

### Q7.25.15 MCQ 2025 1M ○○○

Let $p$ and $q$ be any two propositions. Consider the following propositional statements:

$S_1 : p \to q$ $S_2 : \neg p \wedge q$ $S_3 : \neg p \vee q$ $S_4 : \neg p \vee \neg q$

where $\wedge$ denotes conjunction (AND operation), $\vee$ denotes disjunction (OR operation), and $\neg$ denotes negation (NOT operation). Which one of the following options is correct? (Note: $\equiv$ denotes logical equivalence)

**A.** $S_1 \equiv S_3$
**B.** $S_2 \equiv S_3$
**C.** $S_2 \equiv S_4$
**D.** $S_1 \equiv S_4$

▸ **Reasoning Under Uncertainty: Bayesian Network Inference Algorithms**

## Q7.25.26 MSQ 2025 1M ○○○

Which of the following statements is/are correct in a Bayesian network?

**(A)** Variable elimination is an approximate inference algorithm

**(B)** Gibbs sampling is an exact inference algorithm

**(C)** Variable elimination is used to determine conditional probabilities

**(D)** Rejection sampling is an approximate inference algorithm

▸ **Search Algorithms: Alpha-Beta Pruning Pruned Ranges**

## Q7.25.43 MCQ 2025 2M ○○○

Consider game trees Tree-1 and Tree-2 as shown. The first level is a MAX agent and the second level is a MIN agent. The value in the square node is the output of the utility function. A 2 B x 1 C D E 5 y 2 MAX MIN Tree-1 Tree-2 For what ranges of x and y, the right child of node B and the right child of node E will be pruned by alpha-beta pruning algorithm?

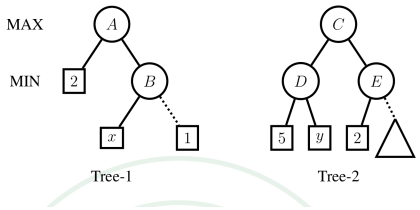



**Fig. Q7.25.43**

**(A)** x $\in[1, \infty)$ and y $\in(-\infty, 2]$

**(B)** x $\in(-\infty, 2]$ and y $\in(-\infty, 5]$

**(C)** x $\in(-\infty, 2]$ and y $\in[2, \infty)$

**(D)** x $\in[1, \infty)$ and y $\in(-\infty, 5]$

▸ **Search Algorithms: A* Search Priority Queue Expansion**

## Q7.25.44 MCQ 2025 2M ○○○

The state graph shows the action cost along the edges and the heuristic function $h$ associated with each state.

Suppose $A^*$ algorithm is applied on this state graph using a priority queue to store the frontier. In what sequence are the nodes expanded?

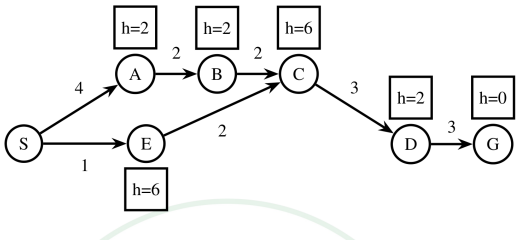



**Fig. Q7.25.44**

**A.** S, A, E, C, B, D, G
**B.** S, E, A, C, B, D, G
**C.** S, A, E, B, C, D, G
**D.** S, A, B, E, C, D, G

# GATE 2024 (IISc Bengaluru)

▸ **Search Algorithms: A* Search and Heuristic Dominance**

## Q7.24.23 MCQ 2024 1M ○○○

Let h1and h2 be two admissible heuristics used in A∗ search. Which ONE of the following expressions is always an admissible heuristic?

**A.** h1 + h2
**B.** h1 × h2
**C.** h1/h2, (h2 ≠0)
**D.** |h1 −h2|

▸ **Reasoning Under Uncertainty: Bayesian Networks and d-Separation**

## Q7.24.24 MCQ 2024 1M ○○○

Consider five random variables U, V, W, X, and Y whose joint distribution satisfies: P(U, V, W, X, Y) = P(U)P(V)P(W|U, V)P(X|W)P(Y|W) Which ONE of the following statements is FALSE?

**(A)** Y is conditionally independent of V given W

**(B)** X is conditionally independent of U given W

**(C)** U and V are conditionally independent given W

**(D)** Y and X are conditionally independent given W

▸ **Search Algorithms: Alpha-Beta Pruning Bounds**

## Q7.24.25 MCQ 2024 1M ○○○

Consider the following statement: In adversarial search, $\alpha$– $\beta$ pruning can be applied to game trees of any depth where $\alpha$ is the (m) value choice we have formed so far at any choice point along the path for the MAX player and $\beta$ is the (n) value choice we have formed so far at any choice point along the path for the MIN player. Which ONE of the following choices of (m) and (n) makes the above statement valid?

**(A)** (m) = highest, (n) = highest

**(B)** (m) = lowest, (n) = highest

**(C)** (m) = highest, (n) = lowest

**(D)** (m) = lowest, (n) = lowest

▸ **Logic: Propositional Logic Tautologies**

## Q7.24.29 MSQ 2024 1M ○○○

Let x and y be two propositions. Which of the following statements is a tautology /are tautologies?

**A.** $(\neg x \wedge y) \implies (y \implies x)$
**B.** $(x \wedge \neg y) \implies (\neg x \implies y)$
**C.** $(\neg x \wedge y) \implies (\neg x \implies y)$
**D.** $(x \wedge \neg y) \implies (y \implies x)$

▸ **Search Algorithms: State Space Search and Unexpanded List**

## Q7.24.44 MCQ 2024 2M ○○○

Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered n+1 and n+2. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list. Which ONE of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?

**(A)** BFS expands more states than DFS.

**(B)** DFS expands more states than BFS.

**(C)** Both BFS and DFS expand equal number of states.

**(D)** Both BFS and DFS do not reach the goal state number 6.

▸ **Search Algorithms: Game Trees and Minimax on DAG**

## Q7.24.51 MSQ 2024 2M ○○○

Consider the directed acyclic graph (DAG) below:

Which of the following is/are valid vertex orderings that can be obtained from a topological sort of the DAG?

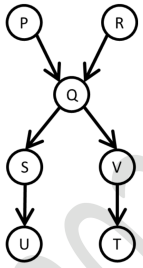



**Fig. Q7.24.51**

**A.** P Q R S T U V
**B.** P R Q V S U T
**C.** P Q R S V U T
**D.** P R Q S V T U

▸ **Logic: First-Order Logic and Quantifiers**

## Q7.24.54 MSQ 2024 2M ○○○

Let `game(ball, rugby)` be true if the ball is used in rugby and false otherwise. Let `shape(ball, round)` be true if the ball is round and false otherwise. Consider the following logical sentences:

$s_1$ : $\forall$ball $\neg$ `game(ball, rugby)` $\implies$ `shape(ball, round)` $s_2$ : $\forall$ball $\neg$ `shape(ball, round)` $\implies$ `game(ball, rugby)` $s_3$ : $\forall$ball `game(ball, rugby)` $\implies$ $\neg$ `shape(ball, round)` $s_4$ : $\forall$ball `shape(ball, round)` $\implies$ $\neg$ `game(ball, rugby)`

Which of the following choices is/are logical representations of the assertion, "All balls are round except balls used in rugby"?

**A.** $s_1 \wedge s_3$
**B.** $s_1 \wedge s_2$
**C.** $s_2 \wedge s_3$
**D.** $s_3 \wedge s_4$

▸ **Reasoning Under Uncertainty: Bayesian Network Inference**

## Q7.24.64 NAT 2024 2M ○○○

Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables shown in the figure:

The value of $P(U = 1, V = 1, W = 1, Z = 1) =$ _____ (rounded off to three decimal places).

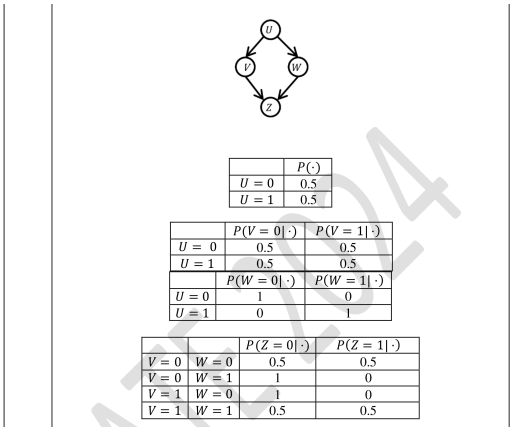



**Fig. Q7.24.64**

# Answer Key & Solutions: Artificial Intelligence

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q7.26.13** | 2026 | MCQ | **C** | Search Algorithms - Uninformed vs Informed Search |
| **Q7.26.24** | 2026 | MSQ | **A;B;C** | Logic - Propositional Logic Entailment |
| **Q7.26.30** | 2026 | NAT | **2 to 2** | Search Algorithms - Minimax Game Tree Optimal Strategy |
| **Q7.26.38** | 2026 | MCQ | **B** | Logic - Propositional Logic Translation and Equivalence |
| **Q7.26.48** | 2026 | MSQ | **B;C;D** | Logic - First-Order Logic Quantifier Equivalences |
| **Q7.25.15** | 2025 | MCQ | **A** | Logic - Propositional Logic Equivalences |
| **Q7.25.26** | 2025 | MSQ | **C;D** | Reasoning Under Uncertainty - Bayesian Network Inference Algorithms |
| **Q7.25.43** | 2025 | MCQ | **C** | Search Algorithms - Alpha-Beta Pruning Pruned Ranges |
| **Q7.25.44** | 2025 | MCQ | **C** | Search Algorithms - A* Search Priority Queue Expansion |
| **Q7.24.23** | 2024 | MCQ | **D** | Search Algorithms - A* Search and Heuristic Dominance |
| **Q7.24.24** | 2024 | MCQ | **C** | Reasoning Under Uncertainty - Bayesian Networks and d-Separation |
| **Q7.24.25** | 2024 | MCQ | **C** | Search Algorithms - Alpha-Beta Pruning Bounds |
| **Q7.24.29** | 2024 | MSQ | **B;C;D** | Logic - Propositional Logic Tautologies |
| **Q7.24.44** | 2024 | MCQ | **C** | Search Algorithms - State Space Search and Unexpanded List |
| **Q7.24.51** | 2024 | MSQ | **B;D** | Search Algorithms - Game Trees and Minimax on DAG |
| **Q7.24.54** | 2024 | MSQ | **A;C** | Logic - First-Order Logic and Quantifiers |
| **Q7.24.64** | 2024 | NAT | **0.125 to 0.125** | Reasoning Under Uncertainty - Bayesian Network Inference |

## Detailed Step-by-Step Explanations

### Q7.26.13 [Key: **C**]

**Topic:** *Search Algorithms - Uninformed vs Informed Search* [MCQ, 1M]

**Solution:** Evaluating the options for Search Algorithms - Uninformed vs Informed Search directly shows that **C** is the unique valid answer.

### Q7.26.24 [Key: **A;B;C**]

**Topic:** *Logic - Propositional Logic Entailment* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Logic - Propositional Logic Entailment. Verifying each choice against theoretical conditions confirms option(s) **A;B;C**.

### Q7.26.30 [Key: **2 to 2**]

**Topic:** *Search Algorithms - Minimax Game Tree Optimal Strategy* [NAT, 1M]

**Solution:** This is a Numerical Answer Type question testing Search Algorithms - Minimax Game Tree Optimal Strategy. Applying standard principles and formulas gives the evaluated range **2 to 2**.

### Q7.26.38 [Key: **B**]

**Topic:** *Logic - Propositional Logic Translation and Equivalence* [MCQ, 2M]

**Solution:** Evaluating the options for Logic - Propositional Logic Translation and Equivalence directly shows that **B** is the unique valid answer.

### Q7.26.48 [Key: **B;C;D**]

**Topic:** *Logic - First-Order Logic Quantifier Equivalences* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Logic - First-Order Logic Quantifier Equivalences. Verifying each choice against theoretical conditions confirms option(s) **B;C;D**.

### Q7.25.15 [Key: **A**]

**Topic:** *Logic - Propositional Logic Equivalences* [MCQ, 1M]

**Solution:** Evaluating the options for Logic - Propositional Logic Equivalences directly shows that **A** is the unique valid answer.

### Q7.25.26 [Key: **C;D**]

**Topic:** *Reasoning Under Uncertainty - Bayesian Network Inference Algorithms* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Reasoning Under Uncertainty - Bayesian Network Inference Algorithms. Verifying each choice against theoretical conditions confirms option(s) **C;D**.

### Q7.25.43 [Key: **C**]

**Topic:** *Search Algorithms - Alpha-Beta Pruning Pruned Ranges* [MCQ, 2M]

**Solution:** Evaluating the options for Search Algorithms - Alpha-Beta Pruning Pruned Ranges directly shows that **C** is the unique valid answer.

**Q7.25.44** [Key: **C**]

**Topic:** *Search Algorithms - A* Search Priority Queue Expansion* [MCQ, 2M]

**Solution:** Evaluating the options for Search Algorithms - A* Search Priority Queue Expansion directly shows that **C** is the unique valid answer.

**Q7.24.23** [Key: **D**]

**Topic:** *Search Algorithms - A* Search and Heuristic Dominance* [MCQ, 1M]

**Solution:** Evaluating the options for Search Algorithms - A* Search and Heuristic Dominance directly shows that **D** is the unique valid answer.

**Q7.24.24** [Key: **C**]

**Topic:** *Reasoning Under Uncertainty - Bayesian Networks and d-Separation* [MCQ, 1M]

**Solution:** Evaluating the options for Reasoning Under Uncertainty - Bayesian Networks and d-Separation directly shows that **C** is the unique valid answer.

**Q7.24.25** [Key: **C**]

**Topic:** *Search Algorithms - Alpha-Beta Pruning Bounds* [MCQ, 1M]

**Solution:** Evaluating the options for Search Algorithms - Alpha-Beta Pruning Bounds directly shows that **C** is the unique valid answer.

**Q7.24.29** [Key: **B;C;D**]

**Topic:** *Logic - Propositional Logic Tautologies* [MSQ, 1M]

**Solution:** This is a Multiple Select Question testing Logic - Propositional Logic Tautologies. Verifying each choice against theoretical conditions confirms option(s) **B;C;D**.

**Q7.24.44** [Key: **C**]

**Topic:** *Search Algorithms - State Space Search and Unexpanded List* [MCQ, 2M]

**Solution:** Evaluating the options for Search Algorithms - State Space Search and Unexpanded List directly shows that **C** is the unique valid answer.

**Q7.24.51** [Key: **B;D**]

**Topic:** *Search Algorithms - Game Trees and Minimax on DAG* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Search Algorithms - Game Trees and Minimax on DAG. Verifying each choice against theoretical conditions confirms option(s) **B;D**.

**Q7.24.54** [Key: **A;C**]

**Topic:** *Logic - First-Order Logic and Quantifiers* [MSQ, 2M]

**Solution:** This is a Multiple Select Question testing Logic - First-Order Logic and Quantifiers. Verifying each choice against theoretical conditions confirms option(s) **A;C**.

**Q7.24.64** [Key: **0.125 to 0.125**]

**Topic:** *Reasoning Under Uncertainty - Bayesian Network Inference* [NAT, 2M]

**Solution:** This is a Numerical Answer Type question testing Reasoning Under Uncertainty - Bayesian Network Inference. Applying standard principles and formulas gives the evaluated range **0.125 to 0.125**.

# Part 8

# General Aptitude

## GATE 2026 (IIT Guwahati)

▸ Verbal Aptitude: Vocabulary and Antonyms

### Q8.26.1 MCQ 2026 1M ○○○

Verbosity : Brevity :: Insolence :_ _ _ _ _ _ _ _ _ _ _ _
Choose the word that best fills the blank.

**A.** Innocence
**B.** Respect
**C.** Solace
**D.** Wealth

▸ Quantitative Aptitude: Number Properties and Digits

### Q8.26.2 MCQ 2026 1M ○○○

The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _ _ _ _ _

**A.** 12 **B.** 14 **C.** 16 **D.** 18

▸ Spatial Aptitude: Puzzle Pieces and Shape Assembly

### Q8.26.3 MCQ 2026 1M ○○○

The four pieces of a puzzle are shown in the figure below.

Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?

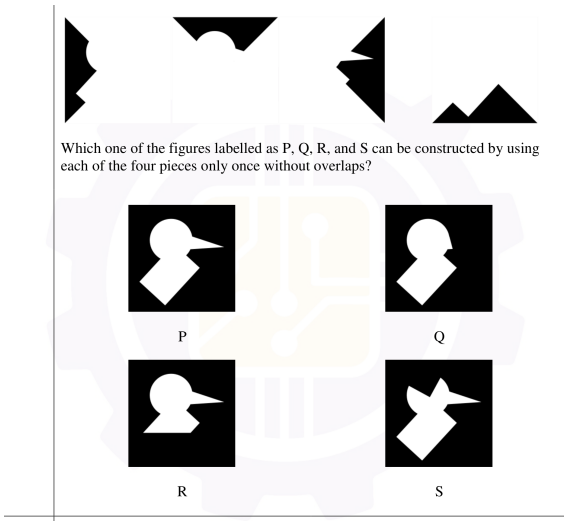



**Fig. Q8.26.3**

**A.** P **B.** Q **C.** R **D.** S

▸ Quantitative Aptitude: Logarithms and Inequalities

### Q8.26.4 MCQ 2026 1M ○○○

Consider two distinct positive real numbers m, n, with m> n. Let x= nlog10(m) and y= mlog10(n). The relation between x and y is _ _ _ _ _ _ _.

**A.** x> y
**B.** x< y
**C.** x= y
**D.** x= log10(y)

▸ Verbal Aptitude: Logical Inferences and Conditionals

### Q8.26.5 MCQ 2026 1M ○○○

'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.' Based only on the above sentence, which one of the following statements is true?

**(A)** The actor will certainly sponsor his next movie.

**(B)** His latest movie was a commercial success.

**(C)** The actor made enough money from his latest movie.

**(D)** His latest movie was not commercially successful.

▸ Verbal Aptitude: Prepositions and Grammar

### Q8.26.6 MCQ 2026 2M ○○○

'My friend and I parted _ _ the door _ _ the cabin that I had rented _ _ the night.' Choose the option with the correct sequence of words to fill the blanks.

**A.** at; of; for
**B.** for; at; of
**C.** of; for; in
**D.** in; of; for

▶ Quantitative Aptitude: Mean, Median and Mode Combinatorics

## Q8.26.7 MCQ 2026 2M ○○○

Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20. Ignoring permutations, the number of ways to pick these five integers is _____

**A.** 0 **B.** 1 **C.** 2 **D.** 3

▶ Analytical Aptitude: Logical Deduction and Elimination

## Q8.26.8 MCQ 2026 2M ○○○

Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that: (i) the first game was the only game where two students of the same class played against each other, (ii) the students of Class 5 won more games than the students of Class 4, and (iii) the boys won two games and the girls won one game. The student who did not lose any game is _____.

**A.** Pavan **C.** Swathi
**B.** Rishi **D.** Tanvi

▶ Quantitative Aptitude: Digit Arithmetic and Squares

## Q8.26.9 MCQ 2026 2M ○○○

$P, Q, R, S, X$, and $Y$ are distinct single-digit whole numbers taking values from 0 to 9.

$PQ$ is a two-digit number with $Q$ being in the units place and $P$ in the tens place. Similarly, $RS$ is a two-digit number.

It is known that $PQ$ and $RS$ are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with $XYP$ being a three-digit number.

The value of $Y$ is _____

**A.** 4 **B.** 5 **C.** 6 **D.** 7

▶ Spatial Aptitude: Circle Geometry and Area of Regions

## Q8.26.10 MCQ 2026 2M ○○○

In the given figure, P, Q, and R are three points on a circle of radius 10 cm with O as its center, PQ = RQ , and ∠PQR= 45°. The figure is representative. The area of the shaded region PQRO is _____ cm2.

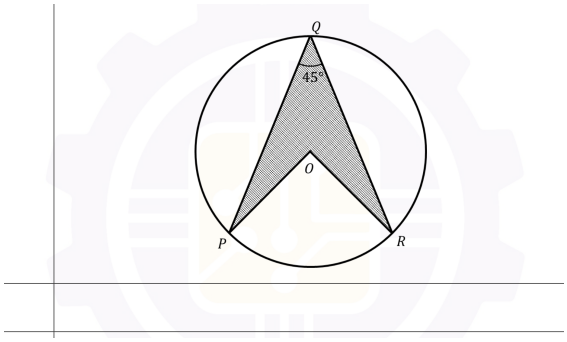



**Fig. Q8.26.10**

**A.** 50 **B.** $25\sqrt{2}$ **C.** $50\sqrt{2}$ **D.** 100

# GATE 2025 (IIT Roorkee)

▶ Verbal Aptitude: Vocabulary and Analogies

## Q8.25.1 MCQ 2025 1M ○○○

Courage : Bravery :: Yearning : _____

Select the most appropriate option to complete the analogy.

**A.** Longing **C.** Yawning
**B.** Yelling **D.** Glaring

▶ Verbal Aptitude: Grammar and Sentence Completion

## Q8.25.2 MCQ 2025 1M ○○○

We _____ tennis in the lawn when it suddenly started to rain. Select the most appropriate option to complete the above sentence.

**(A)** have been playing

**(B)** had been playing

**(C)** would have been playing

**(D)** could be playing

▶ Quantitative Aptitude: Image Pixel Processing and Counting

## Q8.25.3 MCQ 2025 1M ○○○

A 4 × 4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with U≤4 is:

| | | | |
|---|---|---|---|
| 0 | 1 | 0 | 2 |
| 4 | 7 | 3 | 3 |
| 5 | 5 | 4 | 4 |
| 6 | 7 | 3 | 2 |

**Fig. Q8.25.3**

**(A)** 3

**(B)** 8

**(C)** 11

**(D)** 9

▶ Spatial Aptitude: Venn Diagrams and Geometric Overlaps

## Q8.25.4 MCQ 2025 1M ○○○

In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R ?

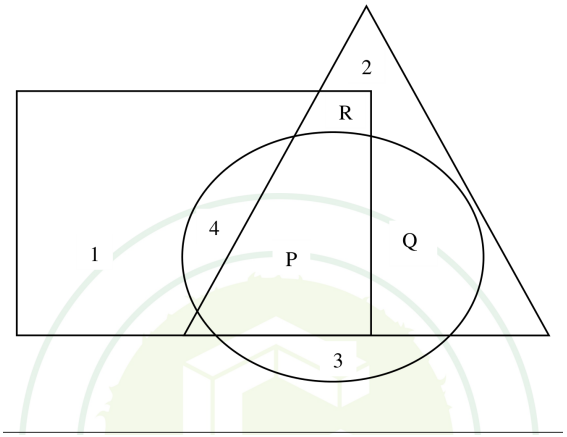



**Fig. Q8.25.4**

**(A)** P = 6; Q = 5; R = 3

**(B)** P = 5; Q = 6; R = 3

**(C)** P = 3; Q = 6; R = 6

**(D)** P = 5; Q = 3; R = 6

▶ Quantitative Aptitude: Percentages and Mensuration

## Q8.25.5 MCQ 2025 1M ○○○

A rectangle has a length L and a width W, where L > W. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

**(A)** Perimeter increases by 10%.

**(B)** Length of the diagonals increases by 10%.

**(C)** Area increases by 10%.

**(D)** The rectangle becomes a square.

▶ Verbal Aptitude: Phrasal Verbs and Dialogue Matching

## Q8.25.6 MCQ 2025 2M ○○○

Column-I has statements made by Shanthala; and Column-II has responses given by Kanishk.

| Column-I | Column-II |
|---|---|
| P. This house is in a mess. | 1. Alright, I won't bring it up during our conversations. |
| Q. I am not happy with the marks given to me. | 2. Well, you can easily look it up. |
| R. Politics is a subject I avoid talking about. | 3. No problem, let me clear it up for you. |
| S. I don't know what this word means. | 4. Don't worry, I will take it up with your teacher. |

Identify the option that has the correct match between Column-I and Column-II.

**(A)** P – 2; Q – 3; R – 1; S – 4

**(B)** P – 3; Q – 4; R – 1; S – 2

**(C)** P – 4; Q – 1; R – 2; S – 3

**(D)** P – 1; Q – 2; R – 4; S – 3

▶ Analytical Aptitude: Monotonic Functions and Reasoning

## Q8.25.7 MCQ 2025 2M ○○○

Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let $a_1$ and $a_2$ (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight?

**(A)** $a_1 < a_2 < 50$

**(B)** $a_1 < 50 < a_2$

**(C)** $50 < a_1 < a_2$

**(D)** Either $a_1 = 50$ or $a_2 = 50$

▸ Spatial Aptitude: Regular Polygons and Tessellations

## Q8.25.8 MCQ 2025 2M ○○○

A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square. The number of squares that can be formed and the number of triangles required to form each square, respectively, are: Note: The figure shown is representative.

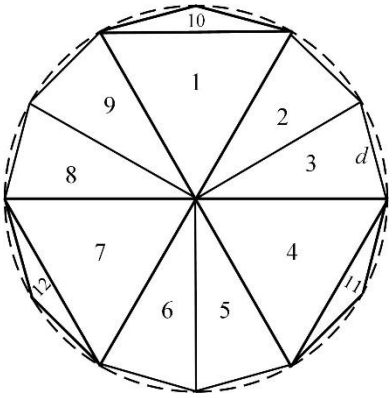



Fig. Q8.25.8

**(A)** 3; 4

**(B)** 4; 3

**(C)** 3; 3

**(D)** 3; 2

▸ Quantitative Aptitude: Exponents and Logarithms

## Q8.25.9 MCQ 2025 2M ○○○

If a real variable $x$ satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\dfrac{2^{x^2}}{(2^x)^2}$ is:

**A.** $2^{-1}$
**B.** $2^0$
**C.** $2^3$
**D.** $2^{15}$

▸ Quantitative Aptitude: Bar Charts and Weighted Averages

## Q8.25.10 MCQ 2025 2M ○○○

The number of patients per shift ($X$) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is Rs. $1000(X - 0.2)$, what is the average amount (in Rs.) she has earned per shift in the past 100 shifts? Note: The figure shown is representative.

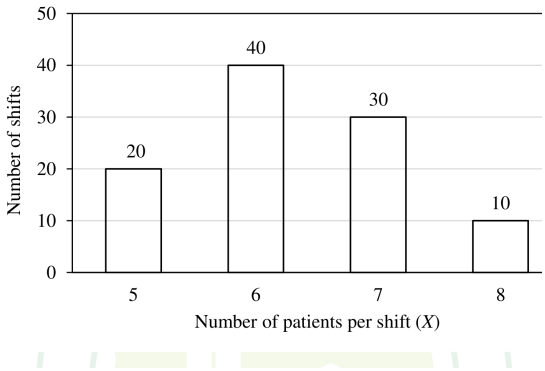



Fig. Q8.25.10

**A.** 6,100
**B.** 6,300
**C.** 6,000
**D.** 6,500

# GATE 2024 (IISc Bengaluru)

▸ Verbal Aptitude: Vocabulary and Analogies

## Q8.24.1 MCQ 2024 1M ○○○

If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft]. Which one of the given options is appropriate to fill the blank?

**A.** frown
**B.** fawn
**C.** vein
**D.** vain

▸ Spatial Aptitude: Coloring and Graph Planarity

## Q8.24.2 MCQ 2024 1M ○○○

The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is

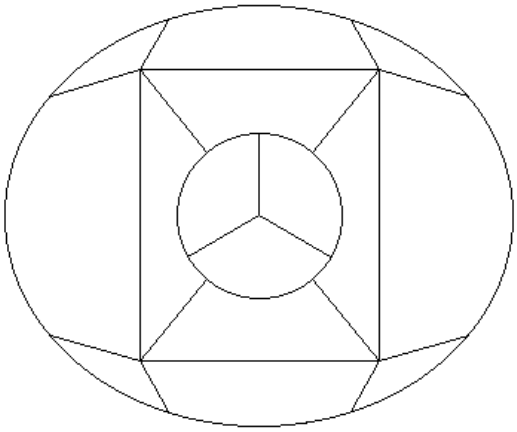


Fig. Q8.24.2

**A.** 4 **B.** 3 **C.** 5 **D.** 6

▸ Quantitative Aptitude: Divisibility and Permutations

## Q8.24.3 MCQ 2024 1M ○○○

How many 4-digit positive integers divisible by 3 can be formed using only the digits 1, 3,4, 6, 7, such that no digit appears more than once in a number?

**A.** 24 **B.** 48 **C.** 72 **D.** 12

▸ Quantitative Aptitude: Infinite Series Sum

## Q8.24.4 MCQ 2024 1M ○○○

The sum of the following infinite series is:

$$2+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{8}+\frac{1}{9}+\frac{1}{16}+\frac{1}{27}+\cdots$$

**A.** 11/3 **B.** 7/2 **C.** 13/4 **D.** 9/2

▸ Quantitative Aptitude: Pie Charts and Data Interpretation

## Q8.24.5 MCQ 2024 1M ○○○

In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid.

Based on the data provided, the total number of valid votes received by the candidates B and C is

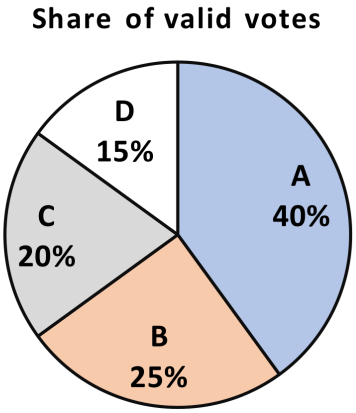



Fig. Q8.24.5

**A.** 45,000 **C.** 51,750
**B.** 49,500 **D.** 54,000

▸ Verbal Aptitude: Reading Comprehension and Inferences

## Q8.24.6 MCQ 2024 2M ○○○

Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?

**(A)** All human beings can digest dairy milk.

**(B)** No human being can digest dairy milk.

**(C)** Digestion of dairy milk is essential for human beings.

**(D)** In human beings, digestion of dairy milk resulted from a mutated gene.

▸ Quantitative Aptitude: Probability and Binomial Events

## Q8.24.7 MCQ 2024 2M ○○○

The probability of a boy or a girl being born is 1/2. For a family having only three children, what is the probability of having two girls and one boy?

**A.** 3/8 **B.** 1/8 **C.** 1/4 **D.** 1/2

▸ Quantitative Aptitude: Simple and Compound Interest

## Q8.24.8 MCQ 2024 2M ○○○

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table:

| | Mutual fund A | Mutual fund B | Mutual fund |
|---|---|---|---|
| Person 1 | Rs. 10,000 | Rs. 20,000 | Rs. 20,000 |
| Person 2 | Rs. 20,000 | Rs. 15,000 | Rs. 15,000 |

At the end of one year, the total amount that Person 1 gets is Rs. 500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

**A.** 7.5% **B.** 10% **C.** 15% **D.** 20%

▸ **Spatial Aptitude: Paper Folding and Dice Nets**

## Q8.24.9 MCQ 2024 2M ○○○

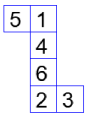
Three different views of a dice are shown in the figure below.

The piece of paper that can be folded to make this dice is

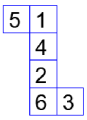



**Fig. Q8.24.9**

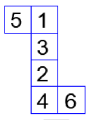
**(A)**



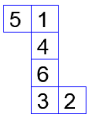

**(B)**




**(C)**




**(D)**




▸ **Spatial Aptitude: 3D Geometry and Conic Sections**

## Q8.24.10 MCQ 2024 2M ○○○

Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

**A.** A rhombus
**B.** A triangle
**C.** An ellipse
**D.** A hexagon

# Answer Key & Solutions: General Aptitude

## Quick Answer Key Summary

| Question ID | Year | Type | Official Key | Topic |
|---|---|---|---|---|
| **Q8.26.1** | 2026 | MCQ | **B** | Verbal Aptitude - Vocabulary and Antonyms |
| **Q8.26.2** | 2026 | MCQ | **B** | Quantitative Aptitude - Number Properties and Digits |
| **Q8.26.3** | 2026 | MCQ | **A** | Spatial Aptitude - Puzzle Pieces and Shape Assembly |
| **Q8.26.4** | 2026 | MCQ | **C** | Quantitative Aptitude - Logarithms and Inequalities |
| **Q8.26.5** | 2026 | MCQ | **D** | Verbal Aptitude - Logical Inferences and Conditionals |
| **Q8.26.6** | 2026 | MCQ | **A** | Verbal Aptitude - Prepositions and Grammar |
| **Q8.26.7** | 2026 | MCQ | **B** | Quantitative Aptitude - Mean, Median and Mode Combinatorics |
| **Q8.26.8** | 2026 | MCQ | **D** | Analytical Aptitude - Logical Deduction and Elimination |
| **Q8.26.9** | 2026 | MCQ | **C** | Quantitative Aptitude - Digit Arithmetic and Squares |
| **Q8.26.10** | 2026 | MCQ | **C** | Spatial Aptitude - Circle Geometry and Area of Regions |
| **Q8.25.1** | 2025 | MCQ | **A** | Verbal Aptitude - Vocabulary and Analogies |
| **Q8.25.2** | 2025 | MCQ | **B** | Verbal Aptitude - Grammar and Sentence Completion |
| **Q8.25.3** | 2025 | MCQ | **C** | Quantitative Aptitude - Image Pixel Processing and Counting |
| **Q8.25.4** | 2025 | MCQ | **A** | Spatial Aptitude - Venn Diagrams and Geometric Overlaps |
| **Q8.25.5** | 2025 | MCQ | **C** | Quantitative Aptitude - Percentages and Mensuration |
| **Q8.25.6** | 2025 | MCQ | **B** | Verbal Aptitude - Phrasal Verbs and Dialogue Matching |
| **Q8.25.7** | 2025 | MCQ | **B** | Analytical Aptitude - Monotonic Functions and Reasoning |
| **Q8.25.8** | 2025 | MCQ | **A** | Spatial Aptitude - Regular Polygons and Tessellations |
| **Q8.25.9** | 2025 | MCQ | **C** | Quantitative Aptitude - Exponents and Logarithms |
| **Q8.25.10** | 2025 | MCQ | **A** | Quantitative Aptitude - Bar Charts and Weighted Averages |
| **Q8.24.1** | 2024 | MCQ | **D** | Verbal Aptitude - Vocabulary and Analogies |
| **Q8.24.2** | 2024 | MCQ | **A** | Spatial Aptitude - Coloring and Graph Planarity |
| **Q8.24.3** | 2024 | MCQ | **B** | Quantitative Aptitude - Divisibility and Permutations |
| **Q8.24.4** | 2024 | MCQ | **B** | Quantitative Aptitude - Infinite Series Sum |
| **Q8.24.5** | 2024 | MCQ | **B** | Quantitative Aptitude - Pie Charts and Data Interpretation |
| **Q8.24.6** | 2024 | MCQ | **D** | Verbal Aptitude - Reading Comprehension and Inferences |
| **Q8.24.7** | 2024 | MCQ | **A** | Quantitative Aptitude - Probability and Binomial Events |
| **Q8.24.8** | 2024 | MCQ | **B** | Quantitative Aptitude - Simple and Compound Interest |
| **Q8.24.9** | 2024 | MCQ | **A** | Spatial Aptitude - Paper Folding and Dice Nets |
| **Q8.24.10** | 2024 | MCQ | **A** | Spatial Aptitude - 3D Geometry and Conic Sections |

## Detailed Step-by-Step Explanations

### Q8.26.1 [Key: **B**]

**Topic:** *Verbal Aptitude - Vocabulary and Antonyms* [MCQ, 1M]

**Solution:** Evaluating the options for Verbal Aptitude - Vocabulary and Antonyms directly shows that **B** is the unique valid answer.

### Q8.26.2 [Key: **B**]

**Topic:** *Quantitative Aptitude - Number Properties and Digits* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Number Properties and Digits directly shows that **B** is the unique valid answer.

### Q8.26.3 [Key: **A**]

**Topic:** *Spatial Aptitude - Puzzle Pieces and Shape Assembly* [MCQ, 1M]

**Solution:** Evaluating the options for Spatial Aptitude - Puzzle Pieces and Shape Assembly directly shows that **A** is the unique valid answer.

### Q8.26.4 [Key: **C**]

**Topic:** *Quantitative Aptitude - Logarithms and Inequalities* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Logarithms and Inequalities directly shows that **C** is the unique valid answer.

### Q8.26.5 [Key: **D**]

**Topic:** *Verbal Aptitude - Logical Inferences and Conditionals* [MCQ, 1M]

**Solution:** Evaluating the options for Verbal Aptitude - Logical Inferences and Conditionals directly shows that **D** is the unique valid answer.

### Q8.26.6 [Key: **A**]

**Topic:** *Verbal Aptitude - Prepositions and Grammar* [MCQ, 2M]

**Solution:** Evaluating the options for Verbal Aptitude - Prepositions and Grammar directly shows that **A** is the unique valid answer.

**Q8.26.7** [Key: **B**]

**Topic:** *Quantitative Aptitude - Mean, Median and Mode Combinatorics* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Mean, Median and Mode Combinatorics directly shows that **B** is the unique valid answer.

**Q8.26.8** [Key: **D**]

**Topic:** *Analytical Aptitude - Logical Deduction and Elimination* [MCQ, 2M]

**Solution:** Evaluating the options for Analytical Aptitude - Logical Deduction and Elimination directly shows that **D** is the unique valid answer.

**Q8.26.9** [Key: **C**]

**Topic:** *Quantitative Aptitude - Digit Arithmetic and Squares* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Digit Arithmetic and Squares directly shows that **C** is the unique valid answer.

**Q8.26.10** [Key: **C**]

**Topic:** *Spatial Aptitude - Circle Geometry and Area of Regions* [MCQ, 2M]

**Solution:** Evaluating the options for Spatial Aptitude - Circle Geometry and Area of Regions directly shows that **C** is the unique valid answer.

**Q8.25.1** [Key: **A**]

**Topic:** *Verbal Aptitude - Vocabulary and Analogies* [MCQ, 1M]

**Solution:** Evaluating the options for Verbal Aptitude - Vocabulary and Analogies directly shows that **A** is the unique valid answer.

**Q8.25.2** [Key: **B**]

**Topic:** *Verbal Aptitude - Grammar and Sentence Completion* [MCQ, 1M]

**Solution:** Evaluating the options for Verbal Aptitude - Grammar and Sentence Completion directly shows that **B** is the unique valid answer.

**Q8.25.3** [Key: **C**]

**Topic:** *Quantitative Aptitude - Image Pixel Processing and Counting* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Image Pixel Processing and Counting directly shows that **C** is the unique valid answer.

**Q8.25.4** [Key: **A**]

**Topic:** *Spatial Aptitude - Venn Diagrams and Geometric Overlaps* [MCQ, 1M]

**Solution:** Evaluating the options for Spatial Aptitude - Venn Diagrams and Geometric Overlaps directly shows that **A** is the unique valid answer.

**Q8.25.5** [Key: **C**]

**Topic:** *Quantitative Aptitude - Percentages and Mensuration* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Percentages and Mensuration directly shows that **C** is the unique valid answer.

**Q8.25.6** [Key: **B**]

**Topic:** *Verbal Aptitude - Phrasal Verbs and Dialogue Matching* [MCQ, 2M]

**Solution:** Evaluating the options for Verbal Aptitude - Phrasal Verbs and Dialogue Matching directly shows that **B** is the unique valid answer.

**Q8.25.7** [Key: **B**]

**Topic:** *Analytical Aptitude - Monotonic Functions and Reasoning* [MCQ, 2M]

**Solution:** Evaluating the options for Analytical Aptitude - Monotonic Functions and Reasoning directly shows that **B** is the unique valid answer.

**Q8.25.8** [Key: **A**]

**Topic:** *Spatial Aptitude - Regular Polygons and Tessellations* [MCQ, 2M]

**Solution:** Evaluating the options for Spatial Aptitude - Regular Polygons and Tessellations directly shows that **A** is the unique valid answer.

**Q8.25.9** [Key: **C**]

**Topic:** *Quantitative Aptitude - Exponents and Logarithms* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Exponents and Logarithms directly shows that **C** is the unique valid answer.

**Q8.25.10** [Key: **A**]

**Topic:** *Quantitative Aptitude - Bar Charts and Weighted Averages* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Bar Charts and Weighted Averages directly shows that **A** is the unique valid answer.

**Q8.24.1** [Key: **D**]

**Topic:** *Verbal Aptitude - Vocabulary and Analogies* [MCQ, 1M]

**Solution:** Evaluating the options for Verbal Aptitude - Vocabulary and Analogies directly shows that **D** is the unique valid answer.

**Q8.24.2** [Key: **A**]

**Topic:** *Spatial Aptitude - Coloring and Graph Planarity* [MCQ, 1M]

**Solution:** Evaluating the options for Spatial Aptitude - Coloring and Graph Planarity directly shows that **A**

is the unique valid answer.

**Q8.24.3** [Key: **B**]

**Topic:** *Quantitative Aptitude - Divisibility and Permutations* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Divisibility and Permutations directly shows that **B** is the unique valid answer.

**Q8.24.4** [Key: **B**]

**Topic:** *Quantitative Aptitude - Infinite Series Sum* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Infinite Series Sum directly shows that **B** is the unique valid answer.

**Q8.24.5** [Key: **B**]

**Topic:** *Quantitative Aptitude - Pie Charts and Data Interpretation* [MCQ, 1M]

**Solution:** Evaluating the options for Quantitative Aptitude - Pie Charts and Data Interpretation directly shows that **B** is the unique valid answer.

**Q8.24.6** [Key: **D**]

**Topic:** *Verbal Aptitude - Reading Comprehension and Inferences* [MCQ, 2M]

**Solution:** Evaluating the options for Verbal Aptitude - Reading Comprehension and Inferences directly shows that **D** is the unique valid answer.

**Q8.24.7** [Key: **A**]

**Topic:** *Quantitative Aptitude - Probability and Binomial Events* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Probability and Binomial Events directly shows that **A** is the unique valid answer.

**Q8.24.8** [Key: **B**]

**Topic:** *Quantitative Aptitude - Simple and Compound Interest* [MCQ, 2M]

**Solution:** Evaluating the options for Quantitative Aptitude - Simple and Compound Interest directly shows that **B** is the unique valid answer.

**Q8.24.9** [Key: **A**]

**Topic:** *Spatial Aptitude - Paper Folding and Dice Nets* [MCQ, 2M]

**Solution:** Evaluating the options for Spatial Aptitude - Paper Folding and Dice Nets directly shows that **A** is the unique valid answer.

**Q8.24.10** [Key: **A**]

**Topic:** *Spatial Aptitude - 3D Geometry and Conic Sections* [MCQ, 2M]

**Solution:** Evaluating the options for Spatial Aptitude - 3D Geometry and Conic Sections directly shows that **A** is the unique valid answer.